

HANDBOOK OF COMBINATORICS

VOLUME I

Edited by
R.L. Graham
M. Grötschel
L. Lovász

NORTH-HOLLAND

HANDBOOK OF COMBINATORICS

Volume 1

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HANDBOOK OF COMBINATORICS

Volume 1

edited by

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Preface

Combinatorics belongs to those areas of mathematics having experienced a most impressive growth in recent years. This growth has been fuelled in large part by the increasing importance of computers, the needs of computer science and demands from applications where discrete models play more and more important roles. But also more classical branches of mathematics have come to recognize that combinatorial structures are essential components of many mathematical theories.

Despite the dynamic state of this development, we feel that the time is ripe for summarizing the current status of the field and for surveying those major results that in our opinion will be of long-term importance. We approached leading experts in all areas of combinatorics to write chapters for this Handbook. The response was overwhelmingly enthusiastic and the result is what you see here.

The intention of the Handbook is to provide the working mathematician or computer scientist with a good overview of basic methods and paradigms, as well as important results and current issues and trends across the broad spectrum of combinatorics. However, our hope is that even specialists can benefit from reading this Handbook, by learning a leading expert's coherent and individual view of the topic.

As the reader will notice by looking at the table of contents, we have structured the Handbook into five sections: Structures, Aspects, Methods, Applications, and Horizons. We feel that viewing the whole field from different perspectives and taking different cross-sections will help to understand the underlying framework of the subject and to see the interrelationships more clearly. As a consequence of this approach, a number of the fundamental results occur in more than one chapter. We believe that this is an asset rather than a shortcoming, since it illustrates different viewpoints and interpretations of the results.

We thank the authors not only for writing the chapters but also for many helpful suggestions on the organization of the book and the presentation of the material. Many colleagues have contributed to the Handbook by reading the initial versions of the chapters and by making proposals with respect to the inclusion of topics and results as well as the structuring of the chapters. We are grateful for the significant help we received.

Even though this Handbook is quite voluminous, it was inevitable that some areas of combinatorics had to be left out or were not covered in the depth they deserved. Nevertheless, we believe that the Handbook of Combinatorics presents a comprehensive and accessible view of the present state of the field and that it will prove to be of lasting value.

Ronald Graham
Martin Grötschel
László Lovász

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Part I

Structures

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CHAPTER 1

Basic Graph Theory: Paths and Circuits

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Our purpose in this chapter is twofold: to introduce the basic concepts and techniques of graph theory, and to develop that part of the theory concerned with paths and circuits. We have attempted to weave these two strands together, proceeding from fundamental notions and elementary observations to more sophisticated concepts and methods, much as one does when engaged in research. One of the attractive features of the subject is the wealth of simply stated but challenging open problems, and we have included a large number of these. It must be emphasized that the literature on paths and circuits is extensive. Thus what is presented here is necessarily a selection. Nevertheless, our hope is that, after studying this chapter, the reader will be well equipped to embark on the chapters dealing with particular aspects of graph theory. Circuits in graphs are treated in depth in the books of Walther and Voss (1974) and Voss (1991). There are also many survey articles on particular aspects of the topic. We recommend, in particular, the articles of Carsten Thomassen. Besides their substantial contributions to the theory of graphs, each includes an informed and stimulating discussion of the question under consideration and its links to related matters.

1. Basic concepts

1.1. Graphs

A graph G consists of a nonempty set $V(G)$ of elements, called *vertices* or *nodes*, and a set $E(G)$ of elements called *edges*, together with a relation of *incidence* that associates with each edge two vertices, called its *ends*. Two graphs G and H are *isomorphic* if there are bijections $\theta : V(G) \rightarrow V(H)$ and $\phi : E(G) \rightarrow E(H)$ such that vertex v and edge e are incident in G if and only if vertex $\theta(v)$ and edge $\phi(e)$ are incident in H . The pair of mappings (θ, ϕ) is an *isomorphism* from G to H .

An edge with identical ends is a *loop* and one with distinct ends is a *link*. Two or more edges with the same pair of ends are *multiple* edges. The two ends of an edge are said to be *joined* by the edge and to be *adjacent* to one another; adjacent vertices are also referred to as *neighbours*. An edge with ends x and y is denoted by the unordered pair xy . The number of vertices of a graph G is called its *order*, the number of edges its *size*; these parameters are denoted by $v(G)$ and $e(G)$, respectively.

A *multigraph* is one with no loops, a *simple* graph one with neither loops nor multiple edges. A graph is *finite* if both its vertex set and its edge set are finite, and *infinite* otherwise. This chapter is devoted entirely to finite graphs. Quite different techniques are needed for dealing with infinite graphs. These are discussed in the survey by Thomassen (1983e) and also in chapter 42 by Hajnal.

Graphs are so named because they can be represented pictorially, a vertex being indicated by a point and an edge by a line joining the points representing its ends; fig. 1 depicts the five *Platonic graphs*, the graphs of the Platonic solids. The graph of the tetrahedron in fig. 1 is an example of a *complete* graph: any two of its vertices

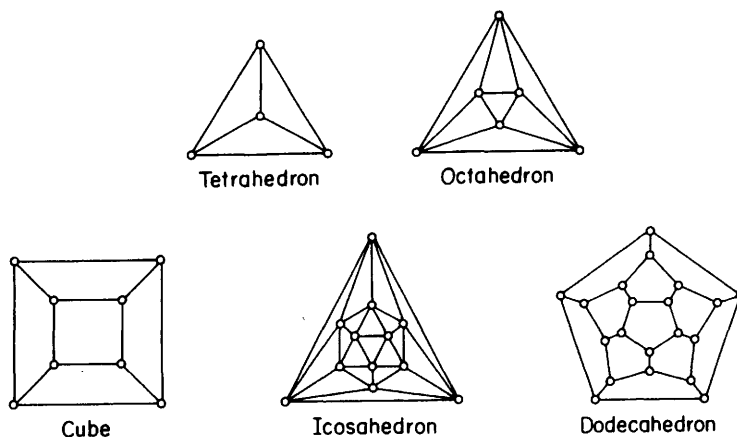


Figure 1. The Platonic graphs.

are joined by an edge. An *empty* graph, by contrast, is one with no edges. The simple complete graph on n vertices is denoted by K_n .

A graph is *bipartite* if its vertex set can be partitioned into subsets X and Y so that each edge has one end in X and one end in Y ; such a partition (X, Y) is called a *bipartition* of the graph. A bipartite graph G with bipartition (X, Y) is denoted by $G(X, Y)$. Such a graph is *balanced* if $|X| = |Y|$ and *complete* if each vertex of X is joined to each vertex of Y ; the graph of the cube in fig. 1 is a balanced bipartite graph. The simple complete bipartite graph $G(X, Y)$ in which $|X| = m$ and $|Y| = n$ is denoted by $K_{m,n}$; if $m = 1$, such a graph is called a *star*. The graphs K_5 and $K_{3,3}$ are known as the *Kuratowski graphs* because they feature in a fundamental theorem of K. Kuratowski (Theorem 5.4); $K_{3,3}$ is also sometimes referred to as the *Thomsen graph*.

It is sometimes convenient to represent the relations of adjacency and incidence in a graph by matrices. The *adjacency matrix* $A(G)$ is the matrix (a_{uv}) whose rows and columns are indexed by the vertices of G , a_{uv} being the number of edges of G joining vertices u and v . The *incidence matrix* $B(G)$ is the matrix (b_{ve}) whose rows are indexed by the vertices of G and columns by the edges of G , b_{ve} being the number of times vertex v and edge e are incident; so $b_{ve} := 2$ if e is a loop at v , $b_{ve} := 1$ if e is a link at v , and $b_{ve} := 0$ otherwise.

As a general rule, the symbol specifying the graph (typically G) is dropped from notation if the context makes clear which graph is being discussed; vertex and edge sets are then denoted simply by V and E , and adjacency and incidence matrices by A and B . By the same token, the notation $G(V, E)$ specifies a graph G with vertex set V and edge set E . An exception to this rule is the frequent use of n for the order, and occasionally m for the size, the letters v and e always denoting vertices and edges, respectively.

The set of neighbours of a vertex v in a graph G is denoted by $N_G(v)$. In a simple graph, the number of neighbours of a vertex is called its *degree*. More generally,

the *degree* of a vertex v in a graph G is the number of edges of G incident with v , a loop counting as two edges. A graph is *d-regular* if each vertex has degree d , and *regular* if it is d -regular for some d . A graph is *even* if each vertex is of even degree. The degree of a vertex v in a graph G is denoted by $d_G(v)$, the minimum, maximum, and average degrees of the vertices of a graph G by $\delta(G)$, $\Delta(G)$, and $d(G)$, respectively. An *isolated vertex* is a vertex of degree zero, a *leaf* one of degree one.

The number of edges in a graph and the degrees of its vertices are linked by a simple identity due to Euler (1736).

Proposition 1.1. *For any graph G ,*

$$\sum_{v \in V} d(v) = 2e(G).$$

Proof (by *counting in two ways*). We consider the incidence matrix $B := (b_{ve})$ of G and compute the sum of its entries in two ways:

$$\sum_{v \in V} d(v) = \sum_{v \in V} \sum_{e \in E} b_{ve} = \sum_{e \in E} \sum_{v \in V} b_{ve} = \sum_{e \in E} 2 = 2e(G). \quad \square$$

We have drawn attention to the proof technique used here, *counting in two ways*, because it is one that is frequently employed in combinatorial arguments. Further techniques will be highlighted as they are encountered; some of these are of general applicability, while others are designed specifically to study paths and circuits in graphs.

Corollary 1.2. *In any graph, the number of vertices of odd degree is even.*

1.2. Subgraphs and minors

Let G and H be graphs. If $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$, and if every edge of H has the same pair of ends in H as it has in G , the graph H is called a *subgraph* of G and the graph G a *supergraph* of H ; these relationships are denoted by $H \subseteq G$ and $G \supseteq H$, respectively. We also say that H is *contained* in G and that G *contains* H . A subgraph or supergraph H of G *spans* G if $V(H) = V(G)$.

The *union* of two subgraphs G_1 and G_2 of G is the subgraph $G_1 \cup G_2$ with vertex set $V(G_1) \cup V(G_2)$ and edge set $E(G_1) \cup E(G_2)$. Likewise the *intersection* of G_1 and G_2 is the subgraph $G_1 \cap G_2$ with vertex set $V(G_1) \cap V(G_2)$ and edge set $E(G_1) \cap E(G_2)$.

If S is a set of edges of a graph $G(V, E)$, the graph whose vertex set is V and whose edge set is $E \setminus S$ is a subgraph of G . Likewise, if S is a set of vertices of G , the subgraph of G whose vertex set is $V \setminus S$ and whose edge set consists of those edges of G with both ends in $V \setminus S$ is a subgraph of G . In both instances, this subgraph is denoted by $G \setminus S$, and is said to be obtained by *deleting* S from G . If H is a subgraph of G , we denote the subgraph $G \setminus V(H)$ simply by $G \setminus H$.

If S is a set of unordered pairs of vertices of $G(V, E)$, the graph whose vertex set is V and whose edge set is $E \cup S$ is a spanning supergraph of G . This graph is denoted by $G + S$, and is said to be obtained by *adding* S to G ; addition and deletion of edges are inverse operations. One may also add a set S of vertices to a graph. In this case, the resulting graph depends on which edges incident to the vertices of S are also added. If all pairs uv with $u \in S$ and $v \in V$ are added, and no others, the resulting graph is denoted by $G + S$.

If S is a set of edges of G , the graph whose vertex set consists of all ends of edges in S and whose edge set is S is a subgraph of G . Likewise, if S is a set of vertices of G , the graph whose vertex set is S and whose edge set consists of those edges of G with both ends in S is a subgraph of G . In both instances, this subgraph is denoted by $G[S]$ and is called the subgraph of G *induced* by S .

Any subgraph of a graph may be obtained by deleting appropriate vertices and edges. Similarly, any spanning subgraph may be obtained by deleting edges, any vertex-induced subgraph by deleting vertices, and any edge-induced subgraph by deleting edges and isolated vertices.

The subgraphs of a graph can be represented algebraically by their incidence matrices, but it turns out to be more fruitful to identify a subgraph with the characteristic function of its edge set. Let E be a set. The *characteristic function* of a subset S of E is the function $\chi_S : E \rightarrow \text{GF}(2)$ defined by

$$\chi_S(e) := \begin{cases} 1 & \text{if } e \in S, \\ 0 & \text{if } e \notin S. \end{cases}$$

The functions χ_S form a vector space over $\text{GF}(2)$. When E is the edge set of a graph G , this vector space is called the *edge space* of G , and may be regarded as the set of edge-induced subgraphs of G , the sum of two such subgraphs $G[S]$ and $G[T]$ being $G[S \Delta T]$, where Δ denotes symmetric difference. Families of subgraphs that are closed under this operation are thus of particular interest. The even subgraphs of G constitute such a family, and form a vector space $\mathcal{C}(G)$ over $\text{GF}(2)$ called the *cycle space* of G . (Originally, even graphs were called *cycles*. To avoid confusion, we have chosen not to adopt this name, because it is also commonly used for what we shall call a circuit.)

As noted earlier, deletion and addition of edges are inverse operations. Another operation closely related to deletion, but in a more subtle way, is that of contraction. If S is a set of edges of a graph G , the graph derived from G by deleting S and identifying each pair of ends of the edges of S is said to be obtained by *contracting* S , and is denoted by G/S . If H is a subgraph of G , we denote the contraction $G/E(H)$ simply by G/H . A *minor* of a graph G is any graph obtainable from G by means of a sequence of vertex deletions, edge deletions and edge contractions. Minors play a very important role in the structural theory of graphs. In particular, the absence of a particular graph as a minor has a profound effect on the properties of a graph. The survey by Robertson and Seymour (1985) gives a good overview of this topic. Minors of graphs are discussed also in chapter 5 by Thomassen.

1.3. Walks

A *walk* in a graph G is a sequence $W := v_0 e_1 v_1 e_2 \cdots e_\ell v_\ell$, where $v_0, v_1, \dots, v_\ell$ are vertices of G , $e_1, e_2, \dots, e_\ell$ are edges of G , and v_{i-1} and v_i are the ends of e_i , $1 \leq i \leq \ell$. In a simple graph, a walk is completely specified by its sequence of vertices, $v_0 v_1 \cdots v_\ell$. The vertex v_0 is the *tail* of W , the vertices $v_1, \dots, v_{\ell-1}$ its *internal vertices*, and the vertex v_ℓ its *head*; the tail and head are often referred to as the *ends* of W . The walk W is described as a v_0 -walk or a $v_0 v_\ell$ -walk if, as is often the case, one wishes to specify explicitly one or both of its ends; one also says that W *connects* v_0 and v_ℓ . The walk W is *open* if $v_0 \neq v_\ell$ and *closed* if $v_0 = v_\ell$. The *length* of W is its number of edges, namely ℓ ; the *parity* of W is the parity of its length.

We say that vertex v_i *precedes* vertex v_j on W if $i < j$, and indicate this relationship by the notation $v_i \prec v_j$. If v_i precedes v_j , the walk $v_i e_{i+1} v_{i+1} \cdots v_{j-1} e_j v_j$ is referred to as the $v_i v_j$ -segment of W , and may be denoted by $W[v_i, v_j]$, $W(v_{i-1}, v_j]$, $W[v_i, v_{j+1})$ or $W(v_{i-1}, v_{j+1})$, as convenient; or by $\overleftarrow{W}[v_j, v_i]$, to indicate traversal of the segment in the opposite sense.

The walk W is a *trail* if its edges $e_1, e_2, \dots, e_\ell$ are distinct, and a *path* if its vertices $v_0, v_1, \dots, v_\ell$ are distinct. Note that a $v_0 v_\ell$ -walk of minimum length is necessarily a $v_0 v_\ell$ -path. A closed trail of positive length whose vertices (apart from its ends) are distinct is called a *circuit*. Dirac (1952) found a link between the length of a longest path or circuit in a graph and the minimum degree.

Theorem 1.3. *Let G be a simple graph of minimum degree δ . Then*

- (i) *G contains a path P of length at least δ ;*
- (ii) *if $\delta \geq 2$, G contains a circuit C of length at least $\delta + 1$.*

Proof (by choosing a maximal path). Let $P := v_0 v_1 \cdots v_\ell$ be a maximal path of G (one that cannot be extended from either end). Then all the neighbours of v_0 lie on P . Setting $k := \max \{i: v_0 v_i \in E\}$, we have $k \geq d(v_0) \geq \delta$. Thus P is of length at least δ and, if $\delta \geq 2$, $C := v_0 v_1 \cdots v_k v_0$ is a circuit of length at least $\delta + 1$. $\square$

A *circuit decomposition* of a graph is a partition of its edge set into circuits. For a graph to have a circuit decomposition, each vertex must clearly be of even degree. Veblen (1912) proved the converse statement.

Theorem 1.4. *A graph is even if and only if it admits a circuit decomposition.*

Proof. A graph with a circuit decomposition is even because each vertex is incident with either zero or two edges of any circuit. On the other hand, a nonempty even graph has at least one circuit, by Theorem 1.3(ii), and the subgraph obtained on deleting the edges of this circuit is even. Sufficiency now follows by induction on $e(G)$. $\square$

1.4. Connection

A graph is *connected* if any two of its vertices are connected by a path. A *component* of a graph G is a maximal connected subgraph of G (that is, a connected

subgraph contained in no other connected subgraph). The number of components of a graph G is denoted by $c(G)$. The vertices and edges of an xy -path form a connected subgraph in which the ends x and y have degree one and the internal vertices have degree two (unless the path is of length zero, in which case $x = y$ has degree zero); similarly, the vertices and edges of a circuit form a connected 2-regular subgraph. The terms *path* and *circuit* will frequently be used to refer to such subgraphs. Circuits are very often called *cycles*; they are also sometimes referred to as *polygons*, short ones being given suggestive geometrical names: *digon*, *triangle*, *quadrilateral*, and so on.

The *distance* $d(x, y)$ between two vertices x and y in a graph is the length of a shortest xy -path in G , if there is one; otherwise their distance is defined to be infinite. The *diameter* of a graph is the maximum distance between its vertices. The *girth* of a graph is the length of a shortest circuit, if there is one; otherwise the girth is defined to be zero.

A connected graph G is *separable* if it contains subgraphs G_1 and G_2 , and a vertex v , such that (a) $G_1 \cup G_2 = G$, (b) $G_1 \cap G_2 = \{v\}$, and (c) $G_i \neq \{v\}$, $i = 1, 2$. Otherwise, G is *nonseparable*. A *block* of a connected graph is a maximal nonseparable subgraph; the blocks of a disconnected graph are the blocks of its components.

A *cut vertex* of a connected graph G is any vertex v satisfying conditions (b) and (c) above for some pair G_1, G_2 of subgraphs of G satisfying (a). The cut vertices of a disconnected graph are defined to be the cut vertices of its components. Thus v is a cut vertex of G if and only if $G \setminus v$ has more components than G , unless v is incident to a loop and at least one other edge. Analogously, we define a *cut edge* of a graph G to be an edge e such that $G \setminus e$ has more components than G ; equivalently, a cut edge is one that belongs to no circuit.

The circuits of a graph lie entirely within its blocks. Thus, where circuits are concerned, nonseparability is a natural condition to impose. Hartman (1983) strengthened Theorem 1.3 (ii) for nonseparable graphs by showing that the circuits of length at least $\delta + 1$ in a simple nonseparable graph G of minimum degree δ generate the cycle space of G , unless δ is odd and $G = K_{\delta+1}$. If such a graph has an odd circuit, it therefore has an odd circuit of length at least $\delta + 1$, unless δ is odd and the graph is $K_{\delta+1}$.

A graph is *acyclic*, or a *forest*, if it contains no circuit or, equivalently, if each of its edges is a cut edge. The following basic property of forests was first noted by Listing (1862).

Proposition 1.5. *Let G be a forest. Then*

$$e(G) - v(G) + c(G) = 0.$$

Proof (by induction). The assertion holds if $e(G) = 0$, since in this case each component is a single vertex. Assume that $e(G) > 0$ and that the assertion holds for all forests on fewer than $e(G)$ edges. Let G' be a subgraph of G obtained by deleting any one edge. Then G' is a forest with $e(G) - 1$ edges, $v(G)$ vertices and $c(G) + 1$

components. By the induction hypothesis, applied to G' ,

$$\begin{aligned} e(G) - v(G) + c(G) &= e(G) - 1 - v(G) + c(G) + 1 \\ &= e(G') - v(G') + c(G') = 0. \end{aligned}$$

Thus the assertion holds for all forests. $\square$

A *tree* is a connected acyclic graph. It follows from Propositions 1.1 and 1.5 that every tree on two or more vertices has at least two vertices of degree one; such vertices are called *leaves*. Any connected graph, when viewed as an assemblage of its blocks, has the structure of a tree. More precisely, the *block cut-vertex tree* of a connected graph G is the tree H whose vertices are the blocks and the cut vertices of G , a block and cut vertex being adjacent in H if the block includes the cut vertex in G : H is connected because G is, and acyclic because a block is a maximal nonseparable subgraph. The blocks of G corresponding to the leaves of H are the *endblocks* of G .

Every connected graph contains a *spanning tree*, a spanning subgraph which is a tree. In fact, the spanning trees are precisely the spanning subgraphs that are connected and are minimal with respect to the property. They are also the spanning subgraphs that are acyclic and are maximal with respect to the property, and thus can be “grown” by starting with one vertex and adding edges and vertices one by one. This latter viewpoint is of fundamental importance in both theoretical and applied contexts, as we shall see shortly in the section on search trees.

Each spanning tree T of a connected graph G determines a basis $\mathcal{C}_T$ of its cycle space $\mathcal{C}$ in a natural way. For each edge e of G not in T , the subgraph $T + e$ contains a unique circuit C_e . The set $\mathcal{C}_T$ of all such circuits is a basis for $\mathcal{C}$: it is independent because C_e is the only circuit in $\mathcal{C}_T$ that includes e , and it generates $\mathcal{C}$ because any even subgraph H of G can be expressed in the form

$$H = \sum_{e \in E(H) \setminus E(T)} C_e.$$

To see this, it suffices to observe that $H + \sum_{e \in E(H) \setminus E(T)} C_e$ is an even subgraph contained in T , and hence is empty by Theorem 1.4.

Because the cycle space of a disconnected graph is the direct sum of the cycle spaces of its components, Proposition 1.5 shows that the dimension of the cycle space of a graph G is given by the formula

$$\dim \mathcal{C}(G) = e(G) - v(G) + c(G).$$

1.5. Extremal graphs

Proposition 1.5 has the following immediate consequence.

Corollary 1.6. *Let G be a graph on n vertices and m edges, where $m \geq n$. Then G contains a circuit.*

Corollary 1.6 is a very simple instance of a result in *extremal graph theory*. It asserts that every graph on n vertices and sufficiently many edges (at least n) contains a circuit. Furthermore, the bound on the number of edges is best possible because there exist graphs on n vertices and $n - 1$ edges that do not contain circuits (namely trees); these graphs (trees) are referred to as *extremal graphs* for the property in question (containing a circuit).

While a graph on n vertices and at least n edges is guaranteed to contain a circuit, the length of the circuit may take on any value between 1 and n . The following theorem, due to Mantel (1907), specifies how many edges are needed to guarantee the existence of a triangle in a simple graph.

Theorem 1.7. *Let G be a simple graph on n vertices and m edges, where $m > n^2/4$. Then G contains a triangle.*

Proof (by contradiction). Suppose that G contains no triangle. Then adjacent vertices have disjoint sets of neighbours, so

$$d(u) + d(v) \leq n, \quad \text{for each edge } uv \in E.$$

Summing over all edges of G , we have

$$\sum_{v \in V} d(v)^2 \leq mn.$$

On the other hand, by Chebyshev's inequality and Proposition 1.1,

$$\sum_{v \in V} d(v)^2 \geq \frac{(\sum_{v \in V} d(v))^2}{n} = \frac{4m^2}{n}.$$

These two inequalities imply that $m \leq n^2/4$, contradicting the hypothesis. $\square$

Equality holds in the above inequalities only if G is regular of degree $n/2$. It follows readily that the extremal graphs in this case are the complete bipartite graphs $K_{n/2, n/2}$. Theorem 1.7 is a special case of Turán's Theorem, the cornerstone of extremal graph theory. This topic is the subject of chapter 23 by Bollobás.

1.6. Search trees

Theorem 1.3, Corollary 1.6 and Theorem 1.7 assert only that certain types of paths and circuits *may* be found in graphs with appropriate properties. They do not address the question, of paramount importance in practical applications, of *how* to find such subgraphs. Of course, an exhaustive search of a finite graph can always, in theory, be completed in finite time. In practice, however, this is not good enough – time is of the essence; to be effective, an algorithm must be efficient. The running time of an algorithm is measured by the total number of basic operations (arithmetic operations, comparisons, and so on) required to implement it, and depends on the size and nature of the input. The *complexity* of a graph-theoretical algorithm is its running time, in the worst case, on graphs of a given

size (meaning, usually, their number of vertices and/or edges). A *polynomial-time* algorithm is one whose complexity is bounded above by a polynomial in the size of the input graph, an *exponential-time* algorithm one whose complexity is bounded below by an exponential function in the size of the input graph. Polynomial-time algorithms are regarded as efficient because their running time grows relatively slowly with the size of the input; this means that fairly large graphs (perhaps with several hundred vertices) can be handled effectively. Exponential-time algorithms, on the other hand, may take inordinately long to process graphs of very modest size (say, twenty vertices).

Spanning trees provide an efficient way to search graphs, and are the key to many polynomial-time algorithms. A *search tree* in a connected graph is a spanning tree whose vertex set $\{v_0, v_1, \dots, v_{n-1}\}$ and edge set $\{e_1, e_2, \dots, e_{n-1}\}$ are ordered in such a way that, for each j , $1 \leq j \leq n-1$, the subgraph with vertex set $\{v_0, v_1, \dots, v_j\}$ and edge set $\{e_1, e_2, \dots, e_j\}$ is a tree; the vertex v_0 is the *root* of the search tree. In other words, a search tree is the *growth process* of a spanning tree, not just the tree itself. Starting at the root v_0 , the tree is grown by adding a vertex v_j and edge $e_j = v_i v_j$ at each stage of the process; we call v_j a *child* of v_i , and v_i the *parent* of v_j , in the tree. In general, there is some freedom as to the choice of v_j and e_j , but if they are chosen appropriately, important structural information about the graph is encoded in the resulting search tree. The most basic type of tree search, breadth-first search, was developed by Dijkstra (1959).

A *breadth-first search tree* is a search tree with vertex set $\{v_0, v_1, \dots, v_{n-1}\}$ and edge set $\{e_1, e_2, \dots, e_{n-1}\}$ where, for each j , $1 \leq j \leq n-1$, v_j and e_j are chosen in such a way that the parent of v_j is the vertex v_i with smallest possible index i . Such a tree can be grown in linear time. The key property of breadth-first search is that the distance in a breadth-first search tree between its root and any other vertex is the same as their distance in the graph itself. Thus, by growing a breadth-first search tree from each vertex, shortest paths between all pairs of vertices in a graph can be found. In particular, the diameter can be determined.

Breadth-first search also enables one to decide easily whether or not a graph is bipartite. Let T be a breadth-first search tree with root x in a graph G , which we assume to be connected. (If it is not, the procedure may be applied to each component in turn.) Let X be the set of vertices whose distance from x is even, and Y the set of vertices whose distance from x is odd. If no two vertices of X and no two vertices of Y are adjacent, (X, Y) is a bipartition of G and so G is bipartite. If, on the other hand, two of these vertices, u and v , are adjacent, consider the ux -path P and vx -path Q in T . Let w denote the first vertex common to P and Q . Then $P[u, w]$ and $Q[v, w]$ are paths of the same parity and so, together with the edge uv , form an odd circuit of G . We conclude that G is not bipartite in this case, since a circuit in a bipartite graph alternates between the sets of its bipartition and thus is necessarily even.

The above algorithm outputs a bipartition if the input graph is bipartite and an odd circuit if it is not. Thus it supplies, as a byproduct, the following characterization of bipartite graphs, due to König (1916).

Theorem 1.8. *A graph is bipartite if and only if it contains no odd circuit.*

If a graph is not bipartite, a shortest odd circuit can be found by growing a breadth-first search tree with root x from each vertex x . It is less easy to find a shortest even circuit, if there is one; an algorithm for doing so is described in section 4 (at the end of the discussion of stability number).

The above examples illustrate the power of breadth-first search. Another highly effective tree search procedure, depth-first search, was developed by Tarjan (1972). A *depth-first search tree* is a search tree with vertex set $\{v_0, v_1, \dots, v_{n-1}\}$ and edge set $\{e_1, e_2, \dots, e_{n-1}\}$ where, for each j , $1 \leq j \leq n-1$, v_j and e_j are chosen in such a way that the parent of v_j is the vertex v_i with largest possible index i . Depth-first search also takes linear time. It can be used to find the blocks and cut vertices of a graph, and has many other important applications. These are described in chapter 28 by Grötschel, and in the book by Aho et al. (1974).

In applications, the edges of a graph often have associated weights, or costs. An *edge-weighted graph* is a graph G together with a *weight function* w that assigns to each edge of G a *weight* $w(e)$. Usually, the range of w is the set of nonnegative real numbers; in some instances, however, a different choice is appropriate: the set of all real numbers, the set of integers, or the set of integers modulo m , for example. The *weight* of an edge-weighted graph is the sum of the weights on its edges. An unweighted graph can be regarded as an edge-weighted graph in which each edge has weight one; its weight is then simply its size. In this way, results about graphs can sometimes be generalized to weighted graphs. Dijkstra (1959) showed, for instance, not only how to find shortest paths in graphs but, more generally, how to find minimum-weight paths in edge-weighted graphs with nonnegative weights.

Most of the theorems to be mentioned in this chapter have proofs that can be adapted fairly easily to yield polynomial-time algorithms. However, there are many basic problems for which polynomial-time algorithms have yet to be found, and indeed might well not exist. In fact, determining which problems are solvable in polynomial time is one of the main preoccupations of theoretical computer scientists. An important class of problems in this regard is the class $\mathcal{NP}$ (standing for *nondeterministic polynomial-time*). We give here an informal definition; a precise treatment can be found in chapter 29, or in the book by Garey and Johnson (1979).

A *decision problem* is a question whose answer is either “yes” or “no”; any object to which the question is addressed is an *instance* of the problem. A decision problem belongs to the class $\mathcal{P}$ if there is a polynomial-time algorithm that solves any instance of the problem in polynomial time. A decision problem belongs to the class $\mathcal{NP}$ if, given any instance of the problem whose answer is ‘yes’, there is a certificate validating this fact which can be checked in polynomial time, and one, moreover, whose length is bounded above by a polynomial in the size of the input; such a certificate is said to be *succinct*. It follows from these definitions that $\mathcal{P} \subseteq \mathcal{NP}$, since a polynomial-time algorithm is, in itself, a succinct certificate.

The problem of determining whether a graph is bipartite belongs to $\mathcal{P}$ and hence also to $\mathcal{NP}$; likewise, the problem of determining whether a graph is nonbipartite belongs to both $\mathcal{P}$ and $\mathcal{NP}$. On the other hand, even though no polynomial-time

algorithm is known for determining whether a graph contains a spanning circuit, this problem belongs to $\mathcal{NP}$, an appropriate certificate being the spanning circuit itself. Verifying that a graph contains no spanning circuit, by contrast, appears to be much harder; indeed, its status as regards membership in $\mathcal{NP}$ is unresolved.

As noted above, $\mathcal{P} \subseteq \mathcal{NP}$. Whether $\mathcal{P} = \mathcal{NP}$ or not is the most fundamental open question in theoretical computer science. The so-called $\mathcal{NP}$ -complete problems, which form a subclass of $\mathcal{NP}$, are of much interest in this regard. These problems, which include the problem of determining whether a graph contains a spanning circuit, are algorithmically the most difficult in $\mathcal{NP}$, in the sense that if any one of them can be solved in polynomial time, then so can every problem in $\mathcal{NP}$ (that is, $\mathcal{P} = \mathcal{NP}$). Several $\mathcal{NP}$ -complete problems will be encountered in this chapter. The $\mathcal{NP}$ -complete problems belong to a larger class of seemingly intractable problems, the class of $\mathcal{NP}$ -hard problems. The concept of $\mathcal{NP}$ -completeness was set on a sound theoretical footing by Cook (1971) and further developed by Karp (1972).

1.7. Directed graphs

A *directed graph*, or *digraph*, D is a graph G in which each edge is assigned a direction, one end being designated its *tail* and the other its *head*. We call D an *orientation* of G , and write $D := \vec{G}$. An edge with tail x and head y is denoted by the ordered pair (x, y) ; we say that x *dominates* y and write $x \rightarrow y$. In diagrams, the direction of an edge is indicated by an arrow pointing towards the head of the edge. Two or more edges with the same tail and head are *multiple edges*. A *strict digraph* is one with neither loops nor multiple edges.

The *associated digraph* $D(G)$ of a graph G is the digraph obtained from G by replacing each edge by two oppositely directed edges with the same ends. An example is the *strict complete digraph*, the associated digraph $D(K_n)$ of the simple complete graph K_n . The *underlying graph* $G(D)$ of a digraph D is the graph obtained from D by ignoring the orientations of its edges. An *oriented graph* is a digraph whose underlying graph is simple. A *tournament* is a digraph whose underlying graph is simple and complete.

The *adjacency matrix* $A(D)$ of a digraph D is the matrix (a_{uv}) whose rows and columns are indexed by the vertices of D , a_{uv} being the number of edges of D with tail u and head v .

Let v be a vertex of a digraph D . The *outdegree* $d_D^+(v)$ of v in D is the number of edges of D whose tail is v , the *indegree* $d_D^-(v)$ the number of edges of D whose head is v . A *sink* is a vertex of outdegree zero, a *source* a vertex of indegree zero. The analogue for digraphs of Proposition 1.1 asserts that, for any digraph D ,

$$\sum_{v \in V} d^-(v) = e(D) = \sum_{v \in V} d^+(v).$$

A *directed walk* in a digraph D is a sequence $W := (v_0, e_1, v_1, \dots, e_\ell, v_\ell)$, where $v_0, v_1, \dots, v_\ell$ are vertices of D , $e_1, \dots, e_\ell$ are edges of D , and v_{i-1} and v_i are the tail and head, respectively, of e_i , $1 \leq i \leq \ell$. The vertices v_0 and v_ℓ are the *tail* and *head*

of W ; when one wishes to refer to them explicitly, W is described as a (v_0, v_ℓ) -walk. The notions of *directed trail*, *directed path*, and *directed circuit* are defined in an analogous manner, as are *directed girth* and *directed diameter*. A digraph is *strong* (or *strongly connected*) if any two vertices x and y are connected by an (x, y) -path. A *strong component* of a digraph D is a maximal strong subdigraph of D .

Graphs can be regarded as a subclass of digraphs, a graph being identified with its associated digraph. This leads to a basic question, one that we shall examine with respect to paths and circuits: *which properties of graphs extend to digraphs?* Another fundamental question has to do with the relationship between a graph and its orientations: *how are the properties of a graph reflected in the properties of its orientations?* This latter question was studied by Chvátal and Thomassen (1978) with regard to girth and diameter. They established the existence of an integer-valued function f and an orientation function $\rightarrow$ such that, for any graph G and any edge e of G that belongs to a circuit of length at most k , the image $\overrightarrow{e}$ of e belongs to a directed circuit of length at most $f(k)$ in $\overrightarrow{G}$. This implies, in particular, that a connected graph with no cut edge has a strongly connected orientation, a result due to Robbins (1939). Chvátal and Thomassen (1978) obtained a similar but sharper result for the diameter.

Theorem 1.9. *Let G be a graph of diameter d with no cut edge. Then G admits an orientation of directed diameter at most $2d^2 + 2d$.*

A digraph is *acyclic* if it contains no directed circuit. An acyclic tournament is called a *transitive tournament*, the one of order n being denoted TT_n . The digraph D^* obtained from D by contracting each strong component to a single vertex is an acyclic digraph, the *condensation* of D ; if D is a tournament, D^* is a transitive tournament. A strong component of D is called *maximal* if it has indegree zero in D^* and *minimal* if it has outdegree zero in D^* .

An *arborescence* is an orientation of a tree in which one vertex, called the *root*, has indegree zero and every other vertex has indegree one. The analogue for digraphs of a breadth-first search tree is a *breadth-first search arborescence*. Shortest directed paths between pairs of vertices in a digraph can be found efficiently by growing such arborescences. In a strong digraph, a directed odd circuit can also be found in this way, if there is one. Consider a breadth-first search arborescence with root x . Let X be the set of vertices whose distance from x is even, and Y the set of vertices whose distance from x is odd. If some edge e joins two vertices in the same set, x is connected to the head y of e by directed paths of both parities. The concatenation of one of these paths with a (y, x) -path is a directed closed walk of odd length, which necessarily contains a directed odd circuit. On the other hand, if each edge has one end in X and the other end in Y , the digraph is bipartite. Thus a digraph has a directed odd circuit if and only if some strong component is not bipartite.

This characterization leads directly to a theorem of Richardson (1953) on the existence of kernels in digraphs. A *kernel* in a digraph D is a set S of pairwise nonadjacent vertices that together dominate every vertex of $V \setminus S$. Many digraphs

fail to have kernels, the simplest being the directed odd circuits. What Richardson proved is that a digraph without a kernel necessarily contains a directed odd circuit.

Theorem 1.10. *Let D be a digraph with no directed odd circuit. Then D has a kernel.*

Proof. By induction on the order of D . If D is strong, then D is bipartite and each class of the bipartition is a kernel of D . If D is not strong, let D_1 be a maximal strong component of D , with vertex set V_1 . By the induction hypothesis, D_1 has a kernel, S_1 . Let V_2 be the set of vertices of D dominated by vertices of S_1 , and define $D_2 := D \setminus (V_1 \cup V_2)$. Again by induction, D_2 has a kernel, S_2 . The set $S_1 \cup S_2$ is a kernel of D . $\square$

Kernels originated in the analysis of games on graphs, a kernel representing a set of winning positions. But they have other applications. We refer the reader to Berge (1985, chapter 14) for further information.

A shortest directed odd circuit is a shortest directed closed odd walk, and such a walk, if one exists, can be found in polynomial time by taking odd powers of the adjacency matrix until a nonzero diagonal entry is encountered. Finding a directed even circuit appears to be a much harder proposition. Friedland (1989) gave a simple necessary and sufficient condition in terms of the adjacency matrix for a digraph to contain such a circuit.

Theorem 1.11. *Let D be a digraph, with adjacency matrix A . Then D has a directed even circuit if and only if*

$$\text{per}(I + A) \neq \det(I + A),$$

where per denotes the permanent and $\det$ the determinant.

However, no polynomial-time algorithm has been developed to determine when this condition is satisfied. On the other hand, the problem of finding a directed even circuit in a digraph (posed by D.H. Younger – see Bermond and Thomassen 1981) has not been proved $\mathcal{NP}$ -complete.

Problem 1.12. Is there a polynomial-time algorithm for finding a directed even circuit in a digraph?

Thomassen (1993) has described such an algorithm for planar digraphs. Besides being a fundamental question, Problem 1.12 has an intriguing knife-edge quality: the slightly harder question of deciding whether there is a directed odd (or even) circuit through a given edge in a digraph is already $\mathcal{NP}$ -complete (Thomassen 1985a).

Problem 1.12 turns out to be equivalent to a question about a special class of digraphs considered by Seymour and Thomassen (1987). To *subdivide* an edge $e := xy$ in a graph is to replace e by a path $P := xvy$, where v is a new vertex. A *subdivision* of a graph is any graph derivable from it by recursively subdividing

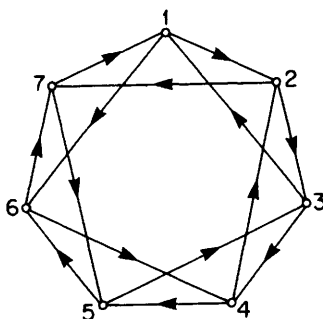


Figure 2. The Koh digraph.

edges. Analogous definitions apply to digraphs, the path P being a directed path in this case. Seymour and Thomassen (1987) characterized the digraphs with the property that every subdivision contains a directed even circuit. This characterization does not, however, lead to a polynomial-time recognition algorithm. Indeed, Seymour and Thomassen showed that the existence of such an algorithm is equivalent to the existence of a polynomial-time algorithm for finding a directed even circuit in a digraph.

Thomassen (1992) found a sufficient condition in terms of the minimum indegree and outdegree for the existence of a directed even circuit in a strong digraph.

Theorem 1.13. *Let D be a strict strong digraph of minimum indegree δ^- and minimum outdegree δ^+ , where $\delta^- \geq 3$ and $\delta^+ \geq 3$. Then D contains a directed even circuit.*

The bound on the degrees in Theorem 1.13 is sharp. Figure 2 shows a strong 2-diregular digraph on seven vertices without directed even circuits, found by Koh (1976).

1.8. Hypergraphs

A *hypergraph* H consists of a set $V(H)$ of elements, called *vertices*, and a family $E(H)$ of subsets of $V(H)$, called *hyperedges* (or simply *edges*, when there is no ambiguity). An example is the *Fano hypergraph*, shown in fig. 3, which has as vertices and edges the seven points and seven lines of the *Fano plane*, the finite projective plane $\text{PG}(2, 2)$ (see, for example, Biggs and White 1979).

Note that the seven 3-subsets defined by the vertices of the Koh digraph (fig. 2) and their outneighbours constitute the edges of the Fano hypergraph; this is an instance of a connection established by Seymour (1974) between strong digraphs without directed even circuits and a certain class of hypergraphs of which the Fano hypergraph is prototypical.

A vertex v and edge e of a hypergraph are *incident* if $v \in e$. The *incidence graph* of a hypergraph H is the bipartite graph with bipartition $(V(H), E(H))$ in which

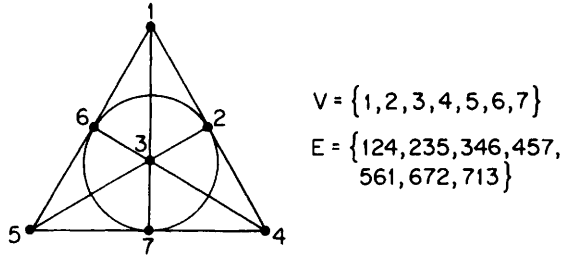


Figure 3. The Fano hypergraph.

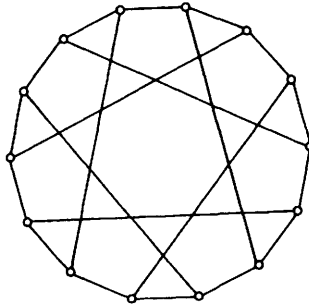


Figure 4. The Heawood graph.

$v \in V(H)$ and $e \in E(H)$ are adjacent if and only if they are incident in H . The incidence graph of the Fano hypergraph is a 3-regular graph on fourteen vertices known as the *Heawood graph*, shown in fig. 4.

With each hypergraph, we may thus associate a bipartite graph. Conversely, to each bipartite graph $G(X, Y)$, there corresponds a hypergraph H whose vertex set is X and whose hyperedges are the sets of neighbours of the vertices of Y . In this way, we can recover a hypergraph from its incidence graph, so the two structures are, in a sense, one and the same. Many other hypergraphs of interest can be associated with a graph G . We have already encountered one, the cycle space $\mathcal{C}(G)$, which may be regarded as the hypergraph whose hyperedges are the edge sets of the even subgraphs of G . Retaining only the minimal non-null elements of $\mathcal{C}(G)$, we obtain the hypergraph whose hyperedges are the edge sets of the circuits of G . This hypergraph has two special properties: (a) no proper subset of a hyperedge is a hyperedge, and (b) if C_1 and C_2 are hyperedges and e is an element of $C_1 \cap C_2$, then $(C_1 \cup C_2) \setminus e$ contains a hyperedge. A hypergraph with property (a) is termed a *clutter*, one with properties (a) and (b) a *matroid*. The matroid defined by the circuits of a graph G is called the *cycle matroid* of G . Although they are a substantial abstraction of graphs, matroids capture perfectly certain graph-theoretical concepts and properties; indeed, they are the appropriate framework for the study of graph minors and for the treatment of a number of significant questions about circuits, forests and spanning trees. They are also of

fundamental importance in the related field of combinatorial optimization. Their properties are discussed in chapters 9–11 and, more fully, in the book by Welsh (1976). Hypergraphs appear in various guises throughout combinatorics. They are treated in several chapters of this handbook, in particular in chapter 7, as well as in the book by Berge (1989).

2. Hamilton paths and circuits in graphs

2.1. Dirac's theorem

For a simple graph G , the minimum degree δ is a parameter of some importance. The larger it is, as a function of n , the more likely it is that G contains a particular subgraph. For instance, Theorem 1.3(ii) implies that if $\delta \geq 2$ then G contains a circuit, and it follows from Proposition 1.1 and Theorem 1.7 that if $\delta > n/2$ then G contains a triangle; yet another illustration is provided by Theorem 1.13. The theorem that initiated the study of long circuits (Theorem 2.1 below) is of this type.

A path or circuit that includes every vertex of the graph is called a *Hamilton path* or *circuit*. A *hamiltonian* graph is one that contains a Hamilton circuit, a *traceable* graph one that contains a Hamilton path. As we noted in section 1, the problem of determining whether a graph contains a Hamilton circuit is $\mathcal{NP}$ -complete. The more general problem of finding a minimum-weight Hamilton circuit in an edge-weighted complete graph is also $\mathcal{NP}$ -complete. Known as the *Travelling Salesman Problem*, this latter question has been the subject of much study. Its many aspects are surveyed in chapters 28 and 35, and in the book edited by Lawler et al. (1985). Whether it is $\mathcal{NP}$ -hard to determine the parity of the number of Hamilton circuits, a question raised by K.A. Berman (personal communication 1986b), is not known.

Dirac (1952) determined how large the minimum degree must be to guarantee the existence of a Hamilton circuit.

Theorem 2.1 (Dirac's Theorem). *Let G be a simple graph of minimum degree δ on n vertices, where $\delta \geq n/2$ and $n \geq 3$. Then G contains a Hamilton circuit.*

We present three proofs of this fundamental result. All are based on an observation that we shall refer to as the Lollipop Lemma (2.2). An (x, y) -lollipop is a graph $C \cup P$, where C is a circuit and P an xy -path such that $C \cap P = \{y\}$; lollipops have also been called *loops* and *lassos*.

Let C be a circuit in a graph G , with a specified orientation, and let x be a vertex of C . The predecessor of x on C is denoted by x^- , the successor of x on C by x^+ . This notation is extended to a set S of vertices of C in the natural way:

$$S^- := \{x^- | x \in S\} \quad \text{and} \quad S^+ := \{x^+ | x \in S\}.$$

Lemma 2.2 (The Lollipop Lemma). *Let $C \cup P$ be an (x, y) -lollipop in a graph G , where C is a circuit of length ℓ and P a path of length m . Then G contains the xy^- -*

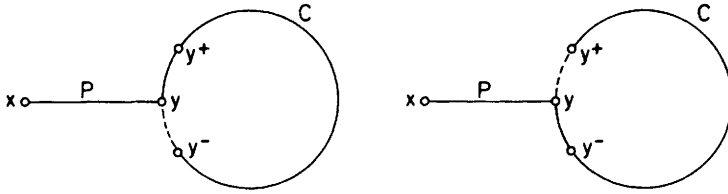


Figure 5. Two paths in a lollipop.

path $PC[y, y^-]$ and the xy^+ -path $\overleftarrow{P}C[y, y^+]$, both of length $\ell + m - 1$. In particular, if C is a longest circuit of G and P is nontrivial, x is nonadjacent to both y^- and y^+ .

The Lollipop Lemma (2.2) is illustrated in fig. 5. Simple though it is, this lemma is the basis of many theorems on paths and circuits. We shall appeal to it frequently, often without explicit reference.

The first proof that we give of Dirac's Theorem (2.1) is essentially the one given by Dirac himself.

Proof 1 (by choosing a longest circuit). Let C be a longest circuit in G , of length ℓ . By Theorem 1.3(ii), $\ell \geq \delta + 1$. If C is a Hamilton circuit, there is nothing to prove, so suppose that $G \setminus C \neq \emptyset$. Let $C \cup P$ be an (x, y) -lollipop, where P is as long as possible. Let m be the length of P . Note that since

$$\delta \geq n - \delta > n - \ell = v(G \setminus C),$$

each vertex of $G \setminus C$ is adjacent to some vertex of C , and so $m \geq 1$. Each vertex of P is a potential neighbour of x . On the other hand, no vertex of $C \setminus y$ whose distance from y along C is m or less can be adjacent to x ; and of the remaining vertices of C , no two consecutive ones can be adjacent to x . We deduce that

$$d(x) \leq m + \frac{\ell - 2m}{2} = \frac{\ell}{2} < \frac{n}{2} \leq \delta,$$

a contradiction. $\square$

The second proof of Dirac's Theorem (2.1) is due to Newman (1958).

Proof 2 (by choosing a longest path). Let P be a longest path in G , of length ℓ . We first show that G has a circuit of length $\ell + 1$. If the ends, x and y , of P are adjacent, $C := P + xy$ is such a circuit. So we may assume that x and y are not adjacent. By the choice of P , all the neighbours of x and y lie on P . Set

$$P := v_0 v_1 \cdots v_\ell,$$

where $v_0 := x$ and $v_\ell := y$, and define

$$X := \{v_i : v_0 v_{i+1} \in E\}, \quad Y := \{v_i : v_i v_\ell \in E\}.$$

We shall prove that $X \cap Y \neq \emptyset$. Since $\delta \geq n/2$,

$$|X| + |Y| = d(v_0) + d(v_\ell) \geq 2\delta \geq n,$$

and, since $y \notin X \cup Y$,

$$|X \cup Y| \leq \ell \leq n - 1.$$

Therefore

$$|X \cap Y| = |X| + |Y| - |X \cup Y| \geq 1.$$

Let $v_k \in X \cap Y$. Then

$$C := P[v_0, v_k]v_kv_\ell \overleftarrow{P}[v_\ell, v_{k+1}]v_{k+1}v_0$$

is a circuit of length $\ell + 1$.

We now show that C is a Hamilton circuit of G . If not, let $C \cup P$ be a lollipop, where P has length m and, as in Proof 1, $m \geq 1$. By the Lollipop Lemma (2.2), G contains a path of length $\ell + m$, contradicting the choice of P . Therefore C is indeed a Hamilton circuit of G . $\square$

The third proof that we give of Dirac's Theorem, due to Bondy (1981), incorporates elements of each of the other two.

Proof 3. Let C be a longest circuit in G . If C is a Hamilton circuit, there is nothing to prove, so suppose that $G \setminus C \neq \emptyset$ and let $C \cup P$ be an (x, y) -lollipop. Set

$$V_1 := V(C), \quad V_2 := V(P) \setminus y, \quad V_3 := V(G) \setminus V(C \cup P),$$

and let $d_i(x)$ and $d_i(y^+)$ denote the numbers of neighbours of x and y^+ , respectively, in V_i , $i = 1, 2, 3$. Since C is a longest circuit, we deduce that

(i) x and y^+ cannot be adjacent to consecutive vertices v and v^+ , respectively, on C ;

(ii) y^+ cannot be adjacent to any vertex of V_2 ;

(iii) x and y^+ cannot be adjacent to a common vertex of V_3 .

By (i),

$$d_1(x) + d_1(y^+) \leq |V_1|,$$

by (ii),

$$d_2(y^+) = 0,$$

and, by (iii),

$$d_3(x) + d_3(y^+) \leq |V_3|.$$

Since

$$d_2(x) \leq |V_2| - 1,$$

we have

$$2\delta \leq d(x) + d(y^+) = \sum_{i=1}^3 (d_i(x) + d_i(y^+)) \leq |V_1| + |V_2| + |V_3| - 1 = n - 1,$$

contradicting the hypothesis. Thus C is indeed a Hamilton circuit of G . $\square$

No inductive proof of Dirac's Theorem (2.1) is known. A false one was concocted by Woodall (1975) in order to illustrate the potential pitfalls of this fundamental technique.

By the simple idea of transforming one problem into another, further results can be derived very easily. We give one small illustration; others can be found in chapter 29 by Shmoys and Tardos.

Corollary 2.3. *Let G be a simple graph of minimum degree δ on n vertices, where $\delta \geq (n-1)/2$. Then G contains a Hamilton path.*

Proof. For $n=1$, the result is trivial. For $n \geq 2$, we form the graph $H := G + v$. This graph H has $n+1$ vertices and minimum degree at least $(n+1)/2$. By Dirac's Theorem (2.1), H contains a Hamilton circuit C . The subgraph $C \setminus v$ is a Hamilton path of G . $\square$

2.2. Generalizations of Dirac's theorem

Dirac's Theorem (2.1) is important because it is amenable to generalization in a great variety of ways. To start with, the hypothesis can be weakened. Proof 2 does not use the full strength of the hypothesis $\delta \geq n/2$, but just the weaker condition that the ends x and y of a longest path satisfy the inequality $d(x) + d(y) \geq n$. Moreover, this latter condition is applied only to nonadjacent vertices x and y . Thus the very same proof yields the following stronger theorem, due to Ore (1960).

Theorem 2.4 (Ore's Theorem). *Let G be a simple graph on n vertices, where $n \geq 3$, in which the degree sum of any two nonadjacent vertices is at least n . Then G contains a Hamilton circuit.*

In fact, if one examines the proof more carefully still, one sees that it is based on the following observation.

Proposition 2.5. *Let G be a simple graph on n vertices, where $n \geq 3$, and let x and y be nonadjacent vertices of G with degree sum at least n . If $G + xy$ is hamiltonian, then so is G .*

This observation leads naturally to the definition of a closure operation on graphs. The *closure* of a graph G on n vertices is the graph derived from G by recursively joining pairs of nonadjacent vertices having degree sum at least n . Repeated application of Proposition 2.5 yields the following result of Bondy and Chvátal (1976).

Lemma 2.6 (The Closure Lemma). *A simple graph G is hamiltonian if and only if its closure is hamiltonian.*

In particular, a graph is hamiltonian if its closure is complete: the graphs satisfying the hypothesis of Dirac's Theorem (2.1) are hamiltonian for this reason. Chvátal (1972) found a wider class of graphs defined in terms of their degree sequences (the widest, in a certain sense) whose closures are complete and which are consequently hamiltonian.

Theorem 2.7. *Let G be a simple graph on n vertices with degree sequence $(d_1, d_2, \dots, d_n)$, where $n \geq 3$ and $d_1 \leq d_2 \leq \dots \leq d_n$. If there is no value of $k \leq (n-1)/2$ such that $d_k \leq k$ and $d_{n-k} < n-k$, then G contains a Hamilton circuit.*

Proof. We shall show that the closure of G is complete; the conclusion will then follow directly from the Closure Lemma (2.6). Suppose, to the contrary, that this is not so. Let H be the closure of G . Then the degree sum of any two nonadjacent vertices of H is at most $n-1$. Let x and y be two such vertices of H whose degree sum is maximum. Without loss of generality, we may assume that $d_H(x) \leq d_H(y)$. Set $d_H(x) := k$. Then $k \leq (n-1)/2$ and $d_H(y) \leq n-k-1$. Denote by X the set of vertices nonadjacent to y in H . Then $|X| = n-1-d_H(y) \geq k$. Also, by the choice of x and y , $d_G(v) \leq d_H(v) \leq k$ for all $v \in X$. Similarly, if Y denotes the set of vertices that are nonadjacent to x in H , we have $|Y| = n-k-1$ and $d_G(v) \leq d_H(v) \leq n-k-1$ for all $v \in Y$. Thus G has at least k vertices of degree at most k (namely those in the set X) and at least $n-k$ vertices of degree at most $n-k-1$ (namely those in the set $Y \cup \{x\}$). This contradicts the hypothesis on G . We deduce that the closure of G is indeed complete, as claimed. $\square$

Tian and Shi (1986) observed that the Closure Lemma (2.6) can also be used to give an easy proof of a generalization of Ore's Theorem (2.4) due to Fan (1984): *if G is a simple graph on n vertices, where $n \geq 3$, in which no two vertices of degree less than $n/2$ have a common neighbour, then G contains a Hamilton circuit.*

Shi (1992) and Bollobás and Brightwell (1993) gave the following elegant generalization of Dirac's Theorem (2.1).

Theorem 2.8. *Let G be a simple graph on n vertices, and let S denote the set of vertices of degree at least $n/2$ in G . If $|S| \geq 3$, G has a circuit that includes every vertex of S .*

Proof. Let C be a circuit that includes as many vertices of S as possible. Suppose that $S \setminus V(C) \neq \emptyset$, and let $C \cup P$ be an (x, y) -lollipop, where $x \in S$. Note that C includes at least two vertices of S because any two nonadjacent vertices of S lie in a 4-circuit of G . Let z be the first vertex of S following y on C . Then, as in Proof 3 of Dirac's Theorem (2.1), the choice of C implies that $d(x) + d(z) \leq n-1$, a contradiction. $\square$

Another way in which Dirac's Theorem (2.1) can be strengthened is by refining its conclusion. Corollary 2.10 below is such a result. A graph on n vertices is

pancyclic if it contains circuits of every length ℓ , $3 \leq \ell \leq n$. In particular, a pancyclic graph is hamiltonian. The following theorem of Bondy (1971a) gives a condition under which the converse holds. The proof presented here is due to C. Thomassen (see Bollobás 1978).

Theorem 2.9. *Let G be a simple hamiltonian graph on n vertices and at least $n^2/4$ edges. Then either G is pancyclic or n is even and $G = K_{n/2, n/2}$.*

Proof (by induction). There is nothing to prove when $n = 3$, so assume that $n \geq 4$. Suppose, first, that G has a circuit of length $n - 1$,

$$C := v_1 v_2 \cdots v_{n-1} v_1.$$

Let $v := V(G) \setminus V(C)$. If $d(v) \leq (n - 1)/2$, then

$$e(G \setminus v) = e(G) - d(v) \geq \frac{n^2}{4} - \frac{n-1}{2} > \frac{(n-1)^2}{4}.$$

By the induction hypothesis, $G \setminus v$ is pancyclic. Since G is hamiltonian, by hypothesis, G is pancyclic too. If $d(v) > (n - 1)/2$, there exists for each ℓ , $3 \leq \ell \leq n$, an index i such that both vv_i and $vv_{i+\ell-2}$ are edges of G , and G contains the circuit $vv_i C[v_i, v_{i+\ell-2}] v_{i+\ell-2} v$, of length ℓ . Thus G is pancyclic in this case too.

Suppose, now, that G has no circuit of length $n - 1$. Let $v_1 v_2 \cdots v_n v_1$ be a Hamilton circuit of G . Then, for all i, j such that $1 \leq i, j \leq n$ and $j \neq i - 1, i$, at most one of the pairs $v_i v_j$ and $v_{i+1} v_{j+2}$ can be an edge of G . Thus

$$d(v_i) + d(v_{i+1}) \leq n.$$

Summing over all i , $1 \leq i \leq n$, we have

$$4e(G) = 2 \sum_{i=1}^n d(v_i) \leq n^2,$$

whence

$$e(G) \leq \frac{n^2}{4}.$$

Equality holds only if n is even and, for all i, j such that $1 \leq i, j \leq n$ and $j \neq i - 1, i$, exactly one of the pairs $v_i v_j$ and $v_{i+1} v_{j+2}$ is an edge of G . It follows easily that $G = K_{n/2, n/2}$ in this case. $\square$

Corollary 2.10. *Let G be a simple graph of minimum degree δ on n vertices, where $\delta \geq n/2$. Then either G is pancyclic or n is even and $G = K_{n/2, n/2}$.*

Corollary 2.10 was refined substantially by Amar et al. (1991), who proved that a simple hamiltonian graph of minimum degree δ on n vertices, where $\delta > 2n/5$ and n is sufficiently large, is either pancyclic or bipartite. Nash-Williams (1971a) strengthened the conclusion of Dirac's Theorem (2.1) in a very different manner.

Theorem 2.11. *Let G be a simple graph of minimum degree δ on n vertices, where $\delta \geq n/2$ and $n \geq 3$. Then G contains at least $\lfloor cn \rfloor$ edge-disjoint Hamilton circuits, where $c = 5/224 \approx 0.0223$.*

Weakening its hypothesis and strengthening its conclusion are two ways to generalize a theorem. A more common way is simply to extend the domain of the hypothesis, including the theorem as a special case. There are many such generalizations of Dirac's Theorem (2.1). We begin with one due to Pósa (1963). A *path system* is a graph each of whose components is a path.

Theorem 2.12. *Let s be a nonnegative integer, G a simple graph of minimum degree δ on n vertices, where $\delta \geq (n+s)/2$ and $n \geq 3$, and S a set of s edges of G that induce a path system. Then G contains a Hamilton circuit that includes every edge of S .*

Proof (by considering a maximal counterexample). Suppose the theorem is false for some value of n , and let G be a maximal graph on n vertices satisfying the hypotheses but not the conclusion (for some set S of edges). Since a complete graph clearly satisfies the conclusion (for any set S), G is not complete. Let x and y be two nonadjacent vertices of G . Then $G + xy$ also satisfies the hypotheses of the theorem (for the set S). By the maximality of G , there is a Hamilton circuit C in $G + xy$ that includes S . Because G does not contain such a circuit, $xy \in E(C)$. Let $P := C \setminus xy$. Then P is a Hamilton path in G that connects x and y and includes S . Let $P := v_1 v_2 \cdots v_n$, where $v_1 := x$ and $v_n := y$. A counting argument similar to the one used in Proof 2 of Dirac's Theorem (2.1) shows that, for some k , $1 \leq k \leq n-1$, $xv_{k+1} \in E$, $v_k y \in E$ and $v_k v_{k+1} \notin S$. But now $P[v_1, v_k]v_k v_n \overleftarrow{P}[v_n, v_{k+1}]v_{k+1}v_1$ is a Hamilton circuit in G that includes S , contradicting our initial supposition and establishing the theorem. $\square$

Note that the above proof, restricted to the case $s = 0$, closely resembles Proof 2 of Dirac's Theorem (2.1) but is more succinct. In the case $s = 1$, a slightly stronger conclusion can be deduced. A graph is *Hamilton-connected* if any two of its vertices are connected by a Hamilton path. Implicit in the work of Erdős and Gallai (1959, Lemma 1.6) is the following theorem.

Corollary 2.13. *Let G be a simple graph of minimum degree δ on n vertices, where $\delta \geq (n+1)/2$ and $n \geq 3$. Then G is Hamilton-connected.*

Proof. That any two adjacent vertices in G are connected by a Hamilton path follows directly from Theorem 2.12. For a nonadjacent pair, it suffices to add an edge e joining them and apply Theorem 2.12 to the graph $G + e$ and the set $S := \{e\}$. $\square$

A Hamilton circuit is one that is as long as possible, passing through each vertex of the graph. Dirac's Theorem (2.1) has been generalized to results asserting the existence of long (but not necessarily Hamilton) circuits. We mention two instances, the first due to Egawa and Miyamoto (1989) and Bollobás and Brightwell (1993), the second to Dirac himself (1952).

Theorem 2.14. *Let k be a positive integer and G a simple graph of minimum degree δ on n vertices, where $\delta \geq n/(k+1)$ and $n \geq 3$. Then G contains a circuit of length at least n/k .*

Theorem 2.15. *Let G be a simple nonseparable graph of minimum degree δ on n vertices, where $n \geq 3$. Then G contains either a circuit of length at least 2δ or a Hamilton circuit.*

Strictly speaking, Theorem 2.15 is not a generalization of Dirac's Theorem (2.1) because of the additional requirement that G be nonseparable. However, the latter property is an easy consequence of the hypothesis of Dirac's Theorem (2.1). Thus we may indeed regard the theorem as a generalization. A similar comment applies to the following result of Bondy and Nash-Williams (see Nash-Williams 1971a, p. 159), where the degree condition of Dirac's Theorem (2.1) is relaxed considerably. This result is a useful adjunct to the Hopping Lemma (6.7), a powerful proof technique developed by Woodall (1973), and will be employed in section 6.

Theorem 2.16. *Let G be a simple nonseparable graph of minimum degree δ on n vertices, where $\delta \geq (n+2)/3$ and $n \geq 3$, and let C be a longest circuit of G . Then no two vertices of $G \setminus C$ are adjacent.*

Theorems 2.15 and 2.16 can both be derived by the longest-circuit technique, the strategy being to examine the structure of the components of $G \setminus C$, where C is a longest circuit. A proof of Theorem 2.15 will be given in section 4 (see Theorem 4.11).

A Hamilton circuit can also be viewed as a spanning walk that is as short as possible, passing through each vertex just once. Jackson and Wormald (1990a) and Pruesse (1990) generalized Dirac's Theorem (2.1) to spanning walks that pass through each vertex no more than k times; such walks are called k -walks.

Theorem 2.17. *Let k be a positive integer and G a simple connected graph of minimum degree δ on n vertices, where $\delta \geq n/(k+1)$ and $n \geq 3$. Then G has a k -walk.*

A closely related concept is that of a k -tree, a spanning tree in which each vertex is of degree at most k ; 2-trees are thus Hamilton paths. It is easy to see that a k -tree admits a k -walk in which each edge of the tree is traversed twice. Jackson and Wormald (1990a) showed, conversely, that a graph which admits a k -walk necessarily contains a $(k+1)$ -tree. Theorem 2.17 thus implies that, for $k \geq 2$, a simple connected graph of minimum degree δ on n vertices, where $\delta \geq n/k$ and $n \geq 3$, has a k -tree. Win (1975) obtained a slightly sharper bound on the minimum degree.

Theorem 2.18. *Let k be an integer, $k \geq 2$, and G a simple connected graph of minimum degree δ on n vertices, where $\delta \geq (n-1)/k$. Then G contains a k -tree.*

Finally, we turn to digraphs and another theorem of Nash-Williams (1969).

Theorem 2.19. *Let D be a strict digraph on n vertices with minimum indegree δ^- and minimum outdegree δ^+ , where $\min\{\delta^-, \delta^+\} \geq n/2$. Then D contains a directed Hamilton circuit.*

This theorem, too, implies Dirac's Theorem (2.1), since if a graph G satisfies the hypothesis of Dirac's Theorem (2.1), its associated digraph $D(G)$ satisfies the hypothesis of Theorem 2.19. We defer a proof to section 3 (see Meyniel's Theorem, 3.11), observing only that the proof techniques we have described thus far fail for digraphs.

The above examples serve to illustrate not only the central role of Dirac's Theorem (2.1) in the study of circuits in graphs but also, more importantly, the process of generalization in mathematics. They can be thought of as basic elements in an ever-growing structure of generalizations of Dirac's Theorem (2.1). And, just as they generalize Dirac's Theorem (2.1), so they too are susceptible to generalization; indeed, Corollary 2.3 (same proof), Corollary 2.10 (similar proof), Theorem 2.12 (same proof), Corollary 2.13 (see Ore 1963), Theorem 2.14 (see Egawa and Miyamoto 1989, or Bollobás and Brightwell, 1993), Theorem 2.15 (Theorem 4.11), Theorem 2.16 (Theorem 4.23, case $k = 2$), Theorem 2.17 (Theorem 4.26), Theorem 2.18 (Theorem 4.27) and Theorem 2.19 (Meyniel's Theorem, 3.11) all admit generalizations of Ore type.

Many other generalizations of Dirac's Theorem (2.1) can be found in the literature; we shall encounter some in later sections. Surveys on hamiltonian graphs appear periodically. Those by Nash-Williams (1975), Lesniak-Foster (1977), Bermond (1978b), Chvátal (1985) and Gould (1991) represent a variety of viewpoints on the topic.

3. Hamilton paths and circuits in digraphs

3.1. Rédei's theorem

In this section, we consider Hamilton paths and circuits in digraphs. The central result here, playing the analogous role to Dirac's Theorem (2.1), is a theorem on tournaments due to Rédei (1934).

Theorem 3.1 (Rédei's Theorem). *Every tournament contains a directed Hamilton path.*

This theorem is an immediate consequence of the following lemma, the basic idea of which is due to D. König and P. Veress (see Rédei 1934).

Lemma 3.2. *Let D be a strict digraph, P an (x, y) -path in D of length ℓ , and $v \in V(D) \setminus V(P)$. Then*

- (i) $d_P(v) \leq \ell + 2$ if there is no (x, y) -path P' in D such that $V(P') = V(P) \cup \{v\}$;
- (ii) $d_P(v) \leq \ell$ if P is a longest directed path in D .

Proof. (i) Let $P := (v_0, v_1, \dots, v_\ell)$, where $v_0 := x$ and $v_\ell := y$, and set

$$X := \{v_i \in V(P) \setminus v_0 : (v, v_i) \in E\},$$

$$Y := \{v_i \in V(P) \setminus v_0 : (v_{i-1}, v) \in E\}.$$

Then

$$X \cup Y \subseteq \{v_1, v_2, \dots, v_\ell\} \quad \text{and} \quad X \cap Y = \emptyset,$$

for if $v_i \in X \cap Y$, the (x, y) -path $P' := (v_0, v_1, \dots, v_{i-1}, v, v_i, \dots, v_\ell)$ contradicts the hypothesis on P . Thus

$$d_P(v) \leq |X| + |Y| + 2 = |X \cup Y| + |X \cap Y| + 2 \leq \ell + 2.$$

(ii) If P is a longest directed path in D , neither (v, v_0) nor (v_ℓ, v) can be an edge of D , and the above argument yields

$$d_P(v) \leq |X| + |Y| \leq \ell. \quad \square$$

Proof of Theorem 3.1. Let T be a tournament and P a longest directed path in T , of length ℓ . If P is not a Hamilton path of T , let $v \in V(T) \setminus V(P)$. Since T is a tournament, $d_P(v) = \ell + 1$. On the other hand, by Lemma 3.2, $d_P(v) \leq \ell$, a contradiction. Thus P is, indeed, a Hamilton path of T . $\square$

3.2. Generalizations of Rédei's theorem

Rédei's Theorem (3.1), like Dirac's Theorem (2.1), admits a number of interesting generalizations. Thomassen (1982), for instance, refined it by imposing a rather natural degree condition on the ends of the Hamilton path.

Theorem 3.3. *Every tournament contains a directed Hamilton path whose tail is a vertex of maximum outdegree and whose head is a vertex of maximum indegree.*

Another generalization is a theorem of Rédei (1934) on the parity of the number of directed Hamilton paths; it was from this result that Rédei's Theorem (3.1) was originally deduced.

Theorem 3.4. *Every tournament contains an odd number of directed Hamilton paths.*

Theorem 3.4 is established by means of a proof technique known as *inclusion-exclusion*, an inversion formula with applications throughout mathematics. The following version of the formula suffices for our purpose; more general versions are described in chapter 21 by Gessel and Stanley. We omit the proof.

Formula 3.5 (Inclusion-Exclusion). *Let Z be a finite set and $f: 2^Z \rightarrow \mathbb{R}$ a real-valued function defined on the subsets of Z . Define the function $g: 2^Z \rightarrow \mathbb{R}$ by*

$$g(X) := \sum_{X \subseteq Y \subseteq Z} f(Y).$$

Then

$$f(X) = \sum_{X \subseteq Y \subseteq Z} (-1)^{|Y|-|X|} g(Y).$$

If D is a strict digraph with vertex set $V := \{1, 2, \dots, n\}$ and edge set E , and if π is a permutation of V , the subset E_π of E defined by

$$E_\pi := E \cap \{(\pi(i), \pi(i+1)) : i = 1, 2, \dots, n-1\}$$

induces a digraph each of whose components is a directed path. Such a digraph we call a *directed-path system*.

Proposition 3.6. *Let D be a strict digraph with vertex set $V := \{1, 2, \dots, n\}$ and edge set E . Denote by Π the set of permutations of V and by $\mathcal{H}$ the set of directed Hamilton paths of D . For each subset X of E , define*

$$f(X) := |\{\pi \in \Pi : X = E_\pi\}| \quad \text{and} \quad h(X) := |\{H \in \mathcal{H} : X \subseteq E(H)\}|.$$

Then

$$f(X) \equiv h(X) \pmod{2}.$$

Proof. For each subset X of E , define $g(X) := |\{\pi \in \Pi : X \subseteq E_\pi\}|$. Then

$$g(X) = \sum_{X \subseteq Y \subseteq E} f(Y),$$

and so, by the Inclusion–Exclusion Formula (3.5),

$$f(X) = \sum_{X \subseteq Y \subseteq E} (-1)^{|Y|-|X|} g(Y).$$

Observe that $g(Y) = r!$ if and only if the spanning subdigraph of D with edge set Y is a directed-path system with r components. Thus $g(Y)$ is odd if and only if Y induces a directed Hamilton path H of D , and

$$f(X) \equiv \sum_{\{H \in \mathcal{H} : X \subseteq E(H)\}} (-1)^{n-1-|X|} \equiv h(X) \pmod{2}. \quad \square$$

Proof of Theorem 3.4. The theorem is valid for transitive tournaments, each of which has precisely one directed Hamilton path. Since any tournament T on n vertices can be derived from the transitive tournament TT_n by reorienting appropriate edges, it suffices to prove that the parity of $h(T)$, the number of directed Hamilton paths in T , is unaltered by the reorientation of any one edge e . Taking $X := \{e\}$ in Proposition 3.6, we have

$$f(e) \equiv h(e) \pmod{2}.$$

Thus, if T' denotes the tournament derived from T by reorienting e ,

$$h(T') = h(T) + f(e) - h(e) \equiv h(T) \pmod{2}. \quad \square$$

A third generalization of Rédei's Theorem (3.1) by Thomason (1986), asserting the existence in all sufficiently large tournaments of oriented Hamilton paths of every type (not just directed Hamilton paths). The following definition makes this notion precise.

Let D be a digraph, and let $P := v_0 e_1 v_1 \cdots e_\ell v_\ell$ be a (not necessarily directed) path in D . The *type* of P is the sequence $(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_\ell)$, where

$$\varepsilon_i := \begin{cases} - & \text{if } e_i \text{ has head } v_{i-1}, \\ + & \text{if } e_i \text{ has head } v_i. \end{cases}$$

The *type* of a circuit $v_1 e_1 \cdots v_\ell e_\ell v_1$ is defined similarly. A directed path or circuit is thus one of type $(+, +, \dots, +)$.

Theorem 3.7. *Let T be a tournament on n vertices, where $n \geq 2^{128}$. Then T contains a Hamilton circuit of each type except, possibly, the directed Hamilton circuit. In particular, T contains a Hamilton path of each type.*

An analogous generalization of Theorem 3.4 was obtained by Forcade (1973), who discovered the remarkable fact that the parity of the number of Hamilton paths of any type in a tournament is independent of the tournament, depending only on the path type. Perhaps more surprising still is the following beautiful corollary of Forcade's theorem.

Theorem 3.8. *Let T be a tournament of order 2^n . Then T contains an odd number of Hamilton paths of each type.*

Excluding the directed Hamilton circuit in Theorem 3.7 is essential; the transitive tournament, for instance, contains no directed circuits whatsoever. Indeed, an obvious necessary condition for the existence of a directed Hamilton circuit in a digraph is that it be strong. Camion (1959) showed that for tournaments this condition is also sufficient.

Theorem 3.9 (Camion's Theorem). *Every strong tournament contains a directed Hamilton circuit.*

Camion's Theorem (3.9) is yet another generalization of Rédei's Theorem (3.1). To see this, one can use a construction similar to that employed to deduce Corollary 2.3 from Dirac's Theorem (2.1). Alternatively, one can apply Camion's Theorem (3.9) to the strong components of the tournament, thereby obtaining a directed Hamilton path in each, and then piece these paths together to form a directed Hamilton path in the entire tournament.

Camion's Theorem (3.9) is proved by means of a longest-circuit technique. The following simple lemma is the basis of this and other theorems on directed circuits in strong digraphs. Let D be a digraph, C a subgraph of D , and X and Y sets of vertices of D . A C -bypass is a directed path of length at least two whose head and tail lie in C and whose internal vertices lie in $D \setminus C$. An (X, Y) -path is a directed path whose tail lies in X and whose head lies in Y .

Lemma 3.10 (The Bypass Lemma). *Let D be a strong nonseparable digraph, and let C be a nontrivial proper subgraph of D . Then D contains a C -bypass.*

Proof. Since D is nonseparable, there is a path in D of length at least two whose tail x and head y belong to C and whose internal vertices belong to $D \setminus C$. Among all such paths, we choose one in which the number of edges directed the wrong way (that is, towards x) is as small as possible. We shall show that this path P is a C -bypass. Assume the contrary, and let (u, v) be an edge of P directed towards x . Since D is strong, there exist in D a (C, u) -path P_u and a (v, C) -path P_v . It cannot be the case that the tail of P_u is y and the head of P_v is x , since then the (y, x) -walk $P_u(u, v)P_v$ would contain a C -bypass, contradicting the choice of P . Without loss of generality, therefore, we may assume that the tail of P_u is z , where $z \neq y$. But now the zy -walk $P_u P[u, y]$ contains a zy -path that contradicts the choice of P . Thus P is, indeed, a C -bypass. $\square$

Proof of Theorem 3.9. Let T be a strong tournament and C a longest directed circuit in T . If C is not a Hamilton circuit, T contains a C -bypass by the Bypass Lemma (3.10). We choose such a path P that the distance along C from its tail x to its head y is as small as possible. By the choice of C , this distance, the length of $C[x, y]$, is no less than the length of P . Let u and v be internal vertices of $C[x, y]$ and P , respectively. Since T is a tournament, one of these vertices dominates the other. If u dominates v , the (u, y) -path $(u, v)P[v, y]$ contradicts the choice of P . If, on the other hand, v dominates u , the (x, u) -path $P[x, v](v, u)$ contradicts the choice of P . We conclude that C is, indeed, a directed Hamilton circuit of T . $\square$

Moon (1968, p. 6) strengthened the conclusion of Camion's Theorem (3.9) by showing that every strong tournament is *vertex pancyclic*: each vertex lies in a directed circuit of every length. This can be proved easily using the Bypass Lemma (3.10) (see Bondy 1977). The Bypass Lemma (3.10) is also effective in establishing a far-reaching simultaneous generalization of Ore's Theorem (2.4), Theorem 2.19 and Camion's Theorem (3.9), due to Meyniel (1973). The proof we present is by Bondy and Thomassen (1977).

Theorem 3.11 (Meyniel's Theorem). *Let D be a strict strong digraph on n vertices, where $n \geq 2$, in which the degree sum of any two nonadjacent vertices is at least $2n - 1$. Then D contains a directed Hamilton circuit.*

Proof. The degree condition implies that D is nonseparable. Let $C := (v_1, v_2, \dots, v_\ell, v_1)$ be a longest directed circuit in D . If C is not a Hamilton circuit, there is a C -bypass in D by the Bypass Lemma (3.10). As in the proof of Camion's Theorem (3.9), we choose such a path P that the distance along C from its tail x to its head y is as small as possible. Let

$$V_1 := V(P) \setminus \{x, y\}, \quad V_2 := V(C(x, y)), \quad V_3 := V(C[y, x]), \quad (3.1)$$

$$V_4 := V(D) \setminus (V_1 \cup V_2 \cup V_3). \quad (3.2)$$

By the definition of a C -bypass, $V_1 \neq \emptyset$, and by the choice of C , $V_2 \neq \emptyset$. Let $u \in V_1$ and $v \in V_2$, and denote by $d_i(u)$ and $d_i(v)$ the numbers of neighbours of u and v , respectively, in V_i , $i = 1, 2, 3, 4$. Then

$$d_1(u) \leq 2|V_1| - 2.$$

By the choice of P , u and v are nonadjacent; moreover,

$$d_1(v) = 0 \quad \text{and} \quad d_2(u) = 0.$$

Also by the choice of P , there is no directed path (u, w, v) or (v, w, u) with $w \in V_4$, and so

$$d_4(u) + d_4(v) \leq 2|V_4|.$$

Finally, by Lemma 3.2 and the choice of C ,

$$d_3(u) \leq |V_3| + 1.$$

Denote by P' a longest (y, x) -path in D such that $V_3 \subseteq V(P') \subseteq V_2 \cup V_3$. Let

$$V'_2 := V_2 \setminus V(P') \quad \text{and} \quad V'_3 := V(P'),$$

and denote by $d'_i(u)$ and $d'_i(v)$ the numbers of neighbours of u and v , respectively, in V'_i , $i = 2, 3$. By the choice of C , $V'_2 \neq \emptyset$. For $v \in V'_2$,

$$d'_2(v) \leq 2|V'_2| - 2 \quad \text{and} \quad d'_3(v) \leq |V'_3| + 1,$$

the latter inequality following from Lemma 3.2 and the choice of P' . Summing the above inequalities yields

$$\begin{aligned} d(u) + d(v) &= d_1(u) + d_1(v) + d_2(u) + d_3(u) + d'_2(v) + d'_3(v) + d_4(u) + d_4(v) \\ &\leq 2|V_1| - 2 + |V_3| + 2|V'_2| + |V'_3| + 2|V_4| \\ &\leq 2|V_1| + 2|V_2| + 2|V_3| + 2|V_4| - 2 = 2n - 2. \end{aligned}$$

This contradicts the hypothesis and establishes the theorem. $\square$

4. Fundamental parameters

In this section, we introduce three important measures of the complexity of a graph: the connectivity, the stability number, and the chromatic number. These parameters are discussed in detail in chapters 2 and 4. Here we highlight their relationship to paths and circuits. We also introduce the edge analogues of the above concepts: the edge connectivity, the matching number, and the edge chromatic number, as well as one parameter of particular relevance to paths and circuits, the toughness.

4.1. Connectivity and edge connectivity

A connected graph is, by definition, one in which any two vertices are linked by a path. Alternatively, a connected graph can be regarded as one whose vertex set admits no partition into two proper subsets, where vertices in different subsets are nonadjacent. These two viewpoints lead to two different ways of measuring the degree of connectedness of a graph.

We denote a graph G with two specified vertices x and y by $G(x, y)$. The members of a collection of xy -paths in a graph $G(x, y)$ are *edge-disjoint* if no two have an edge in common, and *internally-disjoint* if no two have an internal vertex in common. A graph G on at least two vertices is *k-edge-connected* if any two vertices are connected by at least k edge-disjoint paths, and *k-connected* if any two vertices are connected by at least k internally-disjoint paths; a graph on one vertex is defined to be both *k-edge-connected* and *k-connected* for $k = 0, 1$, but not for $k \geq 2$. Thus every graph is 0-connected, a graph is 1-connected if and only if it is connected, and a loopless graph with at least two edges is 2-connected if and only if it is nonseparable. Also, since internally-disjoint paths are edge-disjoint, *k-connected* graphs are *k-edge-connected*.

Analogous definitions apply to digraphs. A digraph D is *strongly k-edge-connected* if any two vertices u and v are connected by at least k edge-disjoint (u, v) -paths, and *strongly k-connected* or *k-strong* if any two vertices u and v are connected by at least k internally-disjoint (u, v) -paths.

The *edge connectivity* of a graph G is the greatest integer k for which G is *k-edge-connected*, the *connectivity* the greatest integer k for which G is *k-connected*. Likewise, in a digraph D , the *strong connectivity* is the greatest integer k for which D is *k-strong*, and the *strong edge connectivity* is the greatest integer k for which D is *strongly k-edge-connected*. The edge connectivity of G is denoted by $\kappa'(G)$, the connectivity of G by $\kappa(G)$, the strong edge connectivity of D by $\kappa'(D)$ and the strong connectivity of D by $\kappa(D)$. These parameters can be computed in polynomial time (see chapter 2).

If X and Y are sets of vertices in a graph G , the set of edges of G with one end in X and the other end in Y is denoted by $[X, Y]$. When $X \cup Y$ is a partition of $V(G)$, the set $[X, Y]$ is called an *edge cut* of G ; if $X, Y \neq \emptyset$, the edge cut $[X, Y]$ is *nontrivial*; if $x \in X$ and $y \in Y$, it *separates* x and y . The edge cut $[X, Y]$ is also called the *coboundary* of X (or of Y). Every nonempty edge cut is a disjoint union of minimal nonempty edge cuts, or *bonds*. In a connected graph G , a set S of edges is a bond if and only if $G \setminus S$ has exactly two components. The edge cuts of a graph G form a vector space $\mathcal{B}(G)$ over $\text{GF}(2)$ called the *bond space* of G . The bond space and cycle space are orthogonal complements in the edge space. Because of this relationship, edge cuts are sometimes called *cocycles* and bonds *cocircuits*. Edge cuts are also referred to as *cuts* or *cutsets*.

Analogously, if $X \cup S \cup Y$ is a partition of V , where $X, Y \neq \emptyset$ and no edge of G has one end in X and the other end in Y , the set S is called a *vertex cut* of G ; thus a set S of vertices in a graph G is a vertex cut if and only if $G \setminus S$ is disconnected. If $x \in X$ and $y \in Y$, we say that the vertex cut S *separates* x and y ;

note that only nonadjacent vertices are separated by vertex cuts. Vertex cuts are also called *articulation sets*.

An important and extremely useful theorem due to Menger (1927) links these concepts.

Theorem 4.1 (Menger's Theorem).

- Edge version: *Let $G(x, y)$ be a graph. Then the maximum number of edge-disjoint xy -paths in G is equal to the minimum number of edges in an edge cut of G separating x and y .*
- Vertex version: *Let $G(x, y)$ be a graph, where x and y are nonadjacent vertices of G . Then the maximum number of internally-disjoint xy -paths in G is equal to the minimum number of vertices in a vertex cut of G separating x and y .*

There is a corresponding version of Menger's Theorem (4.1) for directed paths in digraphs. Proofs of both versions are given in chapter 2: Connectivity and Network Flows. Menger's Theorem (4.1) is the standard tool for dealing with questions involving connectivity. It is frequently used in the following form.

Lemma 4.2 (The Fan Lemma). *Let G be a k -connected graph, where $k \geq 1$, let x be a vertex of G , and let Y be a set of at least k vertices of G , where $x \notin Y$. Then there exist distinct vertices $y_1, y_2, \dots, y_k$ in Y and internally-disjoint paths $P_1, P_2, \dots, P_k$, such that*

- (i) P_i is an xy_i -path, $1 \leq i \leq k$, and
- (ii) $V(P_i) \cap Y = \{y_i\}$, $1 \leq i \leq k$.

Remark 4.3. Menger's Theorem (4.1) likewise guarantees that, given any two k -sets of vertices $X := \{x_1, x_2, \dots, x_k\}$ and $Y := \{y_1, y_2, \dots, y_k\}$ in a k -connected graph G , there are k disjoint paths $P_1, P_2, \dots, P_k$ connecting X and Y in G . It does not, however, guarantee that path P_i can be so chosen as to connect vertices x_i and y_i , $1 \leq i \leq k$. Such a system is called a k -linkage, and a graph is k -linked if it has a k -linkage for arbitrary k -sets X and Y . The 2-linked graphs were characterized by Seymour (1980) and Thomassen (1980c). The corresponding concept for edge-disjoint paths is called a *weak k -linkage*. Thomassen (1980c) conjectured that a k -edge-connected graph is weakly k -linked if k is odd. This would imply that a k -edge-connected graph is weakly $(k - 1)$ -linked if k is even, a result established by Huck (1991). Further information on linkages can be found in chapter 2.

The Fan Lemma (4.2) is a very versatile tool. We give here one illustration of its use, by Dirac (1960).

Theorem 4.4. *Let G be a k -connected graph, where $k \geq 2$, and let S be a set of k vertices in G . Then G contains a circuit C that includes every vertex of S .*

Proof. The theorem is true for $k = 2$, since the union of two internally-disjoint xy -paths is a circuit that includes both x and y . We proceed by induction on k .

Suppose that the theorem holds for $k < n$, where $n \geq 3$. Let G be an n -connected graph, let S be a set of n vertices of G , and let $x \in S$. Put $S' := S \setminus x$. Then S' is a set of $n - 1$ vertices in the n -connected graph G . By the induction hypothesis, there is a circuit C' in G that includes every vertex of S' . If $x \in V(C')$, we may take $C := C'$, and there is nothing more to prove. Suppose, then, that $x \notin V(C')$. Note that the vertices of S' partition C' into $n - 1$ segments. There are two cases. If C' is of length at least n , we apply the Fan Lemma (4.2) with $k := n$ and $Y := V(C')$. By the pigeonhole principle, two of the paths whose existence is guaranteed by the lemma, say P_i and P_j , must terminate at vertices y_i and y_j that lie in the same segment of C' . The desired circuit C can now be obtained from C' by replacing the $y_i y_j$ -segment of C' by the $y_i y_j$ -path $P_i \overleftarrow{P_j}$. If C' is of length $n - 1$, we apply the Fan Lemma (4.2) with $k := n - 1$ and $Y := V(C')$, and use the above argument. $\square$

Remark 4.5. Watkins and Mesner (1967) characterized the extremal graphs for Theorem 4.4, that is, those k -connected graphs in which some set of $k + 1$ vertices is included in no circuit. For $k \geq 3$, they are the k -connected graphs having a set of k vertices whose deletion leaves a graph with more than k components; there are two additional families of extremal graphs when $k = 2$.

Bermond and Lovász (1975) conjectured a digraph analogue of Theorem 4.4 for the case $k = 2$, namely that in a digraph of sufficiently high strong connectivity, any two vertices lie on a common directed circuit. This was disproved by Thomassen (1991). Bondy and Lovász (1981) generalized Theorem 4.4 by showing that, if $k \geq 2$, the circuits through a set S of k vertices in a k -connected graph G generate the cycle space of G (except in certain specified cases); any circuit of G can, moreover, be expressed as the sum of an odd number of circuits through S . This implies that there is an even circuit through S and, if G is not bipartite, also an odd circuit through S .

Egawa et al. (1991) obtained a strong common generalization of Theorem 4.4 and Theorem 2.15.

Theorem 4.6. *Let G be a simple k -connected graph of minimum degree δ , where $k \geq 2$, and let S be a set of k vertices of G . Then G contains either a circuit of length at least 2δ that includes every vertex of S or a Hamilton circuit.*

Establishing the existence of circuits through specified edges is far more challenging. Let $e_1 := x_1 y_1$ and $e_2 := x_2 y_2$ be any two edges in a 2-connected graph G . Form a new graph G' by subdividing e_i by v_i , $i = 1, 2$. Then G' is 2-connected and, by Theorem 4.4, there is a circuit in G' that includes both v_1 and v_2 . Thus, in G , there is a circuit that includes both e_1 and e_2 . This observation cannot be extended to 3-connected graphs; three edges that are incident with the same vertex, or that form an edge cut, do not lie on a common circuit. However, these are the only exceptions, as Adám (1963) proved. Lovász (1974) was led to formulate the following general conjecture.

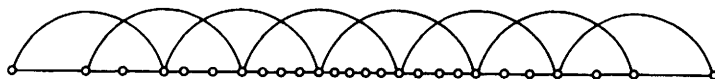


Figure 6. A vine on a path.

Conjecture 4.7. Let G be a k -connected graph, where $k \geq 2$, and let S be a set of k edges in G that induce a path system. Then there is a circuit of G that includes every edge of S , unless k is odd and S is an edge cut of G .

A slightly weaker version of this conjecture, due to Woodall (1977), was verified by Häggkvist and Thomassen (1982). Their proof uses a variant of the Hopping Lemma (6.7), to be discussed in section 6.

Theorem 4.8. Let G be a k -connected graph, where $k \geq 2$, and let S be a set of $k - 1$ edges in G that induce a path system. Then there is a circuit of G that includes every edge of S .

A related conjecture, for 3-regular graphs, was formulated by Holton and Thomassen (1986). The connectivity hypothesis is replaced here by the less stringent one of cyclic edge connectivity. A connected graph G that contains two disjoint circuits is *cyclically k -edge-connected* if it has no bond S of fewer than k edges such that both components of $G \setminus S$ contain circuits.

Conjecture 4.9. Let G be a 3-regular cyclically k -edge-connected graph and let S be a set of $k - 1$ edges in G that induce a path system. Then there is a circuit of G that includes every edge of S .

In Theorem 4.4, the Fan Lemma (4.2) was used to establish the existence of certain circuits in k -connected graphs. Another very useful tool in this context, developed by Dirac (1952), is that of a vine (see fig. 6). A *vine* on a path $P := P[x, y]$ is a set $\mathcal{P} := \{P_i[x_i, y_i] : 1 \leq i \leq m\}$ of internally-disjoint paths such that

- (a) $P_i \cap P = \{x_i, y_i\}$, $1 \leq i \leq m$;
- (b) $x = x_1 \prec x_2 \prec y_1 \preceq x_3 \prec y_2 \preceq x_4 \prec \cdots \preceq x_m \prec y_{m-1} \prec y_m = y$ on P .

Dirac (1952) showed that there is a vine on any path in a 2-connected graph. A generalization of this result was given by Bondy and Locke (1981). Vines $\mathcal{P}_1$ and $\mathcal{P}_2$ on a path $P[x, y]$ are *disjoint* if

- (a) the paths of $\mathcal{P}_1$ and $\mathcal{P}_2$ are internally-disjoint;
- (b) the only vertex that is the tail of a path of $\mathcal{P}_1$ and the tail of a path of $\mathcal{P}_2$ is x ;
- (c) the only vertex that is the head of a path of $\mathcal{P}_1$ and the head of a path of $\mathcal{P}_2$ is y .

For sets of vertices X and Y in a graph, a path whose tail lies in X and whose head lies in Y is called an $[X, Y]$ -path.

Lemma 4.10 (The Vine Lemma). Let P be a path in a k -connected graph G . Then there are $k - 1$ pairwise-disjoint vines $\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_{k-1}$ on P .

Proof. Let $P := v_0 v_1 \cdots v_\ell$. Since G is k -connected, there are k internally-disjoint $v_0 v_\ell$ -paths $Q_1, Q_2, \dots, Q_k$ in G . From each path Q_i , we construct a family $\mathcal{Q}_i$ of paths as follows. If $V(P) \cap V(Q_i) := \{w_1, w_2, \dots, w_m\}$, where $w_j \prec w_{j+1}$ on Q_i , $1 \leq j \leq m-1$, we set

$$\mathcal{Q}_i := \{Q_i[w_j, w_{j+1}]: w_j \prec w_{j+1} \text{ on } P\}.$$

We now define

$$\mathcal{Q} := \cup_{i=1}^k \mathcal{Q}_i.$$

It follows from these definitions that if

$$X_j := \{v_0, v_1, \dots, v_{j-1}\} \text{ and } Y_j := \{v_{j+1}, v_{j+2}, \dots, v_\ell\}, \quad \text{where } 0 < j < \ell,$$

then $\mathcal{Q}_i$ contains an $[X_j, Y_j]$ -path avoiding v_j , for any j such that $v_j \notin V(Q_i)$. Because the paths Q_i are internally-disjoint, at most one of them includes any particular v_j , and so $\mathcal{Q}$ contains at least $k-1$ $[X_j, Y_j]$ -paths avoiding v_j , for each j , $0 < j < \ell$.

We now construct the desired vines inductively. Initially, we select from $\mathcal{Q}$ $k-1$ $[X_1, Y_1]$ -paths $P_1, P_2, \dots, P_{k-1}$ avoiding v_1 and set $\mathcal{P}_i := \{P_i\}$, $1 \leq i \leq k-1$. At each subsequent stage, we modify the families $\mathcal{P}_i$ as follows. Let p_i denote the vertex of $\mathcal{P}_i$ that is furthest from v_0 on P . If $p_1 = p_2 = \dots = p_{k-1} = v_\ell$, the families $\mathcal{P}_i$ are vines on P satisfying the conditions of the lemma. Otherwise, we define $v_j := p_1$, where we assume (relabelling if necessary) that $p_1 \preccurlyeq p_2 \preccurlyeq \dots \preccurlyeq p_{k-1}$. Each $\mathcal{P}_i$, $2 \leq i \leq k-1$, contains at most one $[X_j, Y_j]$ -path avoiding v_j . Therefore, there is at least one more such path Q in $\mathcal{Q} \setminus \cup_{i=1}^{k-1} \mathcal{P}_i$. Let u_j be the tail of Q . We modify the family $\mathcal{P}_1$ by adding Q and deleting each path whose head lies in $P(u_j, v_j]$ except the one whose tail is closest to v_0 ; the families $\mathcal{P}_i$, $2 \leq i \leq k-1$, are not modified. This procedure terminates with $k-1$ pairwise-disjoint vines $\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_{k-1}$ on P . $\square$

Dirac (1952) used the Vine Lemma (4.10) to prove that, if a 2-connected graph has a path of length ℓ , it necessarily has a circuit of length at least $2\ell^{1/2}$. (The graph of fig. 6 is an extremal graph for this theorem, with $\ell = 25$.) The lemma was also applied by Bondy and Locke (1981) and Locke (1982) to establish analogous results for graphs of higher connectivity; for example, if a 3-connected graph has a path of length ℓ , it necessarily has a circuit of length at least $2\ell/5$. Of particular note here is the marked difference in the behaviour of 2-connected and 3-connected graphs; we shall see further instances of this phenomenon shortly. A similar change may well occur in the passage from 3-connected to 4-connected graphs; several intriguing conjectures support this view, although there is little hard evidence.

Dirac (1952) also used the Vine Lemma (4.10) to prove Theorem 2.15, a generalization of his Theorem 2.1. The same proof technique (see Pósa 1963, proof of Theorem 3) yields the corresponding generalization of Ore's Theorem (2.4) first stated explicitly by Bermond (1976) and Linial (1976).

Theorem 4.11. *Let G be a simple 2-connected graph in which the degree sum of any two nonadjacent vertices is at least d . Then G contains either a circuit of length at least d or a Hamilton circuit.*

Proof. Let $P[x, y]$ be a longest path in G and $\mathcal{P} := \{P_i[x_i, y_i]: 1 \leq i \leq m\}$ a vine on P ; note that P_1 and P_m are of length one, by the choice of P . Consider the circuit

$$C := \sum_{i=1}^m C_i,$$

where $C_i := P[x_i, y_i] \overleftarrow{P}_i$, $1 \leq i \leq m$, and assume that $\mathcal{P}$ is chosen so that

- (i) $|\mathcal{P}| = m$ is as small as possible;
- (ii) subject to (i), $|V(C) \cap V(P)|$ is as large as possible. Set $P := v_0 v_1 v_2 \cdots v_\ell$, where $v_0 := x$, $v_k := y_1$ and $v_\ell := y$, and define

$$X := \{v_i: xv_{i+1} \in E\} \quad \text{and} \quad Y := \{v_i: v_i y \in E\}.$$

If $X \cap Y \neq \emptyset$, then (as in Proof 2 of Dirac's Theorem, 2.1) C is a Hamilton circuit of G , since P is a longest path. If $X \cap Y = \emptyset$, then x and y are nonadjacent. Moreover, by the choice of $\mathcal{P}$, C contains every vertex of $(X \cup Y \cup \{y\}) \setminus \{v_{k-1}\}$ and thus is of length at least $d(x) + d(y)$. By hypothesis, this is at least d . $\square$

One consequence of Theorem 4.11 is a frequently used theorem of Erdős and Gallai (1959).

Corollary 4.12. *Let $G(x, y)$ be a simple 2-connected graph such that $d(v) \geq d$, for all $v \neq x, y$. Then there is an xy -path of length at least d in G .*

Proof. Denote by H the graph obtained from G by adding a set S of $d - 1$ vertices and joining them to one another and to both x and y . Then H is a simple nonseparable graph on n vertices, where $n \geq 2d \geq 4$, in which the degree sum of any two nonadjacent vertices is at least $2d$. By Theorem 4.11, H contains a circuit C of length at least $2d$. If C includes vertices of S , $C \setminus S$ is an xy -path in G of length at least d . Otherwise $C \subseteq G$ and, because G is 2-connected, there are disjoint paths P_1 and P_2 in G connecting $\{x, y\}$ and C . In this case, the subgraph $C \cup P_1 \cup P_2$ of G contains two xy -paths, at least one of which is of length at least d . $\square$

Theorem 4.11 is an Ore-type generalization of Theorem 2.15. Voss and Zuluaga (1977) generalized the latter theorem in another direction.

Theorem 4.13. *Let G be a simple nonseparable nonbipartite graph of minimum degree δ on n vertices, where $\delta \leq n/2$. Then G contains both an odd circuit of length at least $2\delta - 1$ and an even circuit of length at least 2δ .*

This clearly implies Theorem 2.15 in the case of nonbipartite graphs. For bipartite graphs, it suffices to add an edge between two vertices in the same part of the bipartition and then apply Theorem 4.13. We propose the following conjecture, a strengthening of one mentioned in Locke (1985). If true, it would imply Theorem 4.13 in the case of 3-connected graphs.

Conjecture 4.14. Let G be a simple 3-connected graph of minimum degree δ on n vertices, where $\delta \leq n/2$. Then the circuits of length at least $2\delta - 1$ generate the cycle space of G ; moreover, every circuit of G can be expressed as the sum of an odd number of circuits of length at least $2\delta - 1$.

We close this discussion of connectivity with an attractive conjecture of S. Smith (see Grötschel 1984).

Conjecture 4.15. In a k -connected graph, where $k \geq 2$, any two longest circuits have at least k vertices in common.

4.2. Stability number

A set S of vertices of a graph G is *stable* if no two elements of S are adjacent in G . The maximum cardinality of a stable set of G is called the *stability number* of G and denoted by $\alpha(G)$. Stable sets are also called *independent* sets and the stability number the *independence number*. Determining whether a graph has stability number at most k is an $\mathcal{NP}$ -complete problem.

A simple and surprising relationship linking the stability number, the connectivity, and the existence of Hamilton circuits was discovered by Chvátal and Erdős (1972).

Theorem 4.16 (The Chvátal–Erdős Theorem). *Let G be a graph on at least three vertices with stability number α and connectivity κ , where $\alpha \leq \kappa$. Then G is hamiltonian.*

The proof of the Chvátal–Erdős Theorem (4.16) is based on the fundamental concept of a bridge, illustrated in fig. 7. Let G be a graph and H a subgraph of G . Consider the equivalence relation $\sim$ on $E(G) \setminus E(H)$ defined by $e \sim f$ if and only if there is a walk internally-disjoint from H whose first edge is e and whose last edge is f . The subgraphs of G induced by the equivalence classes under this relation are the *bridges* of H in G ; isolated vertices of G in $G \setminus H$ are also regarded as bridges of H . A bridge having at most one edge is *degenerate*; all other bridges are *proper*. A *chord* of H is a degenerate bridge linking two distinct vertices of H . If B is a bridge of H in G , the elements of $V(B) \setminus V(H)$ are the *internal vertices* of B and the elements of $V(B) \cap V(H)$ the *vertices of attachment* of B to H . The edges of B incident to its vertices of attachment are its *edges of attachment* to H .

For the proof of the Chvátal–Erdős Theorem (4.16), we shall also need the undirected version of the concept of a bypass, defined for digraphs in section 3. Let G be a graph, and let C be a subgraph of G . A C -*bypass* in G is a path of length at least two whose ends lie in C and whose internal vertices lie in $G \setminus C$.

Proof of Theorem 4.16. Let C be a longest circuit in G . Suppose that C is not a Hamilton circuit. Let B be a proper bridge of C in G and denote by S its set of vertices of attachment to C . Any two vertices of S are connected by a C -bypass contained in B . Because C is a longest circuit, we deduce from the Lollipop Lemma

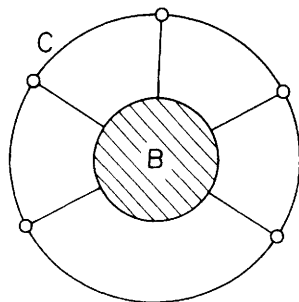


Figure 7. A bridge of a circuit.

(2.2) that S^+ is a stable set of G and that no internal vertex x of B is adjacent to any vertex of S^+ . Thus $S^+ \cup \{x\}$ is a stable set and

$$|S^+| < \alpha.$$

Also, S is a vertex cut of G ($B \setminus S$ being one component of $G \setminus S$) and so

$$|S| \geq \kappa.$$

But now

$$\kappa \leq |S| = |S^+| < \alpha,$$

contradicting the hypothesis. Therefore, C is indeed a Hamilton circuit of G . $\square$

For future reference, we note the following consequence of the Chvátal–Erdős Theorem (4.16), derived by the same technique as was used to establish Corollary 2.3 from Dirac’s Theorem (2.1).

Corollary 4.17. *Let G be a graph with stability number α and connectivity κ , where $\alpha \leq \kappa + 1$. Then G has a Hamilton path.*

Bondy (1978a) observed that the Chvátal–Erdős Theorem (4.16) implies Ore’s Theorem (2.4) and also the following result of Nash-Williams (1971b).

Corollary 4.18. *Let G be a simple d -regular graph on $2d + 1$ vertices, where $d \geq 2$. Then G is hamiltonian.*

The Chvátal–Erdős Theorem (4.16) applies only to graphs G which satisfy $\alpha \leq \kappa$. Kouider (1994) extended it to all 2-connected graphs.

Theorem 4.19. *Let G be a graph with connectivity κ , where $\kappa \geq 2$, and let H be a subgraph of G . Then G has a circuit C such that either $V(H) \subseteq V(C)$ or $\alpha(H \setminus C) \leq \alpha(H) - \kappa(G)$.*

A conjecture of Amar, Fournier, Germa, Häggkvist and Thomassen (see Fournier 1982) on the number of circuits needed to cover the vertex set of a 2-connected graph follows easily from the above theorem. It, too, is an extension of the Chvátal–Erdős Theorem (4.16). A *vertex covering* of a graph or digraph is a family of subgraphs whose union spans the graph or digraph.

Corollary 4.20. *Let G be a graph with stability number α and connectivity κ , where $\kappa \geq 2$. Then G admits a vertex covering by at most $\lceil \alpha/\kappa \rceil$ circuits.*

The 5-circuit is one example of a graph which satisfies the inequality $\alpha \leq \kappa$ of the Chvátal–Erdős Theorem (4.16) but is not pancyclic. There are many others; indeed, if G has this property, so does the composition $G[\bar{K}_r]$. Jackson and Ordaz (1990) conjecture, however, that in all such examples $\alpha = \kappa$.

Conjecture 4.21. *Let G be a graph with stability number α and connectivity κ , where $\alpha \leq \kappa - 1$. Then G is pancyclic.*

The idea of considering the minimum degree sum of two nonadjacent vertices was introduced by Ore (1960). More generally, the minimum degree sum of a stable set of k vertices, denoted σ_k , turns out to be a parameter of relevance to circuits. Bondy (1981) proposed a common generalization of Ore's Theorem (2.4) and the Chvátal–Erdős Theorem (4.16) involving this parameter.

Conjecture 4.22. *Let G be a simple k -connected graph on n vertices such that $\sigma_{k+1} \geq n + k(k - 1)$, and let C be a longest circuit in G . Then $G \setminus C$ contains no path of length $k - 1$.*

Conjecture 4.22 has been verified for $k = 1$ (Ore's Theorem 2.4) and $k = 2$ (a generalization of Theorem 2.16); these are special cases of Theorem 4.23 below, which establishes a weaker form of the conjecture. The case $k = 3$ of Conjecture 4.22 is pertinent to Conjecture 5.20 and can be proved by slightly modifying the proof of Theorem 4.23 (Bondy 1981).

Theorem 4.23. *Let G be a simple k -connected graph on n vertices such that $\sigma_{k+1} \geq n + k(k - 1)$, and let C be a longest circuit in G . Then $G \setminus C$ contains no complete subgraph of order k .*

Bauer et al. (1989) applied the case $k = 2$ of Theorem 4.23 to derive a sufficient condition for a graph to be hamiltonian. This is yet another generalization (of a generalization) of Dirac's Theorem (2.1).

Theorem 4.24. *Let G be a simple graph of connectivity κ on n vertices such that $\sigma_3 \geq n + \kappa$, where $\kappa \geq 2$. Then G contains a Hamilton circuit.*

Another variant of Conjecture 4.22 was established by Fraisse (1986). It concerns k -dominating circuits rather than longest circuits, and also generalizes Ore's Theorem (2.4). A *dominating circuit* of a graph G is a circuit C such that $V(G) \setminus V(C)$ is a stable set of G . More generally, if k is a positive integer, a *k -dominating circuit* of a graph G is a circuit C such that each component of $G \setminus C$ has fewer than k vertices.

Theorem 4.25. *Let G be a simple k -connected graph on n vertices, where $n \geq 3$, such that $\sigma_{k+1} \geq n + k(k-1)$. Then G has a k -dominating circuit.*

The concepts of k -dominating circuit and k -walk are intimately related via a graph product called composition. The *composition* (or *lexicographic product*) of simple graphs G and H is the graph $G[H]$ whose vertex set is $V(G) \times V(H)$, vertices (u_1, v_1) and (u_2, v_2) being adjacent if and only if either $u_1 u_2 \in E(G)$ or both $u_1 = u_2$ and $v_1 v_2 \in E(H)$. Jackson and Wormald (1990a) observed that a graph G has a k -walk if and only if the composition $G[K_k]$ has a k -dominating circuit. Applying Theorem 4.25, they deduced the following common generalization of Theorem 2.17 and Theorem 2.4.

Corollary 4.26. *Let G be a simple connected graph on n vertices such that $\sigma_{k+1} \geq n$. Then G has a k -walk.*

Closely related is a common generalization of Theorem 2.18 and the Ore-type analogue of Corollary 2.3, due to Win (1975).

Theorem 4.27. *Let G be a simple connected graph on n vertices such that $\sigma_k \geq n-1$, where $k \geq 2$. Then G contains a k -tree.*

Fournier and Fraïsse (1985) proved a common generalization of Theorem 4.11 and the Chvátal–Erdős Theorem (4.16) involving degree sums of stable sets of vertices.

Theorem 4.28. *Let G be a simple k -connected graph, where $k \geq 2$, such that $\sigma_{k+1} \geq d$. Then G contains either a circuit of length at least $2d/(k+1)$ or a Hamilton circuit.*

The stability number also plays an important role in digraphs. We define a *stable set* of a digraph to be a stable set of its underlying graph. Gallai and Milgram (1960) found a beautiful link between the stability number of a digraph and partitions of its vertex set into directed paths. A *partition* of a graph G is a family of disjoint subgraphs whose vertices partition $V(G)$. A *path partition* is a partition into paths, and the minimum number of paths in a path partition of a graph G is denoted by $\pi(G)$. Analogous definitions apply to digraphs, a path partition of a digraph being a partition into directed paths.

For a path partition $\mathcal{Q}$ in a digraph D , we denote by $\alpha(\mathcal{Q})$ the cardinality of a maximum stable set whose elements belong to different paths of $\mathcal{Q}$, by $h(\mathcal{Q})$ the set of heads of the paths in $\mathcal{Q}$ and by $t(\mathcal{Q})$ the set of tails of the paths in $\mathcal{Q}$. The following lemma (Bondy 1995) is a slight refinement of one due to Gallai and Milgram (1960).

Lemma 4.29. *Let D be a digraph and $\mathcal{Q}$ a path partition of D such that $|\mathcal{Q}| > \alpha(\mathcal{Q})$. Then there is a path partition $\mathcal{P}$ of D such that $|\mathcal{P}| = |\mathcal{Q}| - 1$, $h(\mathcal{P}) \subset h(\mathcal{Q})$ and $t(\mathcal{P}) \subset t(\mathcal{Q})$.*

Proof (by induction on $v(D)$, the case $v(D) = 1$ holding vacuously). Let $\mathcal{Q}$ be a path partition of D such that $|\mathcal{Q}| > \alpha(\mathcal{Q})$. Then $h(\mathcal{Q})$ is not a stable set, so there exist $y, z \in h(\mathcal{Q})$ such that $(y, z) \in E$. If (z) is a path of $\mathcal{Q}$, we define $\mathcal{P}$ to be the path partition of D obtained from $\mathcal{Q}$ by deleting (z) and extending the path of $\mathcal{Q}$ with head y by the edge (y, z) . Thus we may assume that (z) is not a path of $\mathcal{Q}$. Let x be the predecessor of z on the path of $\mathcal{Q}$ with head z , set $D' := D \setminus z$, and let $\mathcal{Q}'$ be the restriction of $\mathcal{Q}$ to D' . Note that

$$|\mathcal{Q}'| = |\mathcal{Q}| > \alpha(\mathcal{Q}) \geq \alpha(\mathcal{Q}'),$$

$$h(\mathcal{Q}') = (h(\mathcal{Q}) \setminus \{z\}) \cup \{x\} \quad \text{and} \quad t(\mathcal{Q}') = t(\mathcal{Q}).$$

By the induction hypothesis, there is a path partition $\mathcal{P}'$ of D' such that $|\mathcal{P}'| = |\mathcal{Q}'| - 1$, $h(\mathcal{P}') \subset h(\mathcal{Q}')$ and $t(\mathcal{P}') \subset t(\mathcal{Q}')$. If $x \in h(\mathcal{P}')$, we define $\mathcal{P}$ to be the path partition of D obtained from $\mathcal{P}'$ by extending the path of $\mathcal{P}'$ with head x by the edge (x, z) . If $x \notin h(\mathcal{P}')$, then $y \in h(\mathcal{P}')$ and we define $\mathcal{P}$ to be the path partition of D obtained from $\mathcal{P}'$ by extending the path of $\mathcal{P}'$ with head y by the edge (y, z) . In both cases, $|\mathcal{P}| = |\mathcal{Q}| - 1$, $h(\mathcal{P}) \subset h(\mathcal{Q})$ and $t(\mathcal{P}) \subset t(\mathcal{Q})$. $\square$

An immediate consequence of this lemma is the theorem of Gallai and Milgram (1960) alluded to above.

Theorem 4.30 (The Gallai–Milgram Theorem). *Let D be a digraph with stability number α . Then D admits a partition into at most α directed paths. That is,*

$$\pi \leq \alpha.$$

Remark 4.31. The Gallai–Milgram Theorem (4.30) can be viewed as a formula for the stability number of a graph in terms of the path partition numbers of its orientations:

$$\alpha(G) = \max \{ \pi(\vec{G}) : \vec{G} \text{ an orientation of } G \}.$$

To see this, observe that $\alpha(G) \geq \pi(\vec{G})$ for every orientation $\vec{G}$ of G , by the Gallai–Milgram Theorem (4.30), and that the reverse inequality holds when $\vec{G}$ is an orientation of G in which every vertex of some maximum stable set is a source.

The Gallai–Milgram Theorem (4.30) includes a well-known theorem of Dilworth (1950) as a special case.

Theorem 4.32 (Dilworth’s Theorem). *Let $P := (X, \prec)$ be a partially ordered set. Then the minimum number of chains of P into which X can be partitioned is equal to the maximum number of elements in an antichain of P .*

Proof. Denote by $D(P)$ the digraph whose vertex set is X and whose edges are the ordered pairs (u, v) such that $u \prec v$ in P . Chains and antichains in P then correspond to directed paths and stable sets, respectively, in $D(P)$, and the Gallai–Milgram Theorem (4.30) implies the existence of a chain partition of cardinality

no greater than the maximum number of elements in an antichain. The reverse inequality follows from the observation that no two elements of an antichain can belong to a common chain. $\square$

An analogue of the Gallai–Milgram Theorem (4.30) for strong digraphs was proposed by Gallai (1964).

Conjecture 4.33. Let D be a strong digraph with stability number α . Then D admits a vertex covering by at most α directed circuits.

Two special cases of Conjecture 4.33 are easily verified. When D is the associated digraph of a 2-edge-connected graph G , this follows from a result of Pósa (1963) which asserts that any graph admits a partition into at most α circuits, edges and vertices; alternatively, one may apply Corollary 4.20 to an endblock of G and use induction. And when $\alpha = 1$, it is just Camion’s Theorem (3.9). Chen and Manalastas (1983) (see also Bondy 1995) verified the case $\alpha = 2$.

Theorem 4.34. Let D be a strong digraph with stability number two. Then D admits a vertex covering by two directed circuits $\{C_1, C_2\}$, where $C_1 \cap C_2$ is either empty or a directed path of D .

One interesting consequence of Theorem 4.34 is the following.

Corollary 4.35. Let D be a strong digraph with stability number two. Then D contains a directed Hamilton path.

This corollary leads to two conjectures. One, a simultaneous generalization of Corollary 4.35 and the Gallai–Milgram Theorem (4.30), was proposed (in a stronger form) by M. Las Vergnas (see Berge 1983).

Conjecture 4.36. Let D be a strong digraph with stability number α , where $\alpha \geq 2$. Then D admits a partition into at most $\alpha - 1$ directed paths.

The second conjecture, put forward by Jackson and Ordaz (1990), involves a less stringent definition of stability in digraphs. A stable set in a digraph D may equally well be defined as a set S of vertices such that $D[S]$ contains no directed circuit, or such that $D[S]$ contains no directed 2-circuit. The corresponding stability numbers are called the *acyclic stability number* and the *oriented stability number*, respectively, and are denoted by α_1 and α_2 . Clearly, $\alpha \leq \alpha_1 \leq \alpha_2$. The following conjecture would thus generalize Corollary 4.17.

Conjecture 4.37. Let D be a digraph with oriented stability number α_2 and strong connectivity κ , where $\alpha_2 \leq \kappa + 1$. Then D contains a directed Hamilton path.

Remark 4.38. It is not true in general that a k -strong digraph with stability number $k + 1$ contains a directed Hamilton path; while this is so for $k = 0$, by Rédei’s Theorem (3.1), and also for $k = 1$, by Corollary 4.35, it is false for $k = 2$, a counterexample being the digraph derived from the composition of $K_{1,3}$ with K_2 by orienting each copy of $K_{2,2}$ that corresponds to an edge of $K_{1,3}$ as a directed 4-circuit.

Jackson (1987) also made use of the concept of oriented stability number to establish a directed analogue of the Chvátal–Erdős Theorem (4.16).

Theorem 4.39. *Let D be a digraph with strong connectivity κ , where $\kappa \geq 2^{\alpha_2}(\alpha_2 + 2)!$. Then D contains a directed Hamilton circuit.*

Heydemann (1985) likewise proved the directed analogue of Conjecture 4.33 for the oriented stability number.

4.3. Matching number

In this section, we consider the edge analogue of a stable set. A set M of edges of a graph G is a *matching* if no two elements of M are adjacent in G . The matching M is perfect if each vertex of G is incident to an edge of M . The maximum cardinality of a matching of G is called the *matching number* of G and denoted by $\nu(G)$.

The Ore-type generalization of Corollary 2.3 referred to at the end of section 2 asserts that a simple graph on n vertices in which the degree sum of any two nonadjacent vertices is at least $n - 1$ has a Hamilton path. Häggkvist (1979) showed that when n is even, the very same hypothesis implies something much stronger.

Theorem 4.40. *Let G be a simple graph on $2m$ vertices in which the degree sum of any two nonadjacent vertices is at least $2m - 1$. Then each perfect matching of G is contained in a Hamilton path of G .*

Proof. Let $M := \{x_1y_1, x_2y_2, \dots, x_my_m\}$ be a perfect matching of G , where the sets $X := \{x_1, x_2, \dots, x_m\}$ and $Y := \{y_1, y_2, \dots, y_m\}$ are chosen so as to maximize the number of edges in the bipartite subgraph of $G \setminus M$ induced by the bipartition (X, Y) . Denote this subgraph by B . By the choice of X and Y ,

$$\begin{aligned} d_B(x_i) + d_B(y_i) &\geq \frac{1}{2}(d_{G \setminus M}(x_i) + d_{G \setminus M}(y_i)) = \frac{1}{2}(d_G(x_i) + d_G(y_i)) - 1, \\ 1 &\leq i \leq m. \end{aligned}$$

Let D be the digraph obtained from B by first orienting each edge from X to Y and then identifying x_i and y_i , denoting the resulting vertex by v_i , $1 \leq i \leq m$. For any two nonadjacent vertices v_i and v_j of D ,

$$\begin{aligned} d_D(v_i) + d_D(v_j) &= d_B(x_i) + d_B(y_i) + d_B(x_j) + d_B(y_j) \\ &\geq \frac{1}{2}(d_G(x_i) + d_G(y_i)) + \frac{1}{2}(d_G(x_j) + d_G(y_j)) - 2 \geq 2m - 3. \end{aligned}$$

It now follows from Meyniel's Theorem (3.11), by applying the digraph analogue of the trick used to establish Corollary 2.3 from Dirac's Theorem, 2.1, that D has a directed Hamilton path. Without loss of generality, let this path be $(v_1, v_2, \dots, v_m)$. Then $x_1y_1x_2y_2 \cdots x_my_m$ is a Hamilton path of G through each edge of M . $\square$

An analogous theorem on Hamilton circuits containing perfect matchings also due to Häggkvist (1979), was extended by Berman (1983) as follows.

Theorem 4.41. *Let G be a simple graph on n vertices, in which the degree sum of any two nonadjacent vertices is at least $n + 1$, where $n \geq 3$. Then each matching of G is contained in a circuit of G .*

Theorem 4.41 implies Theorem 4.40. Let G satisfy the hypotheses of Theorem 4.40. Form G' from G by adding two new vertices, u and v , and joining each to every vertex of G and also to one another. Define $M' := M \cup \{e\}$, where $e := uv$. Then M' is a perfect matching of G' . Also, the degree sum of any two nonadjacent vertices of G' is at least $2m + 3 = (2m + 2) + 1$. By Theorem 4.41, G' has a Hamilton circuit through each edge of M' , so G has a Hamilton path through each edge of M .

Jackson and Wormald (1990b) characterized the extremal graphs for Theorem 4.41 and derived a strengthening of Dirac's Theorem (2.1) for graphs whose number of vertices is divisible by four.

Theorem 4.42. *Let G be a simple graph of minimum degree δ on n vertices, where $\delta \geq n/2$ and $n \equiv 0 \pmod{4}$. Then each perfect matching of G is contained in a Hamilton circuit of G .*

Matchings enjoy a rich and extensive theory. This is discussed in detail in chapter 3 and in the treatise by Lovász and Plummer (1986). The algorithmic aspects of matchings are also well developed. In particular, a maximum-weight matching in an edge-weighted graph can be found in polynomial time. This leads to an efficient method for finding a shortest path of specified parity between two vertices of a graph, if there is one. The idea, due to J. Edmonds (see Grötschel and Pulleyblank 1981), is most easily explained with the aid of the following definition. The *cartesian product* of simple graphs G and H is the graph $G \square H$ whose vertex set is $V(G) \times V(H)$, vertices (u_1, v_1) and (u_2, v_2) being adjacent if and only if either $u_1 = u_2$ and $v_1v_2 \in E(H)$ or $v_1 = v_2$ and $u_1u_2 \in E(G)$. The cartesian product $G \square K_2$ is referred to as the *prism* over G ; it can be viewed as consisting of two disjoint copies of G , G_1 and G_2 , joined by a perfect matching $M := \{v_1v_2 \mid v \in V(G)\}$, where v_i is the vertex of copy G_i corresponding to vertex v of G , $i = 1, 2$.

There is a natural bijection between the odd xy -paths of G and the perfect matchings of the graph H derived from the prism over G by deleting the vertices x_2 and y_2 of G_2 that correspond to x and y . Moreover, an odd xy -path of length ℓ in G corresponds to a perfect matching of H that includes exactly $n - 1 - \ell$ edges of M . Therefore, if each edge of G_i in H , $i = 1, 2$, is assigned weight $n - 1$, and each edge of M in H is assigned weight n , where $n := v(G)$, a matching of weight $n(n - 1) - \ell$ in H is necessarily a perfect matching of H and corresponds to an odd xy -path of length ℓ in G . Thus a maximum-weight matching in H corresponds to a shortest odd xy -path in G . A shortest even xy -path in G can be found by simply adding a vertex y' and edge yy' to G and finding a shortest odd xy' -path in the resulting graph G' .

4.4. Chromatic number and edge chromatic number

A k -colouring of a graph G is a function $c: V(G) \rightarrow S$, where S is a k -set; commonly, S is the set $\{1, 2, \dots, k\}$. The element $c(v)$ is called the *colour* of vertex v , and the sets $C_i := \{v: c(v) = i\}$, $i \in S$, the *colour classes* of G with respect to c . A k -colouring of a graph is thus a partition of its vertex set into colour classes. A colouring c is *proper* if each colour class of G with respect to c is a stable set of G ; only loopless graphs admit proper colourings, and we implicitly assume here that all graphs are loopless. A graph is k -colourable if it admits a proper k -colouring. A 2-colourable graph is one which is bipartite, and thus can be recognized as such in polynomial time (see section 1.6, Search Trees). On the other hand determining whether a graph is k -colourable when $k \geq 3$ is an $\mathcal{NP}$ -complete problem.

The *chromatic number* of a graph G is the minimum value of k for which G is k -colourable, and is denoted by $\chi(G)$. A graph G is k -chromatic if $\chi(G) = k$, and χ -critical if $\chi(G \setminus v) < \chi(G)$ for all vertices v of G . Two basic properties of χ -critical k -chromatic graphs are easily established: they are nonseparable and have minimum degree at least $k - 1$.

A graph with chromatic number at least three is nonbipartite and so contains an odd circuit. Erdős and Hajnal (1966) extended this observation by showing that a k -chromatic graph, where $k \geq 3$, contains an odd circuit of length at least $k - 1$. A word of caution is in order here. Despite its appearance, the theorem of Erdős and Hajnal is linked only superficially to the chromatic number; indeed, the theorem is an immediate consequence of Theorem 4.13: a k -chromatic graph contains a χ -critical k -chromatic subgraph and thus has minimum degree at least $k - 1$.

A more essential use of the chromatic number was made by Gallai (1968a) and Roy (1967), who discovered a simple relationship between the chromatic number of a digraph and the length of a longest directed path in the digraph, where the *chromatic number* $\chi(D)$ of a digraph D is defined to be the chromatic number of its underlying graph $G(D)$. We denote the number of vertices in a longest directed path of a digraph D by $\lambda(D)$.

Theorem 4.43 (The Gallai–Roy Theorem). *Let D be a digraph with chromatic number χ . Then D contains a directed path of length at least $\chi - 1$. That is,*

$$\lambda \geq \chi.$$

Proof. Let D' be a maximal acyclic subdigraph of D , λ' the number of vertices in a longest directed path of D' , and $\lambda'(v)$ the number of vertices in a longest directed v -path of D' . Noting that $V(D') = V(D)$, define a λ' -colouring c of D by setting $c(v) := \lambda'(v)$, $v \in V(D')$. We claim that c is a proper colouring of D .

Observe, first, that if there is an (x, y) -path P in D' , then $c(x) > c(y)$. For if Q is a y -path in D' , PQ is an x -path in D' , because D' is acyclic. In particular, if $(x, y) \in E(D')$, $c(x) \neq c(y)$. On the other hand, if $(x, y) \in E(D) \setminus E(D')$, then there is an (y, x) -path in D' , by the choice of D' , and again $c(x) \neq c(y)$. $\square$

Remark 4.44. The Gallai–Roy Theorem (4.43) can be regarded as a “dual” of the Gallai–Milgram Theorem (4.30), the roles of directed paths and stable sets

being interchanged; this duality is well understood in the case of acyclic digraphs (Frank 1980; see also chapter 2), but how far it extends to all digraphs is not clear. The Gallai–Roy Theorem (4.43) can also be viewed as a formula for the chromatic number of a graph in terms of the lengths of longest directed paths in orientations of the graph:

$$\chi(G) = \min \{ \lambda(\vec{G}) : \vec{G} \text{ an orientation of } G \}.$$

To see this, observe that $\chi(G) \leq \lambda(\vec{G})$ for every orientation $\vec{G}$ of G , by the Gallai–Roy Theorem (4.43), and that the reverse inequality holds when $\vec{G}$ is the orientation of G in which u dominates v whenever $c(u) < c(v)$, where $c: V \rightarrow \{1, 2, \dots, \chi\}$ is any χ -colouring of G (cf. Remark 4.31).

Laborde et al. (1983) proposed the following rather strong refinement of the Gallai–Roy Theorem (4.43).

Conjecture 4.45. Let D be a nontrivial digraph and $\lambda_1 + \lambda_2$ a partition of $\lambda(D)$ into two positive integers. Then there is a partition (V_1, V_2) of $V(D)$ such that $\lambda(D[V_i]) = \lambda_i$, $i = 1, 2$.

The case $\lambda_2 = 1$ of Conjecture 4.45 asserts that in any digraph there is a stable set meeting every longest directed path; applied recursively, this would yield the Gallai–Roy Theorem (4.43). Interestingly, the dual statement is conjectured by Hahn and Jackson (1990) to be false.

Conjecture 4.46. For any positive integer k , there exists a digraph D with stability number k such that $\alpha(D \setminus \cup_{i=1}^{k-1} P_i) = \alpha(D)$ for any family $\{P_1, P_2, \dots, P_{k-1}\}$ of $k - 1$ disjoint directed paths of D .

A generalization of the Gallai–Roy Theorem (4.43) to strong digraphs was obtained, with the aid of the Bypass Lemma (3.10), by Bondy (1976).

Theorem 4.47. Let D be a strong digraph with chromatic number χ . Then D contains a directed circuit of length at least χ .

Linial (1981) proposed a common generalization of the Gallai–Roy Theorem (4.43) and the Gallai–Milgram Theorem (4.30). In order to state it, we need two definitions. Let D be a digraph and k a positive integer. A path partition $\mathcal{P}$ of D that minimizes the function

$$\sum_{P \in \mathcal{P}} \min\{v(P), k\}$$

is called k -optimal, and the minimum value of this function is denoted by $\pi_k(D)$. In particular, $\pi_1 = \pi$, the minimum number of directed paths in a path partition of D , and $\pi_k = v(D)$ if $k \geq \lambda(D)$. A *partial k -colouring* of D is a family $\mathcal{C}$ of

k pairwise-disjoint stable sets of D . A partial k -colouring $\mathcal{C}$ that maximizes the function

$$|\cup_{C \in \mathcal{C}} C|$$

is called *optimal*, and the maximum value of this function is denoted by $\alpha_k(D)$. In particular, $\alpha_1 = \alpha$, the stability number of D , and $\alpha_k = v(D)$ if and only if $k \geq \chi(D)$.

Conjecture 4.48. For any digraph D and any positive integer k ,

$$\alpha_k \geq \pi_k.$$

Conjecture 4.48 was first posed, as a question, by Linial (1981), and was proved by him for acyclic digraphs; it achieved the status of a conjecture with Berge (1982). When $k = 1$, it is the Gallai–Milgram Theorem (4.30); when $k \geq \lambda(D)$, it is the Gallai–Roy Theorem (4.43). It is unproved for all other values of k . A still stronger conjecture was proposed by Berge (1982). A path partition $\mathcal{P}$ and a partial k -colouring $\mathcal{C}$ are *orthogonal* if every directed path $P \in \mathcal{P}$ meets $\min\{v(P), k\}$ different colour classes of $\mathcal{C}$.

Conjecture 4.49. Let D be a digraph, k a positive integer and $\mathcal{P}$ a k -optimal path partition of D . Then there is a partial k -colouring of D orthogonal to $\mathcal{P}$.

Like its weaker variant, this conjecture is valid for $k = 1$ (Linial 1978) and for $k \geq \lambda(D)$ (Berge 1982). It has also been verified for acyclic digraphs by several authors, some using network flow techniques (see, for example, Aharoni et al. 1985).

The duality between directed paths and stable sets referred to in Remark 4.44 suggests potential duals of Conjectures 4.48 and 4.49. The stronger of these dual conjectures turns out to be false (Aharoni et al. 1985), but the dual of Conjecture 4.48, posed as a question by Linial (1981), is still open. Let D be a digraph and k a positive integer. A *partial k -path partition* of D is a family $\mathcal{P}$ of k pairwise-disjoint directed paths of D . A partial k -path partition $\mathcal{P}$ that maximizes the function

$$|\cup_{P \in \mathcal{P}} V(P)|$$

is called *optimal*, and the maximum value of this function is denoted by $\lambda_k(D)$. In particular, $\lambda_1 = \lambda$, the number of vertices in a longest directed path of D , and $\lambda_k = v(D)$ if and only if $k \geq \pi(D)$. A proper vertex colouring $\mathcal{C}$ of D that minimizes the function

$$\sum_{C \in \mathcal{C}} \min\{|C|, k\}$$

is called *k -optimal*, and the minimum value of this function is denoted by $\chi_k(D)$. In particular, $\chi_1 = \chi$, the chromatic number of D , and $\chi_k = v(D)$ if $k \geq \alpha(D)$.

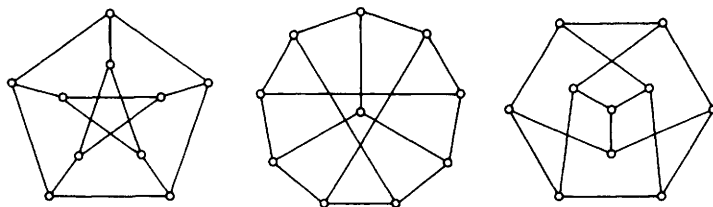


Figure 8. The Petersen graph: three drawings.

Conjecture 4.50. For any digraph D and any positive integer k ,

$$\lambda_k \geq \chi_k.$$

The restriction of Conjecture 4.50 to acyclic digraphs was verified by Hoffman (1983). When $k = 1$, the conjecture is the Gallai–Roy Theorem (4.43); when $k \geq \alpha(D)$, it is the Gallai–Milgram Theorem (4.30). The conjecture is unproved for all other values of k .

We conclude by defining the edge analogue of vertex colouring. A k -edge colouring of a graph G is a function $c: E(G) \rightarrow S$, where S is a k -set. An edge colouring c is *proper* if each colour class of G with respect to c is a matching of G ; only loopless graphs admit proper edge colourings. A graph is k -edge-colourable if it admits a proper k -edge colouring. The *edge chromatic number* of a graph G is the minimum value of k for which G is k -edge-colourable, and is denoted by $\chi'(G)$. The edge chromatic number of a bipartite graph is equal to its maximum degree, and thus can be determined in polynomial time. For graphs in general, however, it is an $\mathcal{NP}$ -complete problem to determine whether a graph is k -edge-colourable when $k \geq 3$. The topic of circuits in edge-coloured graphs is touched upon in Remark 8.39. Edge and vertex colourings are discussed in detail in chapter 4: Colouring, Stable Sets and Perfect Graphs.

4.5. Toughness

We have concentrated up to now on sufficient conditions for the existence of Hamilton circuits. Necessary conditions are unfortunately much harder to come by. A simple one can be derived as follows. Let G be a hamiltonian graph, H a Hamilton circuit in G and S a vertex cut of G . Clearly $H \setminus S$ has at most $|S|$ components. Because $H \setminus S$ is a spanning subgraph of $G \setminus S$, the same is true of $G \setminus S$. Thus $c(G \setminus S) \leq |S|$ for every vertex cut S of G . A graph G with this property is called *tough*. Hamiltonian graphs are tough, as we have just observed, but not every tough graph is hamiltonian, the *Petersen graph* of fig. 8 being a counterexample.

The Petersen graph was, in fact, first constructed by Kempe (1886); it was rediscovered twelve years later by Petersen (1898). It is a graph with many remarkable properties. It is not 3-edge-colourable, and is the smallest simple 3-regular graph with this property. This implies that it is nonhamiltonian, since the edges of a

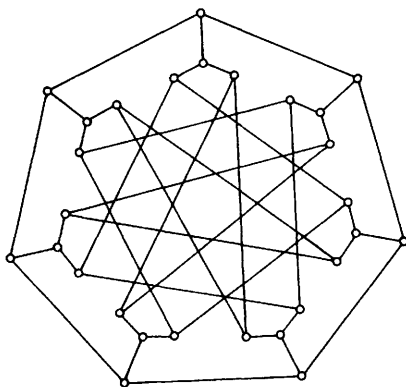


Figure 9. The Coxeter graph.

Hamilton circuit could be properly 2-coloured and the remaining edges assigned a third colour, yielding a proper 3-edge colouring. It is, moreover, maximally nonhamiltonian: the addition of any one edge yields a hamiltonian graph. Another very interesting 3-regular graph is the *Coxeter graph*, discovered by H.S.M. Coxeter (see Tutte 1960) and shown in fig. 9; the beautiful drawing is due to M. Randić (personal communication 1981). This graph, like the Petersen graph, is not only nonhamiltonian (Tutte 1960) but also maximally nonhamiltonian (Clark and Entringer 1983). In addition, it is tough.

Faced with such examples, Chvátal (1973) was prompted to consider a refinement of the concept of toughness. Let t be a positive real number. A graph G is t -tough if

$$c(G \setminus S) \leq \frac{|S|}{t}$$

for every vertex cut S of G . In particular, a 1-tough graph is one that is tough. The *toughness* of a graph G is the largest value of t for which G is t -tough. Chvátal (1973) conjectured the existence of a constant t such that every t -tough graph is hamiltonian. Examples constructed by Enomoto et al. (1985) show that, if such a constant t does indeed exist, $t \geq 2$. A natural conjecture, then, is the following one.

Conjecture 4.51. Every 2-tough graph is hamiltonian.

This conjecture, in a slightly weaker form (restricted to locally-connected graphs), was formulated by Chvátal (1973), who noted that it would imply a theorem of Fleischner (1974) on squares of graphs.

Let G be a simple graph and m a positive integer. The graph with vertex set $V(G)$ in which two vertices are adjacent if and only if their distance in G is at most m is called the m th power of G and denoted by G^m . The *square* of G is the

graph G^2 , the *cube* of G the graph G^3 . A fairly easy induction argument shows that the cube of a connected graph on at least three vertices is hamiltonian (Sekanina 1960). Fleischner proved something much stronger, namely that the square of a 2-connected graph is hamiltonian. Since squares of k -connected graphs are k -tough, Fleischner's theorem is implied by Conjecture 4.51. A simpler proof of Fleischner's theorem was found by Řiha (1991). Underground (1978) noted that the problem of determining whether the square of a graph is hamiltonian is $\mathcal{NP}$ -complete.

Win (1989) established a link between toughness and k -trees.

Theorem 4.52. *Let k be a positive integer. Then every $(1/k)$ -tough graph has a $(k+2)$ -tree.*

A refinement, in terms of k -walks, was conjectured by Jackson and Wormald (1990a).

Conjecture 4.53. *Let k be a positive integer. Then every $(1/k)$ -tough graph has a $(k+1)$ -walk.*

Attractive though Conjectures 4.51 and 4.53 are, the insight they would provide is questionable in light of the fact proved by Bauer et al. (1990) that, for any positive constant t , it is $\mathcal{NP}$ -hard to determine whether a graph is t -tough; each of these conjectures thus relates two properties whose verification is $\mathcal{NP}$ -hard. What can be hoped for, nevertheless, is a polynomial-time algorithm that either finds a Hamilton circuit (respectively, a $(k+1)$ -walk) or else exhibits a set S of vertices showing that the input graph is not 2-tough (respectively, not $(1/k)$ -tough).

Many sufficient conditions for hamiltonicity can be weakened marginally by imposing a toughness condition (see, for instance, the survey by Bauer et al. 1991). Improvement is possible, of course, only if the extremal graphs for the condition in question are themselves not tough; imposing toughness then serves to exclude them. Häggkvist (1992) achieved more remarkable results by imposing a stronger necessary condition, which one might call path-toughness. A graph G is *path-tough* if

$$\pi(G \setminus S) \leq |S|$$

for every nonempty proper subset S of $V(G)$, or equivalently, if each vertex-deleted subgraph is traceable. Hamiltonian graphs are clearly path-tough. Häggkvist's insight was that Dirac's minimum-degree condition can be relaxed substantially for path-tough graphs. His lower bound of $8(n-1)/17$ on the minimum degree was further reduced to $(2n-3)/5$ by Dankelmann et al. (preprint). Schiermeyer (1992) made the following conjecture as to the best possible bound.

Conjecture 4.54. *Let G be a simple path-tough graph of minimum degree δ on n vertices, where $\delta \geq (n-2)/3$ and $n \geq 3$. Then G is either hamiltonian or isomorphic to the Petersen graph.*

Relatively little is known about nonhamiltonian path-tough graphs. Two special classes have been studied, however. A graph G is *hypohamiltonian* if every vertex-deleted subgraph $G \setminus v$ is hamiltonian but G is not, and *hypotractable* if every vertex-deleted subgraph $G \setminus v$ is traceable but G is not. R. Sousselier (see Berge 1963) observed that the Petersen graph (fig. 8) is hypohamiltonian; Bondy (1972) showed likewise that the Coxeter graph (fig. 9) has this property. But there are many other examples. The definitive sources on this topic are Thomassen (1978, 1981a) where a variety of constructions are described, including one for planar hypohamiltonian graphs and another for surprisingly dense hypohamiltonian graphs. Still, a number of basic questions remain. For instance, while it is easy to see that the minimum degree in a hypohamiltonian or hypotractable graph must be at least three, no example is known in which each vertex is of degree at least four.

Problem 4.55. Is there a hypohamiltonian graph of minimum degree at least four?

This problem was posed by Thomassen (1978), who proved, with the aid of the Bridge Lemma (5.12), that no such graph can be planar. Also worth noting is a question of Grötschel (personal communication 1978) on hypotractable graphs.

Problem 4.56. Is there a bipartite hypotractable graph?

The Petersen graph (fig. 8) has girth five, and the Coxeter graph (fig. 9) girth seven. Horák and Širáň (1986) showed how infinite families of maximally nonhamiltonian graphs of girths five and seven can be formed from these graphs, and asked the following question.

Problem 4.57. Is there a maximally nonhamiltonian graph of girth g for all $g \geq 3$?

5. Fundamental classes of graphs and digraphs

We now discuss several classes of graphs and digraphs of particular interest: bipartite graphs, planar graphs, regular graphs, vertex-transitive graphs, line graphs, claw-free graphs, oriented graphs and tournaments. As one would expect, stronger results than are valid for graphs in general often hold for members of these classes. In some instances, the improvements are significant.

5.1. Bipartite graphs

By Theorem 1.8, a graph is bipartite if and only if it contains no odd circuits. It is not surprising, therefore, that in problems relating the existence of circuits to vertex degrees, a good rule of thumb is that the conditions can be weakened by half when the graph in question is bipartite. We illustrate this general principle with two typical examples. The first is a bipartite analogue of Theorem 2.15 and a special case of a result of Jackson (1985).

Theorem 5.1. *Let G be a simple nonseparable balanced bipartite graph of minimum degree δ . Then G contains either a circuit of length at least $4\delta - 2$ or a Hamilton circuit.*

Recall that the closure of a graph G on n vertices is the graph derived from G by recursively joining pairs of nonadjacent vertices having degree sum at least n . There is an analogous concept for bipartite graphs. The *bipartite closure* of a balanced bipartite graph $G(X, Y)$ on $2n$ vertices is the graph derived from G by recursively joining pairs of vertices in different parts whose degree sum is at least $n + 1$. In section 2, it was proved that a simple graph on at least three vertices is hamiltonian if and only if its closure is hamiltonian (Closure Lemma 2.6). The corresponding result for bipartite graphs can be established by a similar argument (Bondy and Chvátal 1976).

Theorem 5.2. *A simple balanced bipartite graph G is hamiltonian if and only if its bipartite closure is hamiltonian.*

5.2. Planar graphs

A *planar embedding* of a graph G is a function ϕ that assigns to each vertex a point in the plane and to each edge a simple curve in the plane, in such a way that the curves $\phi(e_1)$ and $\phi(e_2)$ meet at a point p if and only if $p = \phi(v)$ for some vertex v which is incident in G to the edges e_1 and e_2 . A graph is *planar* if it admits a planar embedding; the Platonic graphs of fig. 1 are the prototypical examples. Note that a graph is planar if and only if it has an embedding in the sphere. A *plane graph* is a pair (G, ϕ) where G is planar and ϕ is a planar embedding of G . The (topological) closures of the connected components of the complement of $\phi(G)$ are the *faces* of G . An edge and face are *incident* if the edge is contained in the face. The edges incident to a face f form a closed walk in which cut edges are each traversed twice. The length of this walk is the *degree* $d(f)$ of f . If the walk is a circuit, it is called a *facial circuit*. Two faces are *adjacent* if they are incident with a common edge; a face which is incident to a cut edge is adjacent to itself. The *boundary* of a face consists of those edges incident with the face and with some other face. The boundary of a face is always an even subgraph.

If G is a plane graph, the *planar dual* of G is the planar graph G^* whose vertices are the faces of G , two vertices being joined by an edge e^* if and only if the corresponding faces of G are incident to a common edge e . The bijection between $E(G)$ and $E(G^*)$ which maps e to e^* has the striking property that it takes circuits to bonds and bonds to circuits. The cycle space of G and the bond space of G^* are therefore isomorphic vector spaces.

In a 2-connected plane graph, the boundary of each face is a circuit, and these facial circuits generate the cycle space; moreover, omitting any one of them results in a cycle basis in which each edge occurs at most twice (because each edge is incident to two faces). MacLane (1937) showed that this property characterizes planar graphs.

Theorem 5.3. *A graph is planar if and only if there is a basis for its cycle space in which each edge occurs at most twice.*

Kuratowski (1930) gave a quite different, and more significant, characterization of planar graphs.

Theorem 5.4 (Kuratowski's Theorem). *A graph is planar if and only if it contains no subdivision of $K_{3,3}$ or K_5 .*

Despite these characterizations, and the existence of linear-time algorithms to test for planarity (see chapter 5), planar graphs remain something of an enigma, on account of the notorious Four-Colour Problem. A *k*-face colouring of a plane graph G with face set $F(G)$ is a function $c: F(G) \rightarrow S$, where S is a k -set. The face colouring c is *proper* if adjacent faces receive different colours; thus a graph has a proper face colouring if and only if it has no cut edge. A plane graph is *k*-face-colourable if it admits a proper k -face colouring. In 1852, F. Guthrie (see Biggs et al. 1976) asked whether every 2-edge-connected plane graph is 4-face-colourable. This innocent question was the prime motivation for much of the research undertaken in graph theory during the hundred years which followed. Known as the *Four-Colour Problem*, and later as the *Four-Colour Conjecture*, it was given an affirmative answer by Appel and Haken (1976). While the basic ideas of their proof are well understood, it is fair to say that the details are still a matter of some dispute, on account of their complexity.

Theorem 5.5 (The Four-Colour Theorem). *Every 2-edge-connected plane graph is 4-face-colourable.*

By duality, an equivalent statement is that every connected loopless planar graph is 4-vertex-colourable. Tait (1880) derived yet another equivalent formulation of the Four-Colour Conjecture: *every 3-connected 3-regular planar graph is 3-edge-colourable*. This settled the question, he firmly believed, it being widely supposed at the time that every 3-connected 3-regular planar graph was hamiltonian (and hence 3-edge-colourable). However, Tutte (1946) showed the latter assertion to be false. Later, Grinberg (1968) gave a simple necessary condition for a plane graph to be hamiltonian and used it to produce a variety of nonhamiltonian 3-connected 3-regular planar graphs, including the cyclically 5-edge-connected one shown in fig. 10. Ironically, Grinberg's approach was familiar already to Kirkman (1881), but Kirkman, sharing Tait's belief that every 3-connected 3-regular planar graph had at least one Hamilton circuit, employed it merely as an aid in searching for these elusive objects.

Theorem 5.6 (Grinberg's Condition). *Let G be a plane graph on n vertices with a Hamilton circuit C . Denote by f'_i and f''_i the numbers of faces of G of degree i contained in the interior and exterior of C , respectively. Then*

$$\sum_{i=1}^n (i-2)(f'_i - f''_i) = 0.$$

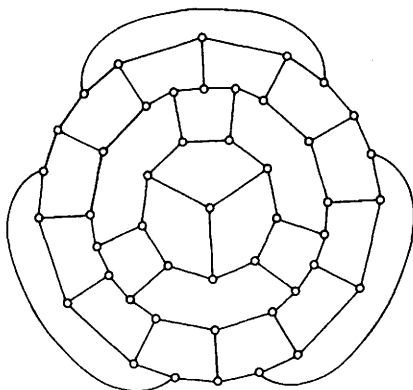


Figure 10. The Grinberg graph.

Proof. Denote by E' the subset of $E(G) \setminus E(C)$ contained in the interior of C . There are $|E'| + 1$ faces of G in the interior of C , and so

$$\sum_{i=1}^n f'_i = |E'| + 1.$$

Each edge in E' lies on the boundary of two faces in the interior of C , and each edge of C lies on the boundary of one such face. Therefore

$$\sum_{i=1}^n i f'_i = 2|E'| + n.$$

Eliminating $|E'|$, we have

$$\sum_{i=1}^n (i - 2) f'_i = n - 2.$$

Similarly,

$$\sum_{i=1}^n (i - 2) f''_i = n - 2.$$

These two equations yield the stated identity. $\square$

To see that the graph of fig. 10 is nonhamiltonian, it suffices to consider the face degrees modulo three and observe that Grinberg's condition cannot hold. It seems hard to construct bipartite counterexamples in this way, and Barnette (1969) has conjectured that there are none. (That the assumption of planarity is necessary here was shown by J.D. Horton; see Bondy and Murty, 1976, p. 240.)

Conjecture 5.7. Every 3-connected 3-regular bipartite planar graph is hamiltonian.

Increasing the connectivity by one, we arrive at a major theorem of Tutte (1956).

Theorem 5.8 (Tutte's Theorem). *Every 4-connected planar graph is hamiltonian.*

Theorem 5.8 is an easy consequence of a lemma about bridges of circuits in planar graphs (the Bridge Lemma, 5.12). Before stating this lemma, we briefly review the important role of bridges in planar graphs. (Bridges were introduced in section 4, before Theorem 4.16.)

A characterization of planar graphs in terms of their bridge structure was given by Tutte (1959). This characterization is the basis of most planarity-testing algorithms. Let G be a graph and C a circuit of G . Two bridges B_1 and B_2 of C in G *overlap* if C cannot be divided into two segments C_1 and C_2 with the property that the vertices of attachment of B_i to C lie on C_i , $i = 1, 2$. The *overlap graph* of G with respect to C has as vertices the bridges of C in G , two bridges being adjacent if they overlap.

Theorem 5.9. *A graph G is planar if and only if, for every circuit C of G , the overlap graph of G with respect to C is bipartite.*

A circuit C in a graph G is *nonseparating* if it has at most one proper bridge, and *peripheral* if it is nonseparating and has no chords. Tutte (1963) showed that in a 3-connected plane graph a circuit is peripheral if and only if it bounds a face. It follows that a 3-connected planar graph has a unique plane embedding (in the sense that there is a unique planar dual) and that its peripheral circuits generate the cycle space. In fact, the latter assertion is true even for nonplanar graphs, as Tutte (1963) proved.

Theorem 5.10. *In a 3-connected graph the peripheral circuits generate the cycle space.*

An insightful exposition of the theory of bridges, leading to a refinement of his original proof of Theorem 5.8, is given by Tutte (1977). Bridges clearly play a very important role in the study of paths and circuits, and it can be argued that their role is central. This is the point of view adopted by Voss (1991) in his treatise on circuits. Of relevance to the study of long circuits is the concept of the span of a bridge, introduced by Voss (1991). Let G be a graph, H a subgraph of G and B a bridge of H in G . The *span* of B (Voss uses the term *length*) is the maximum number of edges in a tree of B whose end vertices are the vertices of attachment of B to H . Voss (1991) made the following conjecture, and proved it for $m \leq 3$.

Conjecture 5.11. Let G be a 2-connected graph, C a longest circuit of G and B_i , $1 \leq i \leq m$, disjoint bridges of C which induce a tree in the overlap graph of G with respect to C . Then

$$\sum_{i=1}^m s(B_i) \leq \ell(C)/2,$$

where $s(B_i)$ denotes the span of B_i and $\ell(C)$ the length of C .

We now return to the statement of the Bridge Lemma.

Lemma 5.12 (The Bridge Lemma). *Let G be a 2-connected plane graph, e an edge of G , C' and C'' the facial circuits of G which include e , and e' any edge of C' . Then there is a circuit C in G including both e and e' such that*

- (i) *each bridge of C in G has either two or three vertices of attachment;*
- (ii) *each bridge of C in G that includes an edge of C' or C'' has exactly two vertices of attachment.*

Let G be a simple 4-connected plane graph, and define e , C' , C'' and e' as in the Bridge Lemma. Then G has a circuit C which includes e and e' and satisfies conditions (i) and (ii). This circuit is necessarily a Hamilton circuit of G . Suppose, to the contrary, that C has a proper bridge. By (i), each bridge of C has at most three vertices of attachment. Because G is 4-connected, C must be a triangle and have exactly one proper bridge B , with three vertices of attachment. Moreover, because G is simple, C can have no degenerate bridge. It follows that $C = C'$ and that B includes an edge of C'' . But this contradicts (ii). Thus every 4-connected planar graph is hamiltonian (Theorem 5.8). Thomassen (1983b) obtained an analogue (and strengthening) of the Bridge Lemma for bridges of paths in plane graphs, and deduced that every 4-connected planar graph is Hamilton-connected. A related class of 4-connected graphs are the graphs of simple 4-polytopes. D.W. Barnette (see Grünbaum 1970, p. 1145) has conjectured that these too are hamiltonian.

While they need not be hamiltonian, 3-connected planar graphs always have long circuits. Jackson and Wormald (1992a) employed the Bridge Lemma (5.12) to prove that every 3-connected planar graph on n vertices has a circuit whose length is of order at least n^c , where $c \approx 0.2$. Examples due to Moon and Moser (1963), constructed from K_4 by recursively inserting a vertex of degree three in each face, show on the other hand that such a graph need not have any circuit whose length is of order more than n^c , where $c := \log 2 / \log 3 \approx 0.63$.

Another property enjoyed by 3-connected planar graphs was discovered by Barnette (1966), who proved that every 3-connected planar graph has a 3-tree. A stronger conjecture, proposed by Jackson and Wormald (1990a), was confirmed by Gao and Richter (1994).

Theorem 5.13. *Every 3-connected planar graph has a 2-walk.*

Properties of graphs embeddable in the plane and other surfaces are discussed in chapter 5: Embeddings and Minors.

5.3. Regular graphs and vertex-transitive graphs

Regular graphs are a particularly fascinating class. The classical examples are the graphs of the Platonic solids (fig. 1). Regular graphs of degree three are often referred to as *cubic* graphs. They appear to capture much of the complexity of graphs in general, and interesting examples abound, the Heawood graph (fig. 4), the Petersen graph (fig. 8), and the Coxeter graph (fig. 9) being three notable ones.

Nonhamiltonian d -regular d -connected graphs exist for all $d \geq 3$. A construction, based on the Petersen graph, was first given by Meredith (1973); moreover, an infinite family of such graphs in which the longest circuit is of length no more than n^c , where n is the order and $c (< 1)$ depends only on d , was described by Jackson and Parsons (1982). Nevertheless, as we have remarked, the imposition of regularity often has surprising consequences. A convincing illustration is provided by the following theorem of C.A.B. Smith (see Tutte 1946) on the parity of the number of Hamilton circuits in a 3-regular graph.

Theorem 5.14 (Smith's Theorem). *Let G be a 3-regular graph and e an edge of G . Then the number of Hamilton circuits of G containing e is even.*

Smith's Theorem (5.14) implies that a 3-regular graph with one Hamilton circuit has at least three such circuits. But, as we shall see, the proof is not constructive, and Chrobak and Poljak (1988) pose the intriguing problem of efficiently finding a second Hamilton circuit in such a graph. The proof we give makes use of the concept of an even 2-factor. A d -factor of a graph is a d -regular spanning subgraph. A 2-factor is *even* if each of its components is an even circuit.

Proof. Consider the bipartite graph $H(X, Y)$, where X is the set of even 2-factors of G containing e , Y is the set of 3-edge colourings of G , and an even 2-factor F containing e is joined to a 3-edge colouring (E_1, E_2, E_3) if and only if $F = E_i \cup E_j$ for some $\{i, j\} \subset \{1, 2, 3\}$. A vertex F of X with c components has degree 2^{c-1} in H , and each vertex of Y has degree two. Thus the vertices of H of odd degree are precisely the Hamilton circuits of G containing e . By Corollary 1.2, their number is even. $\square$

The hamiltonian problem remains $\mathcal{NP}$ -complete even when restricted to 3-connected 3-regular graphs. Nevertheless, almost all such graphs are indeed hamiltonian, by a theorem of Robinson and Wormald (1992), who also established the following more general result (Robinson and Wormald 1991).

Theorem 5.15. *For every integer $d \geq 3$, almost all d -regular graphs are hamiltonian.*

Surprisingly, planarity changes the picture completely, as was shown by Richmond et al. (1985).

Theorem 5.16. *Almost all 3-connected 3-regular planar graphs are not hamiltonian.*

Theorems 5.15 and 5.16 are probabilistic statements. For their precise meaning, we refer the reader to chapter 6: Random Graphs, or to the book of Bollobás (1985).

Jackson (1980a) obtained an analogue of Dirac's Theorem (2.1) for regular graphs. The effect of regularity here is quite striking.

Theorem 5.17 (Jackson's Theorem). *Let G be a 2-connected d -regular graph on n vertices, where $d \geq n/3$. Then G is hamiltonian.*

The bound on d , already close to best possible, was reduced to $(n/3) - 1$ for all $d \geq 3$ by Hilbig (1986); there are two exceptions, the Petersen graph (fig. 8) and the graph obtained from it by *expanding* one vertex to a triangle (that is, deleting the vertex, adding a triangle, and joining the three former neighbours of the vertex to the three vertices of the triangle by a matching).

Jackson's Theorem (5.17) indicates that regular graphs can be expected to contain considerably longer paths and circuits than graphs of the same minimum degree, and one may ask how long a path or circuit there must be in a 2-connected d -regular graph on n vertices. The following conjecture, due to Bondy (1978b), includes Jackson's Theorem (5.17) as a special case.

Conjecture 5.18. *Let G be a 2-connected d -regular graph on n vertices, where $d \geq n/r$, $r \geq 3$, and n is sufficiently large. Then G contains a circuit of length at least $2n/(r-1)$.*

This conjecture is of interest only when r is small compared to n ; when r is of order n , regularity plays essentially no role and can be replaced by a condition on the maximum degree. Voss (1991) showed that a 2-connected graph of maximum degree Δ on n vertices, where $3 \leq \Delta \leq n-2$, has a circuit of length at least $4 \log_{\Delta-1} n - c$, where c is an appropriate constant, while Lang and Walther (1968) constructed regular examples containing no circuits of length more than $4 \log_{\Delta-1} n + 4$. For bipartite graphs, we have the following theorem of Jackson and Li (1994); Häggkvist (1978a) conjectures that the bound on d can be reduced to $n/6$.

Theorem 5.19. *Let G be a 2-connected d -regular bipartite graph on n vertices, where $d \geq (n/6) + 8$. Then G is hamiltonian.*

As noted earlier, the passage from 2-connected to 3-connected graphs often has a substantial effect on circuit structure. Regular graphs are no exception. Fraisse (1986) conjectures that the lower bound on the degree in Jackson's Theorem (5.17) can be reduced significantly for 3-connected regular graphs. The following slightly sharper formulation of Fraisse's conjecture is due to B. Jackson (personal communication 1989).

Conjecture 5.20. *Let G be a 3-connected d -regular graph on n vertices, where $d \geq n/4$. Then G is hamiltonian unless G is either the Petersen graph (fig. 8) or the graph obtained from it by expanding one vertex to a triangle.*

Broersma et al. (preprint) showed that, if $d \geq (n+3)/4$, such a graph has a dominating circuit. Using this fact, they proved that every 3-connected d -regular graph on n vertices, where $d \geq (2n/7) + 2$, is hamiltonian. Surprisingly, raising the connectivity further has little effect. For $d \equiv 0 \pmod{4}$, B. Jackson (see Jackson

et al. 1991) and H.A. Jung (see Min Aung 1989) independently found an example of a nonhamiltonian $(d/2)$ -connected d -regular graph on $4d + 1$ vertices.

Of relevance to Conjecture 5.20 is the following theorem of Fan (1985) and Jung (1984).

Theorem 5.21. *Let G be a 3-connected d -regular graph. Then G contains either a circuit of length at least $3d$ or a Hamilton circuit.*

When d is small, this theorem is weak. Jackson (1986) proved that a 3-connected 3-regular graph on n vertices contains a circuit whose length is of order at least n^c , where $c := \log_2(1 + \sqrt{5}) - 1 \approx 0.69$. Examples based on the Petersen graph (fig. 8) show, on the other hand, that such a graph need not have any circuit whose length is of order more than n^c , where $c := \log 8 / \log 9 \approx 0.96$. However, these constructions work because the resulting graphs are not cyclically 4-edge-connected; they have many 3-edge cuts, which serve to restrict the possible routes taken by a circuit. The following conjecture is due to the author (see Jackson 1986), and is implied by successively stronger conjectures of Thomassen (1986), Matthews and Sumner (1984) and Jackson (1992) (see section 5.4, below).

Conjecture 5.22. *Let G be a 3-regular cyclically 4-edge-connected graph on n vertices. Then G contains a circuit of length at least cn , where c is a positive absolute constant.*

The question of circuits through specified vertices in regular graphs was considered by Holton et al. (1982) and by Kelmans and Lomonosov (1982a). Both groups of authors proved the following theorem on 3-regular graphs.

Theorem 5.23. *Let G be a 3-connected 3-regular graph, and let S be a set of nine vertices of G . Then there is a circuit of G that includes every vertex of S .*

The Petersen graph (fig. 8) is 3-connected and 3-regular but has no circuit including all ten vertices. Ellingham et al. (1984) showed that any 3-connected 3-regular graph having a set S of ten vertices not lying on a common circuit necessarily contains a subdivision of the Petersen graph in which S is the set of vertices of degree three (see, also, Kelmans and Lomonosov 1982b). One may ask the following general question.

Problem 5.24. Does there exist a constant $c > 1$ such that, for every d -regular d -connected graph G and for every set S of $\lfloor cd \rfloor$ vertices of G , where $d \geq 2$, there is a circuit of G that includes every vertex of S ?

Regular digraphs, also, have been studied with regard to circuit structure. For instance, Thomassen (1981b) generalized Corollary 4.18 to digraphs.

Theorem 5.25. *Let D be a strict d -diregular digraph on $2d + 1$ vertices, where $d \geq 1$. Then either D has a directed Hamilton circuit or D is isomorphic to one of two exceptional digraphs of orders five and seven.*

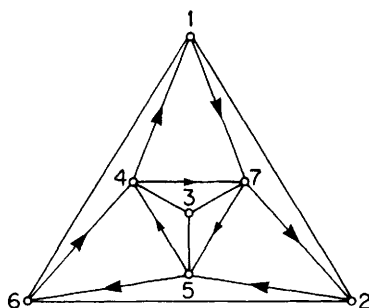


Figure 11. A digraph analogue of the Petersen graph.

The larger of the two exceptional digraphs in Theorem 5.25 is shown in fig. 11. Found by Bondy (1978b), it has several properties reminiscent of the Petersen graph (fig. 8), being 3-diregular, maximally nonhamiltonian, and also hypohamiltonian. Moreover, as R. Häggkvist (personal communication 1978b) observed, the sets of outneighbours of its seven vertices constitute the edges of the Fano hypergraph (fig. 3).

In this chapter, we have chosen to concentrate on longest circuits in graphs. It is appropriate here, however, to say a word about the girth of regular graphs. Regular graphs of diameter k and girth $2k + 1$ are called *Moore graphs*, regular graphs of given girth with the minimum number of vertices *cages*. The Petersen graph (fig. 8) is a Moore graph and also a cage; the Heawood graph (fig. 4) is a cage, too. In general, members of both these classes tend to have much symmetry, and are mainly studied by algebraic methods. We refer the reader to chapter 31 or to the survey by Wong (1982) for further details.

The directed analogue of the cage appears to have a far simpler structure. For $d \geq 1$ and $k \geq 2$, consider the digraph whose vertices are the integers modulo $d(k - 1) + 1$, (i, j) being an edge if and only if $1 \leq j - i \leq d$. This digraph is d -diregular and has girth k . Behzad et al. (1970) conjectured that no digraph with these properties has fewer vertices.

Conjecture 5.26. Let D be a strict d -diregular digraph on n vertices. Then D has girth at most $\lceil n/d \rceil$.

We conclude this section by noting that several of the results mentioned here do not depend on strict regularity for their conclusions but rather on bounds on the minimum and/or maximum degrees. For instance, Jackson (1980a) gave an extension of Theorem 5.17 to almost-regular graphs, and Jackson and Wormald (1993) generalized the theorem of Jackson (1986) on longest circuits in 3-connected 3-regular graphs to graphs of arbitrary maximum degree as follows.

Theorem 5.27. Let G be a 3-connected graph of maximum degree Δ on n vertices. Then G has a circuit of length at least $n^c/2$, where $c := (1 + \log_2 3 + 2 \log_2 \Delta)^{-1}$.

Caccetta and Häggkvist (1978) proposed a generalization of Conjecture 5.26.

Conjecture 5.28. Let D be a strict digraph of minimum outdegree δ^+ on n vertices. Then D has girth at most $\lceil n/\delta^+ \rceil$.

This conjecture has been verified for $\delta^+ \leq 5$ (see Hoáng and Reed 1987) and a near-optimal upper bound, of $(n/\delta^+) + 2500$, was established by Chvátal and Szemerédi (1983); the constant has since been reduced, but not eliminated. Surprisingly, the case when $\delta^+ \geq n/3$ is still unresolved.

An important subclass of regular graphs, with correspondingly stronger properties, are those that are vertex-transitive. An *automorphism* of a graph is an isomorphism from the graph to itself. A graph is *vertex-transitive* if its automorphism group is transitive on its vertices. Vertex-transitive graphs can be constructed in a variety of ways. Two particularly interesting families are the Kneser graphs and the Levi graphs.

Let m be a positive integer and S an n -set, where $n > 2m$. The simple graph whose vertices are the m -subsets of S , two being adjacent if they are disjoint, is called a *Kneser graph* and denoted by $KG(m, n)$. Such graphs, first defined by Kneser (1955), are clearly vertex-transitive. The Petersen graph (fig. 8) is $KG(2, 5)$; the Coxeter graph (fig. 9) can be derived from $KG(3, 7)$ by deleting the seven edges of a Fano hypergraph (fig. 3).

Let $PG(2, n)$ be a finite projective plane of order n , with point set P and line set L . The incidence graph of the hypergraph with vertex set P and edge set L is called the *Levi graph* of $PG(2, n)$ and denoted by $LG(2, n)$; the Heawood graph (fig. 4) is $LG(2, 2)$. If $n = q$, a prime power, the Levi graph $LG(2, q)$ is vertex-transitive (see, for example, Biggs and White 1979).

Lovász (1970) considered the hamiltonicity of vertex-transitive graphs, and asked the following question.

Problem 5.29. Does every connected vertex-transitive graph have a Hamilton path?

This is so for graphs on p , $2p$, $3p$, $4p$, $5p$, p^2 , $2p^2$ and p^3 vertices, where p is prime (see Alspach 1981). In fact, besides K_1 and K_2 , only four simple connected vertex-transitive graphs without Hamilton circuits are known: the Petersen graph (fig. 8), the Coxeter graph (fig. 9), and the 3-regular graphs derived from these by expanding each vertex to a triangle. C. Thomassen (see Bermond 1978b) conjectures that there are only finitely many such graphs. Babai (1979) has shown that every connected vertex-transitive graph on n vertices has a circuit whose length is of order at least $n^{1/2}$.

Groups give rise to vertex-transitive graphs in a natural way. Let Γ be a group and X a set of elements of Γ . The digraph $D(\Gamma, X)$ with vertex set Γ and edge set $\{(g, gx): g \in \Gamma, x \in X\}$ is the *Cayley digraph* of Γ with respect to X . If $X^{-1} = X$, $D(\Gamma, X)$ is the associated digraph of a simple graph $G(\Gamma, X)$, called the *Cayley graph* of Γ with respect to X . The graph $G(\Gamma, X)$ and digraph $D(\Gamma, X)$ are loopless if and only if $1 \notin X$ and connected if and only if X generates Γ . When Γ is a

cyclic group, the resulting Cayley graph or digraph is called a *circulant* or *directed circulant*, respectively. The Koh digraph (fig. 2) is a directed circulant on the group $\Gamma := \mathbb{Z}_7$ with generating set $X := \{1, 5\}$. T.D. Parsons (see Witte and Gallian 1984) proposed the following conjecture.

Conjecture 5.30. Every Cayley graph has a Hamilton circuit.

A review of progress on this conjecture is given in the survey by Witte and Gallian (1984). Vertex-transitive graphs and other highly symmetric graphs are discussed in chapter 27.

5.4. Line graphs and claw-free graphs

The *line graph* of a graph G is the graph with vertex set $E(G)$, two vertices being adjacent if and only if they are adjacent (as edges) in G . (Line graphs should more properly be called *edge graphs*, but the name is by now too well entrenched to be changed.) The line graph of G is denoted by $L(G)$.

The question of hamiltonicity of line graphs is intimately related to the notion of an *Euler tour*, a closed trail that includes every edge of the graph. A graph is *eulerian* if it admits an Euler tour. Such graphs were characterized, not surprisingly, by Euler (1736).

Theorem 5.31. *A graph admits an Euler tour if and only if it is connected and even.*

A proof of Theorem 5.31 was given only much later, by Hierholzer (1873). Hierholzer's proof yields a linear-time algorithm for finding an Euler tour in an eulerian graph. Harary and Nash-Williams (1965) made the key observation linking Hamilton circuits in line graphs to Euler tours.

Proposition 5.32. *Let G be a graph on at least three edges. Then $L(G)$ is hamiltonian if and only if G has a dominating eulerian subgraph.*

Jaeger (1979) established a sufficient condition for a graph to contain a spanning (and hence dominating) eulerian subgraph.

Theorem 5.33. *Every 4-edge-connected graph has a spanning eulerian subgraph.*

As Thomassen (1986) observed, it follows from Theorems 5.32 and 5.33 that line graphs of 4-edge-connected graphs are hamiltonian. By similar methods, Zhan (1991) showed that 7-connected line graphs are Hamilton-connected. These results lend support to the following conjecture of Thomassen (1986).

Conjecture 5.34. Every 4-connected line graph is hamiltonian.

A basic property of line graphs is that they contain no induced subgraph isomorphic to $K_{1,3}$. Such a subgraph is referred to as a *claw*, and a graph which contains

no claw is called *claw-free*. The original motivation for studying claw-free graphs was that they include line graphs as a proper subclass; Minty (1980) and Sbihi (1980) described polynomial-time algorithms for finding a maximum stable set in a claw-free graph, thereby extending Edmonds's maximum matching algorithm (1965), and a third such algorithm can be found in Lovász and Plummer (1986, pp. 471–480). The evidence now, however, is that claw-free graphs are of interest in their own right.

In a claw-free graph, the subgraph induced by the neighbours of a vertex has stability number one or two. If this subgraph is disconnected, it necessarily consists of two disjoint cliques; if it is connected, it contains a Hamilton path by Corollary 4.17. Claw-free graphs therefore include two quite distinct subclasses: those in which each neighbourhood induces a disconnected subgraph (for instance, line graphs of triangle-free simple graphs of minimum degree at least two), and those in which each neighbourhood induces a connected subgraph (the octahedron and icosahedron of fig. 1 being two examples). Graphs with this latter property are called *locally-connected*.

Graphs in the former class do not necessarily have Hamilton circuits; indeed, if G is an arbitrary simple 3-regular graph, the graph H obtained from G by expanding each vertex to a triangle is hamiltonian if and only if G is hamiltonian; moreover, G and H have the same connectivity. Since it is $\mathcal{NP}$ -hard to determine whether an arbitrary 3-connected 3-regular graph is hamiltonian, we deduce that it is $\mathcal{NP}$ -hard to determine whether an arbitrary 3-connected 3-regular claw-free graph is hamiltonian. On the other hand, Oberly and Sumner (1979) proved that every connected, locally-connected claw-free graph is hamiltonian.

One may ask how long a circuit a k -connected claw-free graph on n vertices must contain. For $k = 2$, an upper bound can be derived from the work of Lang and Walther (1968), who constructed 2-connected 3-regular graphs on n vertices with no circuit of length more than $4 \log_2 n + 4$ (see the discussion following Conjecture 5.18). By expanding each vertex to a triangle, one obtains 2-connected claw-free graphs on $3n$ vertices with no circuit of length more than $12 \log_2 n + 12$. Jackson and Wormald (personal communication 1992b) proved that this lower bound is of the right order of magnitude. They also obtained an analogous result for 3-connected claw-free graphs.

Theorem 5.35. *Let G be a 3-connected claw-free graph on n vertices. Then G contains a circuit of length at least n^c , where c is a positive constant.*

A similar construction shows that this theorem is sharp, too, apart from the value of the constant c . The case of 4-connected graphs was addressed by Matthews and Sumner (1984), who proposed the following conjecture, a strengthening of Conjecture 5.34.

Conjecture 5.36. Every 4-connected claw-free graph is hamiltonian.

A still stronger conjecture was put forward by Jackson (1992). A *Tutte circuit* is a circuit of length at least four each of whose bridges has at most three vertices

of attachment. The Bridge Lemma (5.12) guarantees the existence of such circuits in 2-connected planar graphs on at least four vertices. Jackson conjectures their existence in 2-connected claw-free graphs as well.

Conjecture 5.37. Every 2-connected claw-free graph on at least four vertices contains a Tutte circuit.

A similar strengthening of Theorem 5.33 was also proposed by Jackson (1992).

Conjecture 5.38. Every simple 2-edge-connected graph on at least four edges contains a connected even subgraph on at least four edges each of whose bridges has at most three edges of attachment.

Conjecture 5.38 would imply that every 2-connected 3-regular graph on at least four vertices has a Tutte circuit, and this would imply the truth of Conjecture 5.22 (with $c = 3/4$).

Li (1988) proved that the bound on the degree of regularity in Jackson's Theorem (5.17) can be reduced even further when the graphs are assumed also to be claw-free.

Theorem 5.39. Let G be a 2-connected d -regular claw-free graph on n vertices, where $d \geq n/5$ and $d \geq 10$. Then G is hamiltonian.

5.5. Oriented graphs and tournaments

Whereas two edges (one in each direction) may link a pair of vertices in a strict digraph, at most one edge is permitted in an oriented graph. For this reason, and by analogy with the situation that obtains in bipartite graphs, one might expect conditions for the existence of directed paths and circuits in oriented graphs to be roughly half as demanding as their digraph counterparts. This is sometimes the case, but there are exceptions. Moreover, best possible results are much harder to come by. Jackson (1980b) showed that an oriented graph with minimum indegree and outdegree k has a directed path of length at least $2k$. However, the corresponding question for directed circuits, also considered by Jackson (1980b), remains unresolved.

Conjecture 5.40. Let D be a strong oriented graph with minimum indegree and outdegree at least k . Then D has a directed circuit of length at least $2k + 1$.

This conjecture is sharp when k is small compared to the order of D , but not when k is large. Häggkvist (1993) proved that an oriented graph on n vertices with minimum indegree and outdegree at least cn , where $c := (1/2) - 2^{-18} \approx 0.499996$, has a directed Hamilton circuit, and conjectured that a much smaller constant will do.

Conjecture 5.41. Let D be an oriented graph of minimum outdegree δ^+ on n vertices, where $\delta^+ \geq (3n - 2)/8$. Then D contains a directed Hamilton circuit.

An analogous conjecture for diregular oriented graphs was proposed by Jackson (1981b).

Conjecture 5.42. Every k -diregular oriented graph on at most $4k + 1$ vertices, where $k \neq 2$, contains a directed Hamilton circuit.

Evidently, questions about oriented graphs are much easier to come by than are answers, and we mention here just one more. Due to Thomassen (1987a), it concerns a property which resembles maximal nonhamiltonicity but is meaningful only for oriented graphs.

Problem 5.43. Is there a nonhamiltonian oriented graph in which the reversal of any edge results in a hamiltonian graph?

Remark 5.44. There do exist oriented graphs with the slightly weaker property that reversing any edge increases the length of a longest directed circuit; one such family, found by Thomassen (1987b), are the oriented graphs $C_5 \square C_{10k+7}$, $k \geq 0$.

While oriented graphs in general are poorly understood, the theory of tournaments is rich and highly developed, as the beautiful theorems of Rédei (3.4), Thomason (3.7) and Forcade (3.8) on Hamilton paths and circuits in tournaments demonstrate. Two directed analogues of Hamilton-connectedness were considered by Thomassen (1980a).

A digraph is *weakly Hamilton-connected* if any two vertices are connected by a directed Hamilton path (in at least one direction) and *strongly Hamilton-connected* if any two vertices are connected by directed Hamilton paths in both directions. Thomassen (1980a) characterized the weakly Hamilton-connected tournaments and proved that 4-connected tournaments are strongly Hamilton-connected. He also showed that each edge of a 3-connected tournament lies in a directed Hamilton circuit. Thomassen's methods were adapted by Bang-Jensen et al. (1992), resulting in a polynomial-time algorithm for finding, in a tournament, a directed Hamilton path with prescribed head and tail, if there is one. In particular, the algorithm can be applied to determine whether a tournament is weakly or strongly Hamilton-connected and whether there is a directed Hamilton circuit through a prescribed edge in a tournament. Bang-Jensen et al. (1992) also made the following conjecture.

Conjecture 5.45. For each fixed k , there exists a polynomial-time algorithm for deciding if there is a directed Hamilton circuit through k prescribed edges in a tournament.

Bang-Jensen and Thomassen (1992) proved that the above problem is $\mathcal{NP}$ -complete if k is not fixed, and they described a polynomial-time algorithm for finding a directed circuit in a tournament through two prescribed edges, if there is one. This latter problem is $\mathcal{NP}$ -complete for digraphs in general, by a result of Fortune et al. (1980).

We close this brief discussion of oriented graphs by defining the bipartite analogue of a tournament and noting a bipartite analogue of Camion's Theorem (3.9) due to Gutin (1984) and, independently, Häggkvist and Manoussakis (1989). A *bipartite tournament* is an orientation of a simple complete bipartite graph. A *directed d -factor* of a digraph is a d -diregular spanning subgraph.

Theorem 5.46. *A strong bipartite tournament has a directed Hamilton circuit if and only if it has a directed 1-factor.*

The proof by Häggkvist and Manoussakis (1989) of Theorem 5.46 makes use of a neat trick. The bipartite tournament $T(V_1, V_2)$ is transformed into a 2-edge-coloured complete graph G on $V_1 \cup V_2$ by assigning colour i to each edge of $G[V_i]$ and to each edge of T whose tail lies in V_i , $i = 1, 2$. Noting now that T has a directed Hamilton circuit if and only if G has a Hamilton circuit whose edges alternate in colour, Häggkvist and Manoussakis (1989) apply a necessary and sufficient condition due to Bánkfalvi and Bánkfalvi (1968) for a 2-edge-coloured complete graph to possess such a circuit.

6. Special proof techniques for paths and circuits

Thus far, a variety of proof techniques have been introduced. Some, such as counting in two ways (Proposition 1.1), induction (Proposition 1.5), contradiction (Theorem 1.7), search trees (Theorem 1.8), maximal counterexample (Theorem 2.12), and inclusion–exclusion (Theorem 3.4) are of general applicability, while others are expressly designed to handle questions on paths and circuits. Among the latter are the use of maximal paths (Theorem 1.3), longest circuits (Theorem 2.1, Proof 1), longest paths (Theorem 2.1, Proof 2), lollipops (Theorem 2.1), the closure (Theorem 2.7), bypasses (Theorem 3.9) and vines (Theorem 4.11). In this section, we discuss three additional proof techniques specific to paths and circuits.

6.1. Thomason's lemma

Let G be a graph and x a vertex of G . Recall that an x -path is a path in G with tail x . If P is an x -path in G whose head y is adjacent to a vertex z of P other than its immediate predecessor on P , then $P + yz$ is an (x, z) -lollipop and thus, by the Lollipop Lemma (2.2), contains a second spanning x -path Q . The x -paths P and Q are said to be related by an *elementary exchange*. The graph whose vertices are the longest x -paths of G , two such paths being adjacent if and only if they are related by an elementary exchange, is called the *x -path graph* of G . Thomason (1978) proved a basic lemma on x -paths with the aid of this concept.

Lemma 6.1 (Thomason's Lemma). *Let G be a connected graph on at least two vertices, and let x be a vertex of G . Then the number of longest x -paths of G that terminate in a vertex of even degree is even.*

Proof. Let H be the x -path graph of G and P a longest x -path of G . If P terminates in y ,

$$d_H(P) = d_G(y) - 1.$$

Thus y is of even degree in G if and only if P is of odd degree in H . Since the number of vertices of odd degree in H is even, by Corollary 1.2, the number of longest x -paths in G that terminate in a vertex of even degree is also even. $\square$

Corollary 6.2. *Let G be a graph on at least three vertices, and let x and y be two vertices of G . If each vertex of G other than x and y is of odd degree, the number of Hamilton xy -paths in G is even.*

Proof. We may assume that G has at least one Hamilton xy -path; otherwise, the conclusion is trivial. The longest x -paths of $G \setminus y$ are then Hamilton paths of $G \setminus y$, and each Hamilton xy -path of G is an extension of such a path. Let P be a longest x -path of $G \setminus y$ terminating in z . If z is of odd degree in $G \setminus y$, the number of edges between y and z is even, since z is of odd degree in G . Thus P gives rise to an even number of Hamilton xy -paths in this case. On the other hand, if z is of even degree in $G \setminus y$, then P gives rise to an odd number of Hamilton xy -paths; by Thomason's Lemma (6.1), however, the number of such paths P is even. Hence the total number of Hamilton xy -paths in G is also even. $\square$

Smith's Theorem (5.14) is an immediate consequence of Corollary 6.2; it suffices to choose u and v as the ends of the given edge e . Thomason (1978) also applied his Lemma (6.1) to derive a generalization of the theorem of J. Bosák and A. Kotzig (see Bosák 1967) asserting that every 3-regular bipartite graph of order at least four has an even number of Hamilton circuits.

The x -path graph has other applications. Hendry (1984, 1986) considered the class of graphs in which there is a unique Hamilton path from one vertex x to every other vertex. He showed that the x -path graph of such a graph is a forest with $d(x)/2$ components, and deduced that every such graph on n vertices has precisely $3(n-1)/2$ edges.

6.2. Pósa's lemma

The naive approach to finding a long path in a graph is to grow a maximal path P and consider the paths obtainable from P by means of elementary exchanges (applied at either end); if one of these is not maximal, it can be extended to a maximal path longer than P . The procedure may then be repeated. While this approach fails badly on certain graphs, Pósa (1976) proved that it works remarkably well on most graphs of sufficiently high edge density. His argument hinges on the following lemma.

Lemma 6.3 (Pósa's Lemma). *Let G be a graph, P a longest path in G and $\mathcal{P}$ the set of all paths of G obtainable from P by elementary exchanges. Denote by S the*

set of ends of paths in $\mathcal{P}$ and by S^- and S^+ the sets of vertices immediately preceding and following the vertices of S on P , respectively. Then

$$N(S) \subseteq S^- \cup S \cup S^+.$$

Proof. Let $x \in S$ and $y \in V(G) \setminus (S^- \cup S \cup S^+)$, and consider an x -path Q in $\mathcal{P}$. If $y \in V(G) \setminus V(P)$, x and y cannot be adjacent because Q is a longest path in G . Suppose, then, that $y \in V(P) \setminus (S^- \cup S \cup S^+)$. Then y has the same neighbours on each path in $\mathcal{P}$, because an elementary exchange that removed one of these neighbours would, at the same time, establish either it or y as an element of S . If x and y were adjacent, an elementary exchange applied to Q would thus yield a path whose head was a neighbour of y on P , a contradiction. Thus, in both cases, x and y are nonadjacent. $\square$

Pósa (1976) used a simple consequence of his Lemma 6.3, most easily stated using the concept of the degree of a set of vertices. Let S be a set of vertices in a graph G . The *degree* of S in G , denoted by $d(S)$, is the number of neighbours of S in $G \setminus S$, that is,

$$d(S) := |N(S) \setminus S|.$$

(A priori, it might seem more natural to define the degree of a set S of vertices as the total number of neighbours of S , $|N(S)|$, but the definition given here appears to be the right one. Note that the two definitions coincide when S is a stable set.)

Corollary 6.4. *Let G be a graph and k and ℓ integers, where $k \geq -1$, $\ell \geq 1$ and $k \leq \ell$. If $d(S) \geq 2|S| + k$ for every nonempty set S of vertices such that $|S| \leq \lceil (\ell - k + 1)/3 \rceil$, then G contains a path of length ℓ .*

Proof. Let P be a longest path in G and define S as in Pósa's Lemma (6.3). Then $N(S) \setminus S \subseteq S^- \cup S^+$, and so $d(S) \leq 2|S| - 2$. By the hypothesis of Corollary 6.4, $|S| > \lceil (\ell - k + 1)/3 \rceil$. Let S' be a subset of S such that $|S'| = \lceil (\ell - k + 1)/3 \rceil$. Then $d(S') \geq 2|S'| + k$. Since $V(P) \supseteq S' \cup N(S')$,

$$v(P) \geq |S' \cup N(S')| = |S'| + d(S') \geq 3|S'| + k \geq \ell + 1.$$

Thus P is of length at least ℓ . $\square$

A graph on n vertices in which $d(S) \geq c|S|$ for all sets S of at most $\lceil dn \rceil$ vertices is called a (c, d) -*expander*. The case $k = 0$ of Corollary 6.4 asserts that $(2, 1/3)$ -expanders possess Hamilton paths. Expanders have many useful properties (see, for example, the survey article by Bien 1989). Ways of constructing them are discussed in chapter 32. Note that the case $k = \ell - 2$ of Corollary 6.4 is simply Theorem 1.3(i).

Using Corollary 6.4, Pósa proved that a random graph with n vertices and $cn \log n$ edges, where c is a sufficiently large constant, is almost surely hamiltonian. Apart from the value of the constant c , this result is best possible. Pósa's bound was sharpened fully by Komlós and Szemerédi (1983). Corollary 6.4 has

been invoked to help resolve a number of other problems on paths and circuits in random graphs. We limit ourselves here to one remarkable application by Beck (1983); further applications can be found in the stimulating survey article by Frieze (1989).

Theorem 6.5. *There exists a family of graphs $\{G_\ell: \ell \geq 3\}$ such that $e(G_\ell) \leq 900\ell$ and every subgraph of G_ℓ with at least $e(G_\ell)/2$ edges contains a path of length ℓ .*

Remark 6.6. Theorem 6.5 implies that in any 2-edge-colouring of such a graph G , there is a monochromatic circuit of length m . Statements of this type, asserting the existence of a monochromatic configuration in an edge-coloured graph or hypergraph, are studied under the collective name of Ramsey Theory. Chapter 25 by Nešetřil is devoted to this topic, as is the book by Graham et al. (1980).

6.3. Woodall's hopping lemma

Pósa's Lemma (6.3) demonstrates the effectiveness of iterative procedures in establishing sufficient conditions for Hamilton circuits; so too does the Closure Lemma (2.6). Here, we discuss a powerful technique of this kind, developed by Woodall (1973).

Lemma 6.7 (The Hopping Lemma). *Let G be a graph and C a longest circuit of G such that $G \setminus C$ has as few components as possible. Suppose that some component of $G \setminus C$ is an isolated vertex, u . Put $X_0 := N(u)$ and define, recursively, sets X_i , $i \geq 1$, by*

$$X_i := N((X_{i-1}^- \cap X_{i-1}^+) \cup \{u\}),$$

where S^- and S^+ , respectively, denote the sets of vertices immediately preceding and following the vertices of S on C with respect to a fixed orientation. Set

$$X := \lim_{i \rightarrow \infty} X_i.$$

Then (i) $X \subset V(C)$, and (ii) $X \cap X^+ = \emptyset$.

Corollary 6.8. *Let G , C , u and X be defined as in the Hopping Lemma (6.7) and set*

$$Y := X^- \cap X^+.$$

Then (i) $N(Y \cup \{u\}) = X$, (ii) $Y \cup \{u\}$ is a stable set, and (iii) $v(C) \geq 3|X| - |Y|$.

Proof. Assertion (i) follows directly from the definitions of X and Y . Assertion (ii) is implied by assertion (i), part (ii) of the Hopping Lemma (6.7) and the definition of Y . To prove (iii), consider the partition of $V(C)$ induced by the $|X|$ disjoint segments of C whose tails belong to X and whose heads belong to X^- . By part (ii) of the Hopping Lemma (6.7) each such segment has at least two vertices. Moreover, by the definition of Y , a segment has exactly two vertices if and only if it includes a vertex of Y . Therefore

$$v(C) \geq 3(|X| - |Y|) + 2|Y| = 3|X| - |Y|. \quad \square$$

We give one illustration, due to Woodall (1978a), of the use of the Hopping Lemma (6.7). [The following statement is slightly stronger than the one given by Woodall, who considered only the case $\ell = n$ and imposed the condition $d(S) \geq (n + |S| - 1)/3$ on all nonempty sets S of vertices. The proof, however, is the same.]

Theorem 6.9. *Let G be a simple 2-connected graph of minimum degree δ on n vertices, where $\delta \geq (n + 2)/3$, and let ℓ be a positive integer. If $d(S) \geq (\ell + |S| - 1)/3$ for every stable set S of vertices of G such that $2 \leq |S| \leq (\ell + 1)/2$, then G contains a circuit of length at least ℓ .*

Proof. Let C be a longest circuit of G . Suppose that C is of length less than ℓ . By Theorem 2.16, $V(G) \setminus V(C)$ is a stable set. Let $u \in V(G) \setminus V(C)$, and define the sets X and Y as in the Hopping Lemma (6.7) and Corollary 6.8. By part (ii) of Corollary 6.8, $Y \cup \{u\}$ is a stable set. Since $Y \subset V(C)$, we deduce that

$$|Y \cup \{u\}| \leq \frac{1}{2}v(C) + 1 \leq \frac{1}{2}(\ell + 1).$$

By parts (i) and (iii) of Corollary 6.8,

$$d(Y \cup \{u\}) = |X| \leq \frac{1}{3}(v(C) + |Y|) \leq \frac{1}{3}(\ell + |Y| - 1),$$

contradicting the hypothesis, with $S := Y \cup \{u\}$. $\square$

Theorem 6.9 implies that a simple 2-connected graph on n vertices in which the minimum degree is at least $(n + 2)/3$ and the degree of any stable set of two vertices is at least $n/2$ is hamiltonian. It thus generalizes Dirac's Theorem (2.1) and, for graphs of minimum degree at least $(n + 2)/3$, Ore's Theorem (2.4). The Petersen graph (fig. 8) is an example of a simple nonhamiltonian 2-connected graph on n vertices in which the degree of any stable set of two vertices is at least $n/2$. Broersma et al. (1993) characterized all such graphs, thereby generalizing Ore's Theorem (2.4) fully. Of related interest is a generalization of Dirac's Theorem (2.1) by Häggkvist (1985b), restated here in terms of degrees of sets of vertices.

Theorem 6.10. *Let G be a simple graph of minimum degree δ on n vertices. If there is a positive integer $k \leq \delta$ such that $d(S) \geq n - \delta$ for every set S of k vertices, then G has a Hamilton circuit.*

The Hopping Lemma (6.7) (and variants of it) has diverse applications, among them the theorems of Häggkvist and Thomassen (4.8) and Jackson (5.17) referred to earlier. Suitably generalized, it may well be the appropriate tool to handle open questions such as Conjectures 4.7, 4.9, 4.22, 4.51, 5.11, 5.18, 5.20, 5.22 and 5.36 and Problems 4.55 and 5.24, as well as others to be discussed in later sections, such as Problem 7.14. The following conjectures, due to Thomassen (personal communication 1976; see also Alspach and Godsil 1985, p. 466) and to Fleischner and Jackson (1989), respectively, should also be mentioned in this context.

Conjecture 6.11. *Let C be a longest circuit in a 3-connected graph. Then C has at least one chord.*

Conjecture 6.12. Let G be a 3-regular cyclically 4-edge-connected graph. Then G has a dominating circuit.

Conjecture 6.12 is implied by Conjecture 5.36 and implies Conjecture 5.22. Fleischer and Jackson (1989) have shown that it is equivalent to the conjecture of Thomassen (1986) that every 4-connected line graph is hamiltonian, Fouquet and Thuillier (1990) that it is equivalent to the assertion that every 3-regular cyclically 4-edge-connected graph has a dominating circuit containing two specified non-adjacent edges. (Fouquet and Thuillier 1990, establish a similar strengthening of Conjecture 5.7.) Conjecture 6.12 is true for planar graphs by the Bridge Lemma (5.12).

The toughness is a measure of how well-knit a graph is. Another such measure is the *binding number*, introduced by Woodall (1973). This is the largest value of t such that $|N(S)| \geq t|S|$ for all sets S of vertices such that $N(S) \neq V$. Unlike the toughness, the binding number can be computed in polynomial time, using network flow techniques (Cunningham 1990). Woodall (1973) applied the Hopping Lemma (6.7) to prove that a loopless graph is hamiltonian if its binding number is at least $3/2$. His conjecture that such a graph is pancyclic under the same condition was verified by Shi (1987).

Theorem 6.13. A loopless graph with binding number at least $3/2$ is pancyclic.

7. Lengths of circuits

Most of the discussion so far has centred on long circuits, and we have been content to establish lower bounds on the lengths of circuits in graphs with given properties, rather than their exact values (except, of course, in the case of Hamilton circuits). Here, we consider questions of the latter type. We discuss conditions on the minimum degree and the number of edges guaranteeing the existence of various types of circuits.

Two basic lemmas, both due to Erdős (1965a,b), prove to be very useful in attacking extremal questions such as these. Roughly speaking, the first asserts that a graph with many edges has a subgraph of large minimum degree, the second that a graph of large minimum degree has a bipartite spanning subgraph of large minimum degree.

Lemma 7.1. Let G be a graph on n vertices and more than $(c-1)n$ edges, where c is a positive integer. Then G has a subgraph H of minimum degree at least c .

Proof. Any minimal subgraph H with more than $(c-1)v(H)$ edges has the required property: if H had a vertex v of degree at most $c-1$, the subgraph $H \setminus v$ would contradict the choice of H . $\square$

Lemma 7.2. Every graph G has a bipartite spanning subgraph B such that $d_B(v) \geq d_G(v)/2$ for all vertices v .

Proof. Any bipartite spanning subgraph $B(X, Y)$ with the maximum number of edges has the required property: if B had a vertex v of degree less than $d_G(v)/2$, where without loss of generality $v \in X$, the bipartite spanning subgraph with bipartition $(X \setminus v, Y \cup \{v\})$ would contradict the choice of B . $\square$

These lemmas are often invoked in proofs of extremal theorems. If a best possible result is not sought, but only one in which the number of edges is of the right order of magnitude, their use serves to restrict the class of graphs under consideration to bipartite graphs of large degree, and such graphs are easier to treat. One reason for this is that breadth-first search trees, which are very useful in the analysis of circuit structure, take on a particular form in bipartite graphs of large degree: the sets of vertices equidistant from the root are stable and initially tend to grow rapidly in size with increasing distance from the root.

7.1. Circuits of given length modulo k

The existence of odd and even circuits has already been covered in some detail (see, for example, Theorems 1.8 and 4.13 and the discussions thereon). Barefoot et al. (1991) considered the case of circuits of length 0 modulo 3. They proved that simple 3-regular graphs necessarily have such circuits, and made a stronger conjecture, proved by Chen and Saito (1994).

Theorem 7.3. *Every simple graph of minimum degree at least three contains a circuit of length 0 modulo 3.*

A still more general conjecture was formulated by N. Dean (see Chen and Saito 1994).

Conjecture 7.4. *Every simple graph of minimum degree at least k , where $k \geq 3$, contains a circuit of length 0 modulo k .*

The case $k = 4$ of this conjecture was verified by Dean et al. (1993). Other circuit lengths modulo 3 were considered by Dean et al. (1991), who showed that a simple 2-connected graph of minimum degree at least three contains a circuit of length 1 modulo 3 unless it is the Petersen graph (fig. 8), and a circuit of length 2 modulo 3 unless it is K_4 or $K_{3,3}$ (see also Saito 1992).

Thomassen (1983c) proved that every simple graph of minimum degree at least $2\ell(k+1)$ contains a circuit of even length ℓ modulo k . Note that a bipartite graph has no circuit of odd length modulo k if k is even, no matter how large its minimum degree. On the other hand, for odd k , Bollobás (1977) established the existence of a function f such that every simple graph with minimum degree at least $f(k)$ contains circuits of all lengths modulo k ; Thomassen (1983c) likewise established the existence of a function g such that every simple 2-connected nonbipartite graph with minimum degree at least $g(k)$ contains circuits of all lengths modulo k , and speculated as to the best possible such function g .

Conjecture 7.5. *Every simple 2-connected nonbipartite graph of minimum degree at least $k+2$ contains circuits of all lengths ℓ modulo k .*

A degree condition for the existence of a directed circuit of length 0 modulo k in a strict digraph was given by Alon and Linial (1989). The same condition guarantees, more generally, a directed circuit of weight 0 modulo k in a graph with integral edge weights. Alon (1991) has asked whether there exists a polynomial-time algorithm for finding such a circuit; the proof of Alon and Linial, which uses probabilistic methods, does not supply one.

7.2. Circuits of given length

The complete bipartite graph $K_{n/2, n/2}$ has $n^2/4$ edges and no odd circuits. Theorem 1.7 tells us that any simple graph on n vertices and more than this number of edges contains a triangle. Bondy (1971b), with the aid of Theorem 2.9, showed that any such graph contains circuits of all lengths ℓ , $3 \leq \ell \leq \lfloor (n+3)/2 \rfloor$. This number of edges is extremal for odd ℓ but not for even ℓ . In the range $\lfloor (n+3)/2 \rfloor \leq \ell \leq n$, Woodall (1972) determined the exact bound, namely $\binom{\ell-1}{2} = \binom{n-\ell+2}{2}$; the connected graph with blocks $K_{\ell-1}$ and $K_{n-\ell+2}$ has this number of edges but no circuit of length ℓ .

The situation as regards short even circuits is less clear. Reiman (1958) observed that only $n^{3/2}/2 + n/4$ edges are needed to guarantee the existence of a circuit of length four; a simple graph G with this number of edges satisfies the inequality

$$\sum_{v \in V} \binom{d(v)}{2} > \binom{n}{2},$$

and thus has two vertices with a pair of neighbours in common, that is, a 4-circuit. By exploiting the structure of breadth-first search trees, Bondy and Simonovits (1974) showed, more generally, that a simple graph on n vertices and more than $c n^{1+1/\ell}$ edges, where $c := c(\ell)$ is a function only of ℓ , contains a 2ℓ -circuit. Apart from the value of c , this number of edges is probably extremal; it is known to be so for $\ell = 2, 3$ and 5 . For instance, the Levi graph $\text{LG}(2, q)$ of a finite projective plane of order q has $2(q^2 + q + 1)$ vertices, $(q+1)(q^2 + q + 1)$ edges and no 4-circuit. Brown (1966) and Erdős et al. (1966b) constructed denser examples in the case $\ell = 2$, also based on finite projective planes, and Füredi (1983, preprint) proved that these examples are precisely the extremal graphs when $n = q^2 + q + 1$ and $q \notin \{1, 7, 9, 11, 13\}$. Benson (1966) settled the cases $\ell = 3$ and $\ell = 5$, also by constructing graphs from finite geometries. Conceptually simpler examples, with fewer edges but still of the right order of magnitude, were constructed by Wenger (1991) for the same three values of ℓ .

Problem 7.6. For each integer $\ell \geq 2$ and for infinitely many values of n , construct a simple graph on n vertices and $c n^{1+1/\ell}$ edges, where c is a constant depending only on ℓ , containing no 2ℓ -circuit.

By Lemma 7.2, the extremal problem is no easier for bipartite graphs. For digraphs, on the other hand, the extremal question was completely settled by Häggkvist and Thomassen (1976). They determined the number of edges needed

in a digraph on n vertices to guarantee the existence of a directed circuit of length ℓ , and also derived bounds that are close to best possible for the restriction of this problem to strong digraphs. These and many other extremal questions on circuits in digraphs are discussed in the survey by Bermond and Thomassen (1981) and in the book by Bollobás (1978).

7.3. Circuits of many lengths

The proof of Theorem 1.3 shows that a simple graph of minimum degree k has at least $k - 1$ different circuit lengths; this is best possible, in view of the complete graph K_{k+1} . A more sensitive measure of the richness of circuit lengths in a graph – the sum of their reciprocals – was proposed by Erdős and Hajnal (1966) (see also Erdős 1975). Defining

$$\mathcal{L}(G) := \sum \{1/\ell : G \text{ has a circuit of length } \ell\},$$

and noting that

$$\mathcal{L}(K_{k+1}) \approx \log k + \gamma - \frac{3}{2} \quad \text{and} \quad \mathcal{L}(K_{k,k}) \approx \frac{1}{2}(\log k + \gamma - \frac{3}{2}),$$

where γ is the Mascheroni constant ($\gamma \approx 0.5772$), they asked if there exists a constant c such that the sum of the reciprocals of the circuit lengths in a simple graph of minimum degree k is at least $c \log k$, regardless of the order of the graph. An affirmative answer was provided by Gyarfás et al. (1984). Note that, by virtue of Lemma 7.1, the same remarkable property holds (with a different constant) in a graph of average degree k . Sparse graphs, with average degree $1 + (1/k)$, where k is sufficiently large, exhibit a similar behaviour. Gyarfás et al. (1985) proved that the sum of the reciprocals of the circuit lengths in such a graph is at least $1/(300k \log k)$.

7.4. Circuits of all lengths

Pancyclic graphs, those that have circuits of all possible lengths, were introduced in section 2. Sufficient conditions for hamiltonicity frequently imply pancyclicity. Corollary 2.10 is a typical instance of this phenomenon, which provides convincing evidence that it is hard to capture the essence of hamiltonicity. This is not surprising, of course, hamiltonicity being an $\mathcal{NP}$ -complete property.

A digraph analogue of Theorem 2.9 was established by Häggkvist and Thomassen (1976), and Corollary 2.10 was generalized to digraphs by Thomassen (1977). A bipartite analogue of Theorem 2.9 was considered by Entringer and Schmeichel (1988). Using similar methods, they proved that a hamiltonian bipartite graph on $2n$ vertices and more than $n^2/2$ edges is *bipancyclic*, that is, has circuits of every even length 2ℓ , $2 \leq \ell \leq n$. This bound on the number of edges is not thought to be best possible, however, and could conceivably be reduced by a factor of two. Specifically, one has the following conjecture of Mitchem and Schmeichel (1985).

Conjecture 7.7. Let G be a simple hamiltonian bipartite graph on $2n$ vertices and at least $n^2/4 + n + 1$ edges. Then G is bipancyclic.

Mitchem and Schmeichel (1985) also proposed a related conjecture, which can be regarded as the bipartite analogue of Corollary 2.10.

Conjecture 7.8. Let G be a simple hamiltonian bipartite graph of minimum degree δ on $2n$ vertices, where $\delta^2 - \delta \geq n$. Then G is bipancyclic.

The Levi graphs of finite projective planes (see sections 5.3 and 7.2) show that the lower bound on δ here is necessary.

We turn now to pancyclicity in tournaments, recalling that every strong tournament is vertex pancyclic. A digraph on n vertices is *edge- ℓ -cyclic* if each edge lies in a directed circuit of length ℓ , and *edge pancyclic* if each edge lies in a directed circuit of each length ℓ , $3 \leq \ell \leq n$. Wu et al. (1981) proved that every edge-3-cyclic tournament on n vertices is edge- ℓ -cyclic for all ℓ , $3 \leq \ell \leq \lfloor (n+5)/2 \rfloor$. This somewhat surprising result was extended by Tian et al. (1982).

Theorem 7.9. *An edge-3-cyclic tournament is edge pancyclic unless it belongs to one of two specified families.*

One corollary of Theorem 7.9 is the theorem of Alspach (1967) that regular tournaments are edge pancyclic. A number of other properties stronger than pancyclicity have been studied. Faudree and Schelp (1975) define a graph on n vertices to be *panconnected* if any two vertices are connected by paths of every length ℓ , $d \leq \ell \leq n-1$, where d is their distance. Hendry (1990) calls a circuit C *extendable* if there is a circuit C' such that $v(C') = v(C) + 1$ and $V(C') \supset V(C)$, and a graph G *circuit extendable* if G contains at least one circuit and every nonspanning circuit of G is extendable. He also introduces the concept of a *pancyclic ordering*, a linear ordering $v_1, v_2, \dots, v_n$ of the vertices of a graph such that the subgraph induced by $\{v_1, v_2, \dots, v_i\}$ is hamiltonian for all i , $3 \leq i \leq n$.

7.5. The number of circuits

Results on the existence of circuits lead naturally to questions about their number, and to some intriguing conjectures. It is lower bounds on the number of circuits that are of principal interest. We consider, first, two problems on the total number of circuits.

The number of circuits in 3-connected 3-regular graphs was investigated by Barefoot et al. (1986), who observed that the total number of circuits in a 3-regular graph on n vertices which has no circuit of length more than n^c , where $c < 1$, is necessarily subexponential in n . Such graphs exist (see the discussion following Theorem 5.21). Barefoot et al. (1986) conjectured, however, that the number of circuits is always moderately large.

Conjecture 7.10. The number of circuits in a 3-connected 3-regular graph on n vertices is superpolynomial in n .

A long-standing conjecture on the number of directed circuits in a digraph is the following one, due to Adám (1964).

Conjecture 7.11. Let D be a strict digraph with at least one directed circuit. Then D has an edge whose reversal reduces the total number of directed circuits.

Although not explicit in Adám's formulation, the requirement that D be strict is essential. Examples of digraphs with multiple edges in which the reversal of any edge increases the number of circuits have been described by Thomassen (1987b) and by Grinberg (1987). Thomassen's examples are based on the existence of digraphs such as those described in Remark 5.44, in which the reversal of any edge increases the length of a longest directed circuit. Multiplying each edge suitably many times yields a digraph with the desired property.

Theorems asserting the existence of Hamilton circuits in graphs are numerous, prominent among them being those of Dirac (2.1), Ore (2.4), Chvátal and Erdős (4.16), Jackson (5.17), and Tutte (5.8). In each case, one may ask how many Hamilton circuits there must be. Surprisingly, of these theorems, only Tutte's has been investigated from this aspect (although the theorems of Dirac and Ore have been studied with regard to the number of edge-disjoint Hamilton circuits; see the discussion following Theorem 8.8). Hakimi et al. (1979) showed that a 4-connected planar graph on n vertices has at least $n/\log_2 n$ Hamilton circuits and conjectured that this bound can be improved to $2(n-2)(n-4)$.

A tournament has at least one directed Hamilton path, by Rédei's Theorem (3.1), but need not have more. Moon (1972) showed, however, that a strong tournament on n vertices has at least c^{n-1} directed Hamilton paths, where $c := 6^{1/4} \approx 1.565$. Thomassen (1985b) examined the same question for directed Hamilton circuits. By Camion's Theorem (3.9), a strong tournament has at least one directed Hamilton circuit but, again, need have no more. Thomassen nevertheless found an exponential lower bound in terms of the minimum outdegree: a strong tournament with minimum outdegree at least $3k+3$ has at least $4^k k!$ directed Hamilton circuits. Thomassen (1980b) also obtained an analogue of Moon's theorem, proving that a 2-strong tournament on n vertices has at least $c^{n/32-1} \approx 1.014^n$ directed Hamilton circuits. Thomassen (personal communication 1990) remarks that no construction is known of a tournament with more than $(n-1)!/2^n$ directed Hamilton circuits, although an elementary counting argument establishes the existence of such a tournament. A related question posed by Thomassen (personal communication 1990) is the following one.

Problem 7.12. Is there a diregular tournament on n vertices with at least $(n-1)!/2^n$ directed Hamilton circuits?

As already noted, Smith's Theorem (5.14) implies that a 3-regular hamiltonian graph has at least three Hamilton circuits. One may ask which have precisely three. The smallest example is the graph on two vertices joined by three edges, $K_2^{(3)}$. Bosák (1967) observed that any graph obtainable from this graph by recursively expanding vertices to triangles has the same property and conjectured, incorrectly, that there are no others; Ninčák (1974) found a nonplanar counterexample, a generalization of the Petersen graph, and asked whether Bosák's conjecture is true when restricted to planar graphs.

Problem 7.13. Let G be a 3-regular planar hamiltonian graph with precisely three Hamilton circuits. Can G be obtained from $K_2^{(3)}$ by recursively expanding vertices to triangles? Equivalently, if $G \neq K_2^{(3)}$, must G contain a triangle?

If G is a 3-regular graph with precisely three Hamilton circuits H_1, H_2, H_3 , the sets $E_i := E(G) \setminus E(H_i)$, $i = 1, 2, 3$, constitute a proper edge colouring in which the edges of any two colours induce a Hamilton circuit. The graphs possessing such an edge colouring were introduced by Kotzig (1962) under the name of *Hamiltonian graphs*; they are now referred to as *strongly hamiltonian*, *perfectly 1-factorable* or *Hamilton 1-factorable*. A selection of open problems about them is given in Kotzig and Labelle (1979).

It is not known whether a simple 4-regular hamiltonian graph must have at least two Hamilton circuits. Sheehan (1975) conjectures that this is so. Fleischner (1994) has constructed a 4-regular uniquely hamiltonian multigraph; thus the condition that the graph be simple cannot be dropped. Also, Entringer and Swart (1980) have constructed uniquely hamiltonian simple graphs of minimum degree three. No example with minimum degree four is known.

Problem 7.14. Is there a uniquely hamiltonian simple graph of minimum degree four?

8. Packings and coverings by paths and circuits

This is an important and extensive topic, rich in challenging open questions. The concepts of packing and covering are best explained in the context of hypergraphs.

8.1. Packings and coverings of hypergraphs

Recall that a hypergraph H consists of a set $V(H)$ of elements, called vertices, and a family $E(H)$ of subsets of $V(H)$, called edges. A *packing* of H is a set of disjoint edges, a *covering* of H a set of edges whose union is $V(H)$. A *partition* of H is a set of edges that constitutes at the same time a packing and a covering.

Given a hypergraph H , several questions naturally arise. How many edges can there be in a packing of H ? How few edges can there be in a covering of H ? Does H admit a partition? If so, how many or how few edges can there be in a partition of H ? Consideration of the first question leads to the concept of a *transversal* of a hypergraph, a set of vertices that meets every edge in at least one vertex. Packings and transversals are linked in a simple way.

Proposition 8.1. Let H be a hypergraph, $\mathcal{P}$ a packing of H , and T a transversal of H . Then

$$|\mathcal{P}| \leq |T|.$$

Proof. Because T is a transversal, each edge of $\mathcal{P}$ contains at least one vertex of T . On the other hand, because $\mathcal{P}$ is a packing, no two edges in $\mathcal{P}$ contain the same vertex of T . The inequality follows. $\square$

Proposition 8.1 implies the following inequality for any hypergraph:

$$\max\{|\mathcal{P}|: \mathcal{P} \text{ a packing of } H\} \leq \min\{|T|: T \text{ a transversal of } H\}.$$

Hypergraphs H for which equality holds in this inequality are said to have the *minmax property*. Such hypergraphs are of particular import in the field of combinatorial optimization. In order to find a packing of maximum cardinality or a transversal of minimum cardinality in a hypergraph with the minmax property, it suffices to find a packing $\mathcal{P}$ and a transversal T of the same cardinality: Proposition 8.1 guarantees that $\mathcal{P}$ is then maximum and T minimum. Finding such optimal packings and transversals efficiently is, of course, another matter. However, it is a striking fact that almost all packing problems for hypergraphs satisfying the minmax property do indeed admit polynomial-time algorithms. This phenomenon has, as yet, no theoretical foundation.

We now define what we mean by a packing or covering of a graph. Associated with a graph G and a family $\mathcal{F}$ of subgraphs of G are two hypergraphs, one having vertex set $V(G)$ and edge set $\{V(F): F \in \mathcal{F}\}$, the second having vertex set $E(G)$ and edge set $\{E(F): F \in \mathcal{F}\}$. Packings and coverings of these hypergraphs correspond in a natural way to packings and coverings of G by members of $\mathcal{F}$, and lead to the following definitions.

A family $\mathcal{F}$ of subgraphs of a graph G is a *vertex packing* of G if the members of $\mathcal{F}$ are vertex-disjoint, an *edge packing* if they are edge-disjoint, a *vertex covering* if $\cup_{F \in \mathcal{F}} V(F) = V(G)$ and an *edge covering* if $\cup_{F \in \mathcal{F}} E(F) = E(G)$. A *partition* (or *vertex partition*) is a vertex packing which is also a vertex covering, a *decomposition* (or *edge partition*) an edge packing which is also an edge covering. We shall be interested in the case where the family $\mathcal{F}$ consists of paths and/or circuits.

8.2. Packings by paths and circuits

For paths, the vertex-packing problem is trivial unless restrictions are placed on the paths. If they are required to be of length one, we have the maximum matching problem referred to in section 4 (at the end of the discussion of stability number). In this case, the minmax property is valid for bipartite graphs, but not in general – a subtler minmax property holds for all graphs (see chapter 3). If the paths are required to be of any other fixed length, the problem is $\mathcal{NP}$ -complete, as is the vertex-packing problem for circuits of any fixed length (Kirkpatrick and Hell 1983).

Vertex packings by circuits do not satisfy the minmax property: only one circuit can be packed into K_5 , for instance, whereas a transversal requires at least three vertices. Nevertheless, a relationship does exist between the cardinalities of a maximum packing and a minimum transversal.

Theorem 8.2. *Let G be a graph and k a positive integer. Either G has k disjoint circuits or there is a set of at most $ck \log k$ vertices in G meeting all circuits, where c is an absolute constant.*

This theorem, due to Erdős and Pósa (1962), leads to a straightforward polynomial-time algorithm for determining if a graph has k disjoint circuits, for

fixed k . Let $s := \lceil ck \log k \rceil$. One first checks each subgraph $G \setminus S$, where $S \subset V$ and $|S| = s$; there are no more than n^s of these, where n is the order of G . If no such subgraph is a forest, Theorem 8.2 guarantees that G has k disjoint circuits. If, on the other hand, some $G \setminus S$ is a forest, every circuit of G meets S . The circuits of G can then be enumerated directly. If C is a circuit meeting S in a subset R , there are at most n^2 choices for the two edges of C incident to any vertex v of R , and so at most n^{2r} choices for all the edges of C incident to R , where $r := |R|$. Moreover, each such choice gives rise to at most one circuit of G ; if there were two, their symmetric difference would be a nonempty even subgraph of the forest $G \setminus S$. Thus the number of circuits of G is polynomial in n . Each set of k circuits can now be checked individually.

Whether there is also a function f such that every digraph has either k disjoint directed circuits or a set of $f(k)$ vertices meeting all directed circuits is not known. This question was first asked by Gallai (1968b). Thomassen (1983d) noted that the existence of such a function would imply a polynomial-time algorithm for determining whether a digraph has k disjoint directed circuits. Younger (1973) conjectured that $f(2) = 3$. This was confirmed by McCuaig (1993).

Theorem 8.3. *Let D be a digraph. Either D has two disjoint directed circuits or there is a set of at most three vertices in D meeting all directed circuits.*

Corrádi and Hajnal (1963) gave a sharp bound on the minimum degree for a graph to have k disjoint circuits.

Theorem 8.4. *Let k be a positive integer and G a simple graph of minimum degree δ on n vertices, where $\delta \geq 2k$ and $n \geq 3k$. Then G has k disjoint circuits.*

Thomassen (1983a) conjectured that if n is a sufficiently large function of k , the hypothesis of Theorem 8.4 guarantees the existence of k disjoint circuits of the same length. This was proved by Egawa (preprint). Häggkvist (1985c) showed that regular graphs have many disjoint circuits of the same length; there are at least $n/64(\log_2 n)^2$, for instance, in a 3-regular graph on n vertices. For digraphs, Thomassen (1983d) proved the existence of a function f such that any strict digraph of minimum outdegree at least $f(k)$ has k disjoint directed circuits, and further conjectured the existence of a function g such that any strict digraph of minimum outdegree at least g has k disjoint directed circuits of the same length. He also ventured a conjecture as to the smallest value of the function f (see, also, Bermond and Thomassen 1981).

Conjecture 8.5. *Let k be a positive integer and D a strict digraph of minimum outdegree at least $2k - 1$. Then D has k disjoint directed circuits.*

Edge packings by xy -paths in graphs, and by (x, y) -paths in digraphs, satisfy the minmax property, by Menger's Theorem (4.1). Lucchesi and Younger (1978) established that edge packings by circuits also have the minmax property, in planar digraphs.

Theorem 8.6. *Let D be a planar digraph. Then the maximum number of edge-disjoint directed circuits in D is equal to the minimum number of edges of D whose deletion destroys all directed circuits.*

A related minmax equality for planar digraphs was conjectured by Woodall (1978b).

Conjecture 8.7. *Let D be a planar digraph. Then the minimum number of edges in a directed circuit of D (the girth of D) is equal to the maximum number of disjoint sets of edges of D the deletion of each of which destroys all directed circuits.*

Theorem 8.6 and Conjecture 8.7 are special cases of the Lucchesi–Younger Theorem and Woodall's conjecture, respectively. Stated in dual form, in terms of directed bonds rather than directed circuits, the latter apply to all digraphs, not just planar ones. A related result, on edge transversals of maximal directed edge cuts, was obtained by Maamoun (1985).

Theorem 8.8. *In any digraph, there is a directed path or directed circuit whose edges meet every maximal directed edge cut.*

McCuaig (1993) obtained an edge analogue of Theorem 8.3.

Theorem 8.9. *Let D be a digraph. Either D has two edge-disjoint directed circuits or there is a set of at most three edges in D meeting all directed circuits.*

Theorem 2.11 refines Dirac's Theorem (2.1) by giving a lower bound on the number of edge-disjoint Hamilton circuits in a graph satisfying the hypotheses of Dirac's Theorem (2.1). A similar refinement of Ore's Theorem (2.4), but valid only for graphs of relatively small minimum degree, was given by Li (1989). Corollary 4.18 shows that a simple $2d$ -regular graph on at most $4d + 1$ vertices has at least one Hamilton circuit. Jackson (1979) proved that every simple d -regular graph on n vertices, where $d \geq (n - 1)/2$ and $n \geq 14$, contains at least $\lfloor (3d - n + 1)/6 \rfloor$ edge-disjoint Hamilton circuits, deducing this from his Theorem 5.17. Thomassen (1982) conjectured the existence of a function f such that every $f(k)$ -connected tournament has k edge-disjoint directed Hamilton circuits. The case of triangles was examined by Tuza (1984), who made the following conjecture.

Conjecture 8.10. *Let G be a simple graph and k a positive integer. Either G has $k + 1$ edge-disjoint triangles or there is a set of at most $2k$ edges in G meeting all triangles.*

Tuza (1990) made a similar conjecture for directed triangles in tournaments and established a minmax equality for directed triangles in planar oriented graphs (Tuza 1995).

We have omitted mention here of conditions on the number of edges guaranteeing the existence of k disjoint or edge-disjoint circuits. For these, the reader may consult either chapter 23 or the book of Bollobás (1978).

8.3. Coverings by circuits

We have already seen a variety of ways in which theorems on the existence of long circuits can be generalized. Yet another way is to determine how few circuits it takes to cover the vertices or edges of the graph in question. For instance, Theorem 4.47 implies that a strong digraph on n vertices with stability number α has a directed circuit of length at least n/α , while Conjecture 4.33 makes the stronger assertion that α directed circuits suffice to cover the vertex set of the digraph. In the same vein, Kouider and Lonc (preprint) established the following generalization of Theorem 2.14.

Theorem 8.11. *Let k be a positive integer and G a simple graph on n vertices such that $\sigma_{k+1} \geq n$. Then V can be covered by k circuits, edges or vertices.*

When $\alpha(G) \leq k$, this is a consequence of Corollary 4.20. The following conjecture on edge coverings by circuits was proposed by Bondy (1990). If true, it would imply the theorem of Pyber (1985) that the edges of a simple graph on n vertices can be covered by $n - 1$ circuits and edges.

Conjecture 8.12. *Let G be a simple 2-connected graph on n vertices. Then $E(G)$ can be covered by at most $(2n - 1)/3$ circuits.*

Thomassen (1985b) proved that the edges of a diregular tournament on n vertices can be covered by $12n$ directed Hamilton circuits. This result is of interest with regard to a question of P.J. Kelly on decompositions of diregular tournaments into directed Hamilton circuits (see Conjecture 8.25 and the subsequent discussion).

The cardinality $|\mathcal{C}|$ of a circuit covering $\mathcal{C}$ is but one measure of the efficiency of the covering; another is the *length* $\|\mathcal{C}\|$ of the covering, defined by

$$\|\mathcal{C}\| := \sum_{C \in \mathcal{C}} e(C).$$

We denote the minimum length of a circuit covering of a 2-edge-connected graph G by $\text{cc}(G)$.

Thomassen (preprint) conjectured that the problem of determining whether $\text{cc}(G) \leq k$ is $\mathcal{NP}$ -complete. Nevertheless, a good lower bound for $\text{cc}(G)$ can be found in polynomial time by determining a shortest postman tour of G . A *postman tour* of a connected graph G is a closed walk that includes every edge of G . We denote the length of a shortest postman tour of G by $\text{pt}(G)$. A polynomial-time algorithm for finding a minimum-weight postman tour in an edge-weighted graph was developed by Edmonds and Johnson (1973). It is based on the shortest-path algorithm of Dijkstra (1959) and the weighted matching algorithm of Edmonds (1965). The idea is to find shortest paths between all pairs of vertices of odd degree, and then determine a pairing of these vertices for which the set of connecting shortest paths is of minimum total weight. A postman tour that traverses the edges of the selected paths twice and each other edge once is one of minimum weight. If $\mathcal{C}$ is a circuit covering of a 2-edge-connected graph G , the edge-disjoint union

of the circuits in $\mathcal{C}$ is an eulerian spanning supergraph of G and thus admits an Euler tour of length $\|\mathcal{C}\|$. The corresponding tour in G is a postman tour of the same length. It follows that

$$\text{cc}(G) \geq \text{pt}(G).$$

Clearly, $\text{cc}(G) = \text{pt}(G) = e(G)$ if G is eulerian, and $\text{cc}(G) \geq \text{pt}(G) > e(G)$ otherwise. An upper bound for $\text{cc}(G)$ in terms of $e(G)$ was conjectured by J.C. Bermond, B. Jackson and F. Jaeger (F. Jaeger, personal communication, 1989).

Conjecture 8.13. Every 2-edge-connected graph on e edges admits a circuit covering of length at most $7e/5$.

The bound of $7e/5$ is attained by the Petersen graph (fig. 8). Bermond et al. (1983) proved that a 2-edge-connected graph on e edges admits a circuit covering of length at most $5e/3$; this was generalized to edge-weighted graphs by Fan (1990).

8.4. Partitions into paths and circuits

The question of partitioning a graph or digraph into as few paths or circuits as possible is strongly related to – indeed includes – the hamiltonian problem. We have already studied many aspects of this question. Moreover, path partitions of digraphs and their connections to stable sets and colourings were discussed in section 4.

Path partitions of 3-regular graphs were examined by Reed (1989), who proved that any simple 2-connected 3-regular graph on n vertices can be partitioned into at most $n/9$ paths and conjectured a slightly stronger conclusion.

Conjecture 8.14. Let G be a simple 2-connected 3-regular graph on n vertices. Then G admits a partition into at most $n/10$ paths.

One may consider, more specifically, partitions of a graph or digraph into paths or circuits of prescribed length. El-Zahar (1984) proposed an appealing generalization of Dirac's Theorem (2.1) in this vein.

Conjecture 8.15. Let n be a positive integer, $n \geq 3$, and let $n_1 + n_2 + \dots + n_k$, where $n_i \geq 3$, be a partition of n into k parts, s of which are odd. If G is a simple graph of minimum degree δ on n vertices, where $\delta \geq (n + s)/2$, then G admits a partition into k circuits, of lengths $n_1, n_2, \dots, n_k$.

The case $k = 2$ was verified by El-Zahar. The case $n = 3k$ is true by virtue of Theorem 8.4, but would also be a consequence of the following conjecture of L. Pósa (see Erdős 1964).

Conjecture 8.16. Let G be a simple graph of minimum degree δ on n vertices, where $\delta \geq 2n/3$. Then G contains the square of a Hamilton circuit.

Fan and Kierstead (1995) established an asymptotic version of this conjecture by showing that G contains the square of a Hamilton circuit if, for some $\varepsilon > 0$, $\delta \geq (2 + \varepsilon)n/3$ and n is sufficiently large (as a function of ε). They also (Fan and Kierstead, preprint) proved the analogue (and corollary) of Conjecture 8.16 for Hamilton paths.

Theorem 8.17. *Let G be a simple graph of minimum degree δ on n vertices, where $\delta \geq (2n - 1)/3$. Then G contains the square of a Hamilton path.*

Aigner and Brandt (1993) verified a weaker form of Conjecture 8.16 in which the bound $(n + s)/2$ is replaced by $(2n - 1)/3$. This result is implied by Theorem 8.17.

8.5. Decompositions into paths and circuits

A *path decomposition* or *circuit decomposition* is a decomposition into paths or circuits, respectively, a *Hamilton decomposition* a decomposition into Hamilton circuits. Similar definitions apply to digraphs, a path or circuit decomposition connoting a decomposition into directed paths or circuits as the case may be. Circuit decompositions were touched on briefly in section 1. Just as Theorem 3.4 refines Rédei's Theorem (3.1), so Theorem 1.4 was refined by Toida (1973).

Theorem 8.18. *An even graph admits an odd number of circuit decompositions.*

Theorem 8.18 can be proved using Thomason's Lemma (6.1) (Bondy and Halberstam 1986); for a different approach, see Woodall (1990). Another refinement of Theorem 1.4, on decompositions of planar graphs into even circuits, was found by Seymour (1981).

Theorem 8.19. *Let G be a 2-connected planar even graph on an even number of edges. Then G admits a decomposition into even circuits.*

The number of circuits in a decomposition is the subject of a longstanding conjecture of G. Hajós (see Lovász 1968, and Dean 1986).

Conjecture 8.20. Every simple even graph on n vertices admits a decomposition into at most $(n - 1)/2$ circuits.

Hajós's conjecture was verified for planar graphs by Tao (1984). T. Gallai (see Lovász 1968) proposed an analogous conjecture on path decompositions.

Conjecture 8.21. Every simple connected graph on n vertices admits a decomposition into at most $(n + 1)/2$ paths.

Lovász (1968) proved that every simple connected graph on n vertices admits a decomposition into at most $n/2$ paths and circuits. If the graph is odd, such a decomposition must necessarily consist of $n/2$ paths; thus Lovász's theorem implies

Conjecture 8.21 in this case. Donald (1980) refined Lovász's argument, establishing that every simple connected graph on n vertices admits a decomposition into at most $(3n_0/4) + (n_1/2)$ paths, where n_0 and n_1 denote the numbers of vertices of even and odd degree, respectively.

A related question, on decompositions into circuits and edges, was considered by P. Erdős and T. Gallai (P. Erdős, personal communication, 1990; see also Erdős et al. 1966a, where it is discussed without attribution).

Conjecture 8.22. Every simple graph on n vertices admits a decomposition into at most cn circuits and edges, where c is an absolute constant.

Dean (1986) reviews analogues of Conjecture 8.20 for digraphs and poses several conjectures, of which the following is typical.

Conjecture 8.23. Let D be an eulerian oriented graph on n vertices. Then D admits a decomposition into at most $2n/3$ directed circuits.

Theorem 8.8, applied inductively, shows that a strict digraph on n vertices admits a decomposition into at most k directed paths or circuits, where k is the number of edges in a largest directed edge cut, and is thus at most $n^2/4$. This can be viewed as a digraph analogue of the theorem of Lovász noted above. O'Brien (1977) obtained a sharper result, a digraph analogue of Conjecture 8.21.

Theorem 8.24. Let D be a strict digraph on n vertices. Then D admits a decomposition into at most $n^2/4$ directed paths.

Alspach and Pullman (1974) defined the *excess* of a digraph D by the formula

$$\text{ex}(D) := \sum_{v \in V} \max \{d^+(v) - d^-(v), 0\},$$

noting that any decomposition of D into directed paths must have at least this cardinality because at least $\max \{d^+(v) - d^-(v), 0\}$ of the paths in the decomposition have v as their tail. Alspach et al. (1976) conjectured that this bound is tight if D is a tournament of even order. N.J. Pullman (see Reid and Wayland 1987) extended the conjecture to all regular oriented graphs of odd degree.

Conjecture 8.25. Let D be a regular oriented graph of odd degree. Then D admits a decomposition into $\text{ex}(D)$ directed paths.

Häggkvist (1985a) found very general results on circuit decompositions of almost-complete graphs and complete bipartite graphs. The basis of his approach is the following observation.

Proposition 8.26. If G is a path or circuit of length n , the composition $G[2K_1]$ is decomposable into two circuits of length $2n$.

Define a list $\{F_1, F_2, \dots, F_m\}$ of graphs to be *even* if each entry occurs an even number of times, and to be *proper*, for a graph H , if

- (1) $e(H) = \sum_{i=1}^m e(F_i)$, and
- (2) H has a subgraph isomorphic to F_i , $1 \leq i \leq m$.

An immediate consequence of Proposition 8.26 is that, if G has a Hamilton decomposition, $G[2K_1]$ has a decomposition into any proper even list of even 2-factors; in particular, $K_{4n+2} - M$, where M is a perfect matching, and $K_{4n,4n}$ both have this latter property, the first being isomorphic to $K_{2n+1}[2K_1]$, the second to $K_{2n,2n}[2K_1]$. Another (less immediate) consequence is that $K_{4n+2} - M$ admits a decomposition into any proper even list of even circuits not including circuits of length $4n$, and that $K_{4n,4n}$ admits a decomposition into any proper even list of even circuits not including circuits of lengths $8n - 2$, $8n - 4$ or $8n - 6$.

To conclude, we note a conjecture on triangle decompositions due to Nash-Williams (1970).

Conjecture 8.27. Let G be a simple even graph of minimum degree δ on n vertices and m edges, where $\delta \geq 3n/4$. Then G admits a decomposition into triangles provided that $m \equiv 0 \pmod{3}$ and n is sufficiently large.

This appears to be a very hard problem. The result of Gustavsson (1991) that the assertion holds if $\delta \geq (1 - 10^{-24})n$ is therefore significant.

8.6. Decompositions into Hamilton circuits

The decomposition of graphs and digraphs is a topic with a wealth of problems and results; indeed, a chapter could easily be devoted to decompositions of the complete graph alone. In a sense, such questions belong to the realm of combinatorial design (see chapter 14) where algebraic techniques are employed as a rule. However, graph-theoretical methods are often highly effective, as Proposition 8.26 well illustrates. The theory of edge colourings was applied by Hilton (1984) to establish the following elegant result on Hamilton decompositions of complete graphs (subsequently generalized by Nash-Williams (1987) using network flow techniques).

Theorem 8.28. Let r and n be positive integers, $r \leq 2n$, let G and H be complete graphs of orders r and $2n + 1$, respectively, and let $\{E_1, E_2, \dots, E_n\}$ be an n -edge colouring of G . Then there is a Hamilton decomposition $\{H_1, H_2, \dots, H_n\}$ of H such that $E_i \subset H_i$, $1 \leq i \leq n$, if and only if the spanning subgraph of G with edge set E_i is a path system with at most $2n + r - 1$ components, $1 \leq i \leq n$.

That complete graphs of odd order admit Hamilton decompositions is a classical result; one such construction, due to a Monsieur Walecki, is described in the book by Lucas (1891, pp. 161–164). Hilton (1984) explains how Theorem 8.28 can be used to find all such decompositions. An extension of Walecki's theorem to regular graphs of high degree (and at the same time a strengthening of Corollary 4.18 for even d) was proposed by Nash-Williams (1971b).

Conjecture 8.29. Let G be a simple $2d$ -regular graph on at most $4d + 1$ vertices, where $d \geq 1$. Then G admits a Hamilton decomposition.

Hamilton decompositions of cartesian products of graphs are the subject of a conjecture of Bermond (1978a), several special cases of which have been settled by Stong (1991), and a problem of Alspach and Rosenfeld (1986).

Conjecture 8.30. If G and H admit Hamilton decompositions, so does $G \square H$.

Problem 8.31. If G is a simple hamiltonian 3-regular graph, does the prism over G , $G \square K_2$, admit a Hamilton decomposition?

In fact, no simple 2-connected 3-regular graph is known whose prism fails to admit a Hamilton decomposition. Hamilton decompositions of 4-regular graphs were considered also by Thomason (1978), who discovered an analogue of Smith's Theorem (5.14) for such graphs.

Theorem 8.32. Let G be a 4-regular graph, and let e and f be two edges of G . Then the number of Hamilton decompositions of G in which e and f belong to different Hamilton circuits is even.

Berman (1986a) applied methods similar to those of Thomason (1978) to obtain yet another parity theorem: *the number of decompositions of a graph into a Hamilton circuit and a spanning tree containing a specified edge is even* (see also Chrobak and Poljak 1988). A corollary of Theorem 8.32 is that a $2d$ -regular graph with one Hamilton decomposition has at least $3^{d-1}(d-1)!$ such decompositions (Thomason 1978). The question as to which graphs admit Hamilton decompositions has received much study. A graph that can be decomposed into d Hamilton circuits must be regular of degree $2d$ and have edge connectivity $2d$. Petersen (1891) proved that a regular graph of even degree necessarily admits a decomposition into 2-factors. The examples of Meredith (1973) show, however, that a decomposition into Hamilton circuits is not guaranteed. Kotzig (1957) found an elegant necessary and sufficient condition for certain 4-regular graphs to admit Hamilton decompositions.

Theorem 8.33. The line graph of a simple 3-regular graph admits a Hamilton decomposition if and only if the graph is 3-edge-colourable.

Many 4-regular graphs without Hamilton decompositions can be produced using this theorem, the smallest being the line graph of the Petersen graph (fig. 8). But no 2-connected planar example can be so obtained because, by virtue of the Four-Colour Theorem (5.5), every simple 2-connected 3-regular planar graph is 3-edge-colourable. Such examples can, however, be found with the aid of a necessary condition akin to Grinberg's condition (Theorem 5.6), for a 4-regular planar graph to admit a Hamilton decomposition (Bondy and Häggkvist 1981). A partial analogue of Theorem 8.33 for 4-regular graphs was found by Jaeger (1983).

Theorem 8.34. The line graph of a simple 4-regular graph admits a Hamilton decomposition if the graph itself admits a Hamilton decomposition.

The above theorem is not a true analogue of Theorem 8.33 because the condition is not both necessary and sufficient. Indeed, the line graph of any simple 3-connected 4-regular graph has a Hamilton decomposition, by a result of Jackson (1991), whereas some 3-connected 4-regular graphs are not even hamiltonian. Jackson (1991) put forward the following general conjecture.

Conjecture 8.35. The line graph of a simple $2d$ -regular graph admits a Hamilton decomposition if and only if it has edge connectivity $4d - 2$.

Conjecture 8.25 includes a longstanding question of P.J. Kelly (see Moon 1968, p. 7, Exercise 9) as to whether every diregular tournament admits a decomposition into directed Hamilton circuits: deleting a vertex from any such tournament T on $2m + 1$ vertices, one is left with a tournament T' on $2m$ vertices and excess m ; a decomposition of T' into m directed paths is necessarily a decomposition into Hamilton paths, and these can be extended to yield a decomposition of T into directed Hamilton circuits. The corresponding question for bipartite tournaments was put forward by Jackson (1981a).

Conjecture 8.36. Every diregular bipartite tournament admits a decomposition into directed Hamilton circuits.

Hamilton decompositions of graphs and digraphs are surveyed by Alspach et al. (1990).

8.7. Compatible circuit decompositions

We discuss here a question on circuit decompositions of eulerian graphs which has important links to coverings. A *transition* in a graph G is a pair of adjacent edges. If v is a vertex of even degree in G , the edges incident to v can be partitioned into transitions; if v is a vertex of odd degree, all but one of these edges can be so partitioned. Specifying such a partition for each vertex v determines a *transition system* for G . Two sets of transitions are *compatible* if they are disjoint. Transition systems were first studied by Kotzig (1968).

An Euler tour induces a transition system whose elements are the pairs of consecutive edges of the tour. Likewise, a circuit decomposition induces a transition system whose elements are the pairs of consecutive edges of circuits of the decomposition. The tour and decomposition are *compatible* if the systems of transitions they induce are compatible (except at vertices of degree two). Kotzig (1968) proved that to each circuit decomposition of an eulerian graph there corresponds a compatible Euler tour. The converse was conjectured by G. Sabidussi (see Fleischner 1980).

Conjecture 8.37. Let G be an eulerian graph and T an Euler tour of G . Then G admits a circuit decomposition compatible with T .

Fleischner (1980) observed that the conjecture is true if $d(v) \equiv 0 \pmod{4}$ for each vertex v . Colouring the edges of T alternately red and blue induces a decomposition of G into two even subgraphs, one red and one blue. Each of these

subgraphs can be decomposed into circuits, resulting in a circuit decomposition of G compatible with T . Fleischner (1980) also proved a stronger theorem in the case of planar graphs. A transition system $\mathcal{T}$ is *nonseparating* if $G \setminus \{e_1, e_2\}$ is connected for each transition $\{e_1, e_2\} \in \mathcal{T}$.

Theorem 8.38. *Let G be a planar eulerian graph, and let $\mathcal{T}$ be a nonseparating transition system of G . Then G admits a circuit decomposition compatible with $\mathcal{T}$.*

This readily implies Theorem 8.19 in the case of 4-regular graphs, as Fleischner and Frank (1990) remark. As before, one colours the edges of an Euler tour alternately red and blue. Since the number of edges is even, each vertex is then incident with two edges of each colour. The pairs of edges of the same colour define a nonseparating transition system $\mathcal{T}$ of G . Any circuit decomposition compatible with $\mathcal{T}$ consists of circuits whose edges alternate in colour, and which are therefore even.

Remark 8.39. More generally, any edge colouring of a graph induces a set of transitions, namely the pairs of adjacent edges of the same colour; a circuit is then compatible with the edge colouring if no two consecutive edges of the circuit have the same colour. Such compatibility questions have been considered by a number of authors (see, for example, Grossman and Häggkvist 1983). In particular, Fleischner and Frank (1990) generalized Theorem 8.38 to sets of transitions induced by edge colourings in this way. An alternative set of transitions determined by an edge colouring is the set of all pairs of adjacent edges of different colour; in this case, a compatible circuit is one that is monochromatic. Compatibility questions of this latter type are touched upon in Remark 6.6.

A generalization of Dirac's Theorem (2.1) involving the notion of compatibility was proposed by R. Häggkvist (see Hellgren 1988, p. 15).

Conjecture 8.40. Let G be a simple graph of minimum degree δ on n vertices, where $\delta \geq n/2$ and n is sufficiently large, and let $\mathcal{T}$ be a transition system for G . Then G contains a Hamilton circuit compatible with $\mathcal{T}$.

The theory of eulerian graphs is extensive, and we refer the reader to the treatise by Fleischner (1990) for further details.

8.8. Double covers

By Theorem 1.4, every even graph admits a circuit decomposition, but no graph with a vertex of odd degree admits such a decomposition, however. In order to formulate an analogue of Theorem 1.4 valid for all graphs, one is led to the concept of a *circuit double cover*, that is, a collection $\mathcal{C}$ of circuits such that each edge of the graph belongs to exactly two members of $\mathcal{C}$.

A necessary condition for a connected graph to admit a circuit double cover is that it be 2-edge-connected, because each edge must lie in some circuit. Szekeres (1973) and Seymour (1979), motivated by quite different considerations, conjectured that this condition is also sufficient.

Conjecture 8.41 (The Circuit Double Cover Conjecture). Every 2-edge-connected graph admits a circuit double cover.

According to Tutte (personal communication 1992), the Circuit Double Cover Conjecture was already known in the 1950s. It has been studied extensively. A recommended reference is the survey article by Jaeger (1985); additional material can be found in Goddyn (1988), including a proof that the girth of a minimal counterexample is at least ten. The conjecture has close links to several other basic questions. It is implied by the combination of Conjectures 8.37 and 6.12 (Fleischner 1984). Perhaps more surprisingly, it is also implied by Conjecture 8.13 (Jamshy and Tarsi 1992). Moreover, to each 2-connected loopless graph can be associated an eulerian graph that has an even circuit decomposition if and only if the original graph has a circuit double cover (Jaeger 1985).

Several refinements of the Circuit Double Cover Conjecture have been formulated, of which we mention four. First, a conjecture on embeddings, due to Haggard (1977). An *embedding* of a graph G in a surface Σ is a function ϕ that assigns to each vertex a point of Σ and to each edge a simple curve in Σ , in such a way that curves $\phi(e_1)$ and $\phi(e_2)$ meet at point $\phi(v)$ of Σ if and only if edges e_1 and e_2 are incident with vertex v in G . The connected components of $\Sigma \setminus \phi(G)$ are the *faces* of G .

Conjecture 8.42. Every 2-edge-connected graph can be embedded in some surface in such a way that each face is bounded by a circuit.

Conjecture 8.42 implies the Circuit Double Cover Conjecture because the circuits bounding the faces in such an embedding constitute a circuit double cover. A still stronger conjecture is that the surface can be chosen to be orientable and the embedded graph 5-face-colourable (see Jaeger 1985).

Next, a conjecture of Seymour (1979) on edge-weighted graphs. A collection $\mathcal{C}$ of circuits in a graph G induces a weighting w of G , the weight $w(e)$ being the number of circuits of $\mathcal{C}$ that include edge e . Seymour asked which edge-weighted graphs arise in this way, that is, which edge-weighted graphs are *sums of circuits*. One obvious necessary condition is that:

(i) for each edge e , $w(e)$ is a nonnegative integer.

There are also two simple necessary conditions involving edge cuts, both consequences of the fact that a circuit meets an edge cut in an even number of edges:

(ii) for each edge cut S , $\sum_{e \in S} w(e)$ is even;

(iii) for each edge cut S and each edge $f \in S$, $w(f) \leq \sum_{e \in S \setminus f} w(e)$.

Seymour (1979) proved that conditions (i)–(iii) are sufficient for an edge-weighted planar graph to be a sum of circuits. They are not sufficient in general: a Petersen graph (fig. 8) in which the edges of a perfect matching have weight two and the remaining edges have weight one is not a sum of circuits. Alspach et al. (1994) extended Seymour's result by proving that this configuration is present in any edge-weighted graph that is not a sum of circuits.

Theorem 8.43. Let G be an edge-weighted graph whose weight function w satisfies

conditions (i)–(iii) above. Then either G is a weighted sum of circuits or there is a weighting w' of G such that

(a) $w'(e) \leq w(e)$ for each edge e of G ,

(b) $w'(e) \in \{0, 1, 2\}$ for each edge e of G , and

(c) the support of w' can be obtained from the edge-weighted Petersen graph described above by recursively applying the following two operations: subdividing an edge of weight one or two, and replacing an edge of weight two by two multiple edges of weight one.

Seymour (1979) conjectured that conditions (i)–(iii) are sufficient, however, if all weights are even (in which case, condition (ii) is automatically satisfied).

Conjecture 8.44. An edge-weighted graph with nonnegative even weights is a sum of circuits if and only if, for each edge cut S and each edge $f \in S$, $w(f) \leq \sum_{e \in S \setminus f} w(e)$.

Thirdly, an analogue of Conjecture 8.20 for double covers, proposed by Bondy (1990).

Conjecture 8.45. Every simple 2-edge-connected graph on n vertices admits a circuit double cover $\mathcal{C}$ such that $|\mathcal{C}| \leq n - 1$.

Two implications of this conjecture have been established. One, concerning path double covers, was obtained by Li (1990). A *path double cover* of a graph G is a collection $\mathcal{P}$ of paths such that each edge of G belongs to exactly two members of $\mathcal{P}$.

Theorem 8.46. Every simple graph G admits a path double cover $\mathcal{P}$ such that each vertex of G occurs exactly twice as an end of a path of $\mathcal{P}$.

The other is an implication for edge-weighted graphs, proved by Bondy and Fan (1991).

Theorem 8.47. Let G be a simple 2-edge-connected edge-weighted graph on n vertices. Then G contains a circuit of weight at least $2w(G)/(n - 1)$.

Fourthly, a conjecture due to Celmins (1984) and Preissmann (1981) on the minimum number of even subgraphs, rather than circuits, in a double cover.

Conjecture 8.48. Every 2-edge-connected graph admits a double cover by five even subgraphs.

Jaeger (1976) and Kilpatrick (1975) obtained a theorem on coverings by even subgraphs that is pertinent to Conjecture 8.48 and has other interesting implications.

Theorem 8.49. (i) Every 4-edge-connected graph is the union of two even subgraphs.

(ii) Every 2-edge-connected graph is the union of three even subgraphs.

Theorem 8.49(i) implies that every 4-edge-connected graph admits a double cover by three even subgraphs: such a graph is the union of two even subgraphs, and these subgraphs together with their symmetric difference constitute a double cover of the graph. Bermond et al. (1983) noted that Theorem 8.49(ii) likewise implies that every 2-edge-connected graph admits a quadruple cover by seven even subgraphs. This is not thought to be best possible, however. Fulkerson (1971) conjectured that every 2-edge-connected 3-regular graph admits a double cover by six perfect matchings, and Jaeger (1985) observed that Fulkerson's conjecture can be reformulated in terms of even subgraphs as follows.

Conjecture 8.50. Every 2-edge-connected graph admits a quadruple cover by six even subgraphs.

Fan (1992) proved that every 2-edge-connected graph has a sextuple cover by ten even subgraphs; for 3-regular non-3-edge-colourable graphs, this number is least possible. Many results on circuit covers, such as this one, are proved by the methods of integer flows. This topic is discussed in the appendix to chapter 4 by Seymour; another excellent source is the survey by Jaeger (1988).

We close by noting an attractive result of Chen (1971) and T. Gallai (see Lovász 1979, p. 254) akin to the two assertions of Theorem 8.49. An inductive proof due to L. Pósa is described in Lovász (1979); for a very short proof, see Woodall (1990).

Theorem 8.51. Every graph can be decomposed into an even subgraph and an edge cut.

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